Feature Selection for Distribution Shift via Meta-Learner

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Abstract

Distribution shift is ubiquitous in real-world machine learning systems, frequently causing performance degradation and undermining the economic value of deployed models. Although many existing methodologies study this phenomenon from a hypothesis-testing perspective, feature-level characterization and adaptation remain comparatively underexplored. This paper provides a systematic overview of statistical and machine learning methodologies for characterizing distribution shift at the feature level, drawing on techniques ranging from Shapley values and meta-learners rooted in causal inference to random forests. We review the benchmark methodologies and show that parsimonious procedures can perform competitively in real-data applications. For covariate shift, we use a random forest as the criterion for feature selection and find performance competitive with state-of-the-art methods. An optional post-hoc subset-level localization procedure quantifies the distance between two subsets via MMD for covariate shift and PO-risk for concept drift. For feature selection under concept drift, recently proposed meta-learner-based distance estimators achieve strong empirical performance and match state-of-the-art results under complex, highly nonlinear concept drift. We examine the efficiency of variable-importance metrics for characterizing feature-level contributions to distribution shift under overlap, model misspecification, and varying batch sizes, among other conditions. Sensitivity analysis covers overlap and the choice of propensity-score and outcome models in the shift-detection procedure. The methodology is validated on challenging real-world datasets with low signal-to-noise ratios, demonstrating the efficiency and broad applicability of FSDS. We also address deployment considerations, including parallelization strategies, runtime constraints, and an automatic fallback mechanism under poor overlap among batches. The resulting procedure converts statistical and machine learning methodology into an operational diagnostic tool that can be embedded in production ML systems. By producing ranked, interpretable feature outputs, it supports proactive feature-quality monitoring and distribution-shift diagnostics rather than reactive troubleshooting alone.

1 Introduction

Understanding the change of data is of high importance in real-world ML systems, which can be categorized into understanding the distribution shift and understanding the performance degradation. We focus on the first aspect: distribution shift, where the distribution of data evolves as time passes. Despite its practical significance, the feature-level characterization of distribution shift remains a largely under-explored area. Spanning the full spectrum from characterizing data change at the feature level to tracing its origins back to specific subsets of variables, it requires highly computationally efficient and robust characterization strategies for production machine learning systems—both at the variable-importance level and in the feature-selection scenario. Although most existing methods primarily target feature selection, hypothesis testing and online adaptation procedures do provide active insights into how to understand distribution shift from the feature level. We focus on the situation with two batches of data, where the existing batch is $\mathcal{D}_{exist} = \{\mathbf{X}_{exist}, \mathbf{Y}_{exist}\}$ and $\mathcal{D}_{new} = \{\mathbf{X}_{new}, \mathbf{Y}_{new}\}$. We split our discussion into two parts: the first primarily focuses on covariate shift, where the difference in covariate distribution satisfies $P(\mathbf{X}_{new}) \neq P(\mathbf{X}_{exist})$; the second focuses on concept drift, where the difference in conditional dependence satisfies $P(\mathbf{Y}_{new}|\mathbf{X}_{new}) \neq P(\mathbf{Y}_{exist}|\mathbf{X}_{exist})$. Current methodologies related to understanding model performance degradation and distribution shifts can be summarized into the following three streams of research:

- **Hypothesis Testing procedures** to detect whether there's a significant difference between the distribution of the two batches of datasets $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$—either for covariate shift or concept drift, or both. There are many existing methodologies along this direction, including but not limited to C2ST [37], conditional dependence test [35], and the recently proposed RFPerm (Random Forest Permutation Test) and CFPerm (Meta-Learner Objective Function Permutation Test).
- **Understanding the distribution shift at the sample/subgroup level**—especially which and how each of the subgroups is driving the distribution shift or the model performance degradation. The pivotal methods include Pro-FILE [**thiagarajan2021accurate**], ShapShift [**bewley2026shapshift**], and SHIFT [**singh2025experiences**], which aim to localize the change of the data to the subgroup and subset level.
- **Understanding the change of data from the feature level**. Decomposing model performance degradation into different feature levels requires strong prior knowledge about the causal graph or variable dependence graph among variables. On the other hand, distribution-shift characterization techniques include a few existing works [**chen2022estimating**, **zhang2023did**, 52], including but not limited to characterizing concept drift from the feature level and decomposing covariate shift into individual directions. However, accurately and concisely

characterizing distribution shift is a highly under-investigated area without any assumptions. In practice, model-agnostic attribution is definitely preferred over the two-step procedure starting from causal discovery followed by performance-drop decomposition. Our research addresses this gap by developing a model-agnostic procedure that is agnostic to the data-generating process, compatible with arbitrary data modalities, and robust to varying model complexities.

We make an advancement on feature selection under distribution shift by systematically evaluating existing methods against our own, and we use per-feature variable importance as a proxy, with a careful justification of its validity. We begin with two use cases that demonstrate its practical relevance.

1.1 Use Cases

We discuss two applied settings that illustrate the distinction between covariate shift and concept drift: fraud detection and market microstructure.

Fraud detection. In models designed to detect fraudulent behavior under continuously evolving fraud typologies and score distributions, both covariate shift and concept drift are present. Covariate shift arises from changes in the mixture of fraudsters and legitimate users: the marginal distribution of transaction-level, device-level, velocity, and network-based features shifts as the population composition changes. Concept drift arises when the conditional relationship between these features and fraud outcomes changes, for example because a given transaction pattern carries a different fraud probability after a change in fraud strategy. Accurately and parsimoniously identifying the subset of features that contribute significantly to either the fraudster-mixture shift or the fraud-pattern shift is therefore essential for producing interpretable, timely diagnostics that guide downstream model updates and case-review prioritization. The two-arm logic is directly applicable: features ranked highly only by the covariate arm point to changes in population composition and upstream data pipelines; features ranked highly only by the concept-drift arm point to changes in fraudulent behavior; features ranked highly by both indicate joint drivers.

Market microstructure. Structural changes in trading venues—tick-size changes, fragmentation, latency improvements, fee rebates, and the spread of algorithmic execution—alter the marginal distribution of order-book and trade-flow covariates. Depth, quoted spreads, queue imbalance, message rates, trade-size distributions, and order-flow autocorrelation can shift substantially across batches, and the covariate arm will rank these features as covariate-shift drivers. Separately, the same microstructure covariates can change their conditional relationship with future returns or risk outcomes. For example, a given level of order-flow imbalance may have a different price-impact or adverse-selection interpretation after a change in market structure; the price-impact mapping, order-flow toxicity, and short-horizon return predictability may drift even if the marginal distribution of order-book features is stable. The concept-drift arm captures this by flagging features whose permutation or removal materially increases the PO-risk distance. The attribution logic is again comparative: features unique to the covariate arm point to changes in market structure and input composition; features unique to the concept-drift arm point to changes in the price-formation mapping; overlapping features indicate that both the input distribution and the mapping have changed. In production, this separates "the market looks different" from "the market responds differently," which is precisely the distinction needed for feature-level monitoring and model maintenance.

In both settings, the object of interest is a ranked, interpretable feature-level characterization of distribution shift rather than a causal attribution. The feature-level output supports root-cause analysis, monitoring, and model maintenance, and provides business and risk teams with near-real-time evidence on how data changes relate to the operating context at both the system and feature levels.

1.2 Contributions and Organization

In addition to benchmarking existing methods, this paper makes three contributions. First, we provide a systematic review of feature-selection methodologies for distribution shift under both covariate shift and concept drift, covering a broad spectrum from ensemble learners to meta-learner-based distance estimation. Second, we show that a random forest classifier with variable importance achieves competitive performance for covariate-shift feature selection and is more computationally efficient than existing state-of-the-art methods. Third, we demonstrate that meta-learner-based approaches—specifically PermuCATE (permutation variable importance) and GRF (causal forest split variable importance, defined as the proportion of times a feature is split on)—achieve strong empirical performance and attain state-of-the-art results under nonlinear and arbitrary concept drift. We also offer production-level recommendations to machine learning practitioners, covering deployment practices and operational considerations, to facilitate the deployment of feature-level distribution-shift characterization in real-world ML systems.

The remainder of the paper is organized as follows. Section 2 presents the methodologies we employ. Section 3 describes the datasets and experimental settings. Section 4 summarizes and compares performance. Section 7 discusses production-level considerations. Section 6 examines the difficulty of disentangling covariate shift from concept drift, together with the associated empirical studies. Section 8 concludes and outlines future directions. The supplementary

materials provide implementation details in Appendix 9 and describe the complementary meta-learner testing procedure as a feature-selection method in Appendix 10. Appendix 11 presents the auxiliary federated aggregation procedure, justified via divide-and-aggregation across feature sub-blocks and evaluated for its feature-selection efficiency and variable-importance (VIMP) ranking.

2 Methodology

To address this gap in the the research landscape in Statistics and Machine Learning, we provide comprehensive evaluation & comparison for the current strategies and methodologies for **Feature-Selection-for-Distribution-Shift** leveraging a wide variety of current Statistical & Machine Learning methodologies and technologies. The introduction of each of the method we benchmark and compare are listed and discussed below carefully.

2.1 Covariate Shift

Understanding or attributing covariate shift at the feature level is a ubiquitous challenge that almost every data scientist faces in daily work when confronting two batches of data, a problem that has been investigated for many years [48]. A natural question is which variable drives the difference, so that targeted downstream business strategies can be conducted. We benchmark the following methodologies for feature selection via a wide variety of techniques. Traditional methodologies can be categorized into three streams: first, capturing marginal dependence between each individual feature and the response, such as mutual-information methods (cMIM, [26]) and minimal redundancy maximum relevance (MRMR, [64]); second, sparse-learning approaches such as Lasso ([36]) and group-lasso ([87]) that impose regularization on model structure to select features; third, model-based procedures that capture wrapper-based recursive feature-selection procedures, including [15] and [30]. Despite their popularity, these methods predominantly focus on marginal associations and often fail to account for complex, intractable correlation structures among features. Moreover, they are unlikely to perform well in low signal-to-noise ratio settings. Recent advances along this direction primarily focus on leveraging highly flexible deep neural networks ([6] and [7]) with claimed state-of-the-art performance, making the results cleaner and more efficient.

Separately, several methods have been developed to be robust to distributional shifts ([31], [71]). Our objective, however, is distinct: we aim to identify features that contribute most significantly to the shift itself, rather than merely being robust to it. In this work, we propose to leverage one of the most popular, robust off-the-shelf predictive modeling procedures for feature selection—through out-of-bag (OOB) variable importance, on top of the traditional MMD/HSIC statistical measures equipped with the leave-one-covariate-out (LOCO) procedure. We introduce the methodology carefully in this subsection.

First and foremost, we borrow the idea from the traditional two-sample testing procedure MMD (maximum mean discrepancy, [68]) and HSIC (Hilbert–Schmidt Independence Criterion, [1]) in estimating the distance via statistical measures (MMD/HSIC) between the two sets of covariates $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$, equipped with a LOCO (leave-one-covariate-out) procedure to estimate the contribution of each feature to covariate shift. Under covariate shift, if a direction (feature) contributes significantly to the shift, the statistical measure would drop substantially after removing that specific feature. Thus, the ranking of the LOCO difference for this measure can serve as a proxy for its contribution to covariate shift. We leverage the standard hyperparameter setting in their original works, where $\sigma = 1$ for the RBF kernel in MMD. In the benchmark comparison we denote them as **MMD-LOCO** and **HSIC-LOCO**. Additionally, a couple of other variable-importance methodologies are leveraged to evaluate feature-selection efficiency: ShadowVIMP ([58], shadow variable importance 2026), Vita Permutation Variable Importance ([39], variable importance testing), and Boruta ([45], one of the most popular methods for feature selection). All three methods are based on random forests as the component base model for a binary classifier distinguishing $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$, where the first two methods are evaluated through both AUROC (with variable importance) and F-1 score; on the other hand, Boruta only outputs the F-1 score with the binary feature-selection decision.

Delta-Shapley (Δ Shapley value, [8]): We compare the difference of feature-wise Shapley values on the two batches of data. Although no existing work describes this in detail, the absolute difference of feature-wise Shapley values across the two batches can represent their relative difference in individual contribution to that specific predictive performance (the magnitude, as the sign only indicates whether a positive or negative dependence exists). By definition, $\Delta(\text{shap})_j = |\text{shap}_{j,exist} - \text{shap}_{j,new}|$. Although this procedure was originally developed for characterizing a feature's contribution to predictive performance, it can be leveraged to understand each feature's contribution to distribution shift. In the benchmark experiments we denote this procedure as **Delta-Shapley**.

VIMP for the classifier. Borrowing ideas from one stream of hypothesis-testing procedures for distribution shift—building a classifier to distinguish the two batches of data $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$ (like classifier two-sample test, [38], and the **AutoML** two-sample test, [44])—we leverage the variable importance of the binary classifier built to distinguish the two batches of data as a proxy for their contribution to the distribution shift. Random forest ([12]), XGBoost ([14]), and the MLP ([75]) are leveraged, and performance is evaluated on a separate holdout set. No tuning is conducted for any of these procedures to ensure robustness. In the domain classifier we designed with the MLP, a 2–3 layer procedure

is adapted, and we denote this methodology as **Domain-Classifier** (rf-domain).

Localization Network We leverage the two recently proposed procedures for covariate-shift detection ([6] and [7]) for benchmark evaluations. They propose to compress statistical properties into embeddings and train a discriminator to maximize separability between two batches of data. Notably, these procedures do not need fine-tuning: per the discussion in their original papers, the pretrained neural network can potentially be generalized to arbitrary distribution shift. They also significantly reduce computational cost compared with pretraining from scratch.

We want to emphasize that for all of these procedures operating directly on the covariate space, although they can be directly extended to the joint distribution $P(\mathbf{X}, Y)$, they will not provide useful information regarding the feature-level contribution to concept drift—thus they are strictly focused on characterizing each feature's contribution to $P(\mathbf{X})$. They propose to compress the statistical properties of each batch into embeddings and train a neural network for feature-level shift-localization methodologies; we include these as benchmark comparisons in our adversarial perturbation experiments described in Section 3.1.

2.2 Concept Drift

We present the recently proposed meta-learner-based procedures for characterizing concept drift at the feature level. First, **LOCO with R-learner**: we compute the relative importance of each feature X_j in terms of the difference of the R-risk ([60]) before and after removing that feature, primarily targeted for concept drift. Second, **DRPerm**: the pseudo-outcome learner ([41]) with the LOCO procedure (the PO-risk), primarily targeted for concept drift while also, to some degree, covariate shift. Third, **GRF**: causal forest variable importance ([5]) with split-frequency VIMP, primarily targeted for concept drift while also, to some degree, covariate shift. Fourth, **PermuCATE**—conditional permutation variable importance for the DR learner through conditional fitting ([61]), primarily targeted for concept drift while also, to some degree, covariate shift. Enormous flexibility is included in this model selection for both propensity score models and outcome models. We leverage random forest for the outcome model and the conditional fitting model $X_j \sim X_{-j}$ (where X_j is continuous), and logistic regression for the propensity score model and the binary outcome conditional fitting model $X_j \sim X_{-j}$. When the outcome is multi-level, the conditional fitting model here is $X_j \sim X_{-j}$, whereas multinomial logistic regression is leveraged when the cardinality of the feature is larger than or equal to 3 for robustness considerations.

We want to mention that causal interpretation is out of scope for the currently proposed statistical methodology—specifically, the meta-learners we leverage, including R-learner, DR learner, causal forest, and Dragon-Net, are strictly restricted to the scope of distance/deviation estimation rather than drawing detailed causal interpretation. This is a very important reminder for every modeler aiming to leverage our proposed methodology.

The current literature on feature selection for distribution shift is fairly scarce; we benchmark the following methods. First, **SGShift** ([52]) incorporates a couple of methodologies for concept-drift explanation, including but not limited to de-randomized knockoff and sparse linear regression with correctly specified and misspecified component models. We further note that in their paper, the experimental design was fairly simple, injecting only linear concept drift, so the strong performance of penalized linear regression is expected by construction. The methods in their paper include the **l1-spec** (penalized regression with l1 penalty, correctly specified), **l1-misspec** (penalized regression with l1 penalty, misspecified model), and **knockoff** (de-randomized knockoff) procedures. Second, **ARD** (automatic relevance determination, [84]), primarily designed for feature selection and dimensionality reduction; the current implementation incorporates an optimization framework to characterize distribution shift (primarily concept drift, to some degree covariate shift). Additionally, all other methods for feature selection under covariate shift are leveraged for characterizing concept drift crosswise.

3 Datasets & Experiments

In this section we present the datasets and experimental settings we leverage for evaluating the efficiency and robustness of our proposed procedure. First we leverage the linear and nonlinear data-generating processes for evaluating the efficiency of the procedures. Specifically the non-linear data-generating processes are leveraged in the existing batch of data

$$\mathcal{D}_{new} - Y = T * (3X_1^2 + 3X_2^2 + 3X_3^2 + 2X_1X_2 + 2X_2X_3 + 2X_3X_4 + 1.5X_1X_4 + 1.5X_2X_4) + \beta * X + \epsilon$$

with nonlinear shift imposed otherwise the linear shift is imposed on the first 4 covariates where the $\Delta(\beta) = (1, 0.8, 0.6, 0.4)$ for $X_1 - X_4$ respectively. The visualization scripts can be found in the notebook - `https://colab.research.google.com/drive/1w5fKyTqnWoKEfixGnpaZCMSzQg0IOr2-#scrollTo=MuxSjR6uuxcP`.

We observe that many existing benchmarks in the meta-learner and causal inference literature (e.g., IHDP [34]) are overly simplistic for our task. Furthermore, the so-called "counterfactual perturbations" commonly used in multi-modal datasets often fail to rigorously justify true causal effects. To address these gaps, we construct a rigorous evaluation framework by injecting highly complex shifts into a diverse suite of simulated, semi-synthetic, and real-world structured datasets. Specifically, our evaluation covers simulated data, semi-synthetic data via WhyShift [51], real-world structured

panels from TableShift [29], and five low-SNR structured datasets: SECOM [54], TCGA [83, 77], PBMC3k [90], Diabetes130 [73] (an EHR proxy; cf. MIMIC [40]), and TEP [24].

To rigorously evaluate feature selection efficiency, we explicitly control the injection mechanism. For each panel (X, y) with p features, we partition the data into random source/target batches (X_0, Y_0) and (X_1, Y_1), and sample ground-truth indices $S \subset \{1, \ldots, p\}$ with $|S| = K = 10$. The shifts are planted exclusively on these K features using the source statistics $\sigma_j = \mathrm{sd}(X_0)_j$ and $\mu = \mathrm{mean}(X_0)$. We inject two types of shifts:

- **Linear Injection (Covariate Shift):** We keep Y unchanged and apply a linear mean shift to the target covariates on S as $X_1^{(j)} \leftarrow X_1^{(j)} + \mathrm{SNR} \cdot \sigma_j$.
- **Nonlinear Injection (Concept Drift):** We keep X unchanged, draw signs $s_j \in \{\pm 1\}$ with $\beta_j = \mathrm{SNR} \cdot s_j$ on S (and 0 elsewhere), and introduce complex nonlinear structures (e.g., interactions, quadratic, and sinusoidal functions) into the binary labels by drawing $Y_1 \sim \mathrm{Bernoulli}(\sigma(X_{1,z}\beta))$ with $\sigma(t) = (1+e^{-t})^{-1}$, leaving Y_0 unplanted.

Additionally, the WhyShift dataset is leveraged to perform 49 pairwise comparisons by imposing adversarial perturbations. We fix $\mathrm{SNR} = 1.5$ throughout. Each method returns a p-vector of scores; we report selection AUROC against $1_{j \in S}$ and Precision@K, which is equivalent to Recall@K since $|S| = K$.

3.1 Adversarial Perturbation

On top of the relatively simple linear and nonlinear distribution shift imposed on the new batch of data $\mathcal{D}_{new}$, we leverage a couple of adversarial perturbation strategies to inject a variety of shift types. Originally proposed for robustness training, the adversarial procedure can be readily leveraged to inject complex distribution shift for evaluating the efficiency of the **Feature Selection for Distribution Shift** procedure. Additionally, the procedure is highly adaptive to a variety of distribution-shift types, including but not limited to the following five methods: NesterovFGSM, MIFGSM, VMIFGSM, rFGSM, and Jitter. All are evolved variants of the fast gradient sign method (FGSM), designed to generate adversarial examples with improved transferability across black-box models. They address the fundamental limitations of vanilla FGSM—namely, susceptibility to local optima and lack of diversity—by introducing distinct optimization enhancements. MI-FGSM ([2]) incorporates momentum accumulation to stabilize gradient directions and escape sharp local maxima. NI-FGSM ([50]) extends this with a Nesterov-based momentum update procedure that anticipates future gradients, enabling more stable updates. **VMI-FGSM** ([82]) further adds a variance-tuning procedure to reduce gradient variance, preventing the optimizer from being trapped in suboptimal regions. These three methods form a progressive family of momentum-based attacking procedures, each built upon the previous to achieve increasingly reliable and transferable perturbations.

On the other hand, **rFGSM** ([4]) and **Jitter** ([69]) take a randomization-oriented procedure. rFGSM injects a small random perturbation before computing the gradient, diversifying the attack path and mitigating the catastrophic overfitting commonly seen in adversarial training. The Jitter procedure incrementally adds stochastic noise directly to the loss function to escape the local optimum. Together these techniques help enhance the robustness and transferability of the adversarial attack. In the implementation procedure, we incorporate a mask to isolate the subset of features to attack during the gradient update step and the momentum recording step, enforcing the adversarial attack on a certain subset of features. The evaluation protocol is similar to that in the other datasets, with F1 and AUC recorded against the ground-truth feature indices attacked.

In the implementation stage, the neural network is leveraged to predict (Y_{exist}, Y_{new}) using features $(\mathbf{X}_{exist}, \mathbf{X}_{new})$. Most importantly, through the gradient-based procedure and regularization on the 30% of features with the highest gradient $\nabla(X)$, the covariate shift of X dominates the procedure. On the existing batch of data, $X_{\text{perturbed}} = X + \epsilon \cdot \mathrm{sign}(\nabla_X l(X, Y))$ on the top 30% of features. We denote this procedure as **FGSM30**. Apart from imposing perturbation on the top 50% of features in $\mathbf{X}_{exist}$, we impose the gradient attack on Y_{new} in the new batch of data. Here, the gradient-based attack is imposed on the response Y via $\nabla_Y l(X, Y)$, where $Y_{\text{perturbed}} = Y + \epsilon \cdot \mathrm{sign}(\nabla_Y l(X, Y))$. We denote this procedure as **FGSM50**. Under this circumstance, both highly nonlinear concept drift and covariate shift exist. A variety of **OpenML** ([9]) datasets are leveraged for model evaluation. The details and scales of the datasets we leverage here are displayed in Table 14.

We want to mention that to inject synthetic shift, the FGSM procedure is indeed imposed on the regression problem where the neural network is fitted to predict (Y_{exist}, Y_{new}) using $(\mathbf{X}_{exist}, \mathbf{X}_{new})$; then after imposing the FGSM attack on 30% of the features in $\mathcal{D}_{new}$, we indeed inject complex concept drift, efficiently mimicking real-world scenarios with intractable perturbations on the conditional dependence of $Y|\mathbf{X}$. To summarize, although subtleties remain, most adversarial perturbation procedures primarily inject covariate shift more than concept drift, unless the regressor is specifically attacked on one of the batches of data extensively through a momentum-based procedure; they mainly serve as benchmark comparison methods for covariate-shift feature-selection procedures.

3.2 Overlap Sensitivity Analysis

Poor overlap between the treatment group and control group is known to severely degrade the estimation quality of both propensity score and outcome models in the causal inference regime; in our meta-causal-learner framework, it is also very important to demonstrate the robustness of feature-selection efficiency under poor overlap. We assess the impact of this overlap issue on feature selection regardless of whether causal interpretation is yielded. This challenge has received ongoing attention across both the causal inference and machine learning communities, with recent work addressing high-dimensional settings ([22]), overlap quantification via o-values ([46]), and representation learning under weak overlap ([19]).

Under our framework of causal-learner-as-distance-estimation, poor overlap between the supports of $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ would, in principle, undermine the quality of distance estimation—a well-known challenge in the causal inference literature. Thus we conduct numerical studies for the lack of overlap between the two batches of data through the repliclust package ([89]) with flexible specification of the overlap ratio between $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ (ranging from 0.01 to 0.5 in our experiments); the AUC for the logistic regression built to distinguish the two batches of data $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$ is recorded for quantifying feature-selection efficiency across various settings. As demonstrated in Tables 30 and 30, the non-causal methods are fairly robust with respect to distribution shift, which aligns perfectly with their design. Moreover, as demonstrated in Figures 10 (concept drift) and 11 (covariate shift), the covariate-shift feature-selection methods, ranging from RF-classifier and FSL-Net to DataFix, are very robust across different overlap proportions; the penalized regression and knockoff procedures are also fairly robust under poor overlap. On the other hand, the tree-based model as the outcome model and propensity score model is very concise and helpful for robust estimation. Thus it aligns with the theoretical support in meta-learner methods, where, to ensure efficiency in practice, at least a moderate degree of overlap between the two batches of data (groups) is required and highly desired.

Similarly, for the high-dimensional dataset with more than 300 covariates, the eigen-shift is implemented on top of the SVD procedure whereas the degree of overlap can be characterized by the AUROC for the logistic regression model fit to distinguish the two batches of datasets where we borrow the ideas from the Classifier Two-Sample Test([38]). The logistic regression is leveraged to characterize the degree of linear separation to avoid the extra sensitivity introduced by this procedure. Moreover, the model here is highly flexible in terms of the dataset itself - providing opportunities to characterize the overlap instead of the traditional statistical measure for distance estimation.

In our framework, we treat the objective function of the meta-learner (pseudo-outcome risk as well as the R-risk) as a tool for distance estimation between two batches of data. Empirically, we observe that tree-based outcome models, including random forests and causal forests, exhibit strong robustness even under poor overlap. This is largely due to their recursive partitioning structure, which adapts to local covariate distributions and mitigates the curse of weak common support. This robustness makes them a practical and defensible default choice in real-world deployments, where overlap conditions are often unknown or changing over time. We emphasize that the robustness of distance estimation can be substantially enhanced by choosing a robust outcome model specification—specifically, random forests or causal forests instead of neural networks or support vector machines—not for treatment-effect estimation or interpretation, but rather only for reliable distance measurement.

3.3 Outcome Model and Propensity Score Model

A critical practical consideration in both machine learning modeling and meta-learner frameworks from causal inference is the appropriate choice of the outcome model; selecting parsimonious yet efficient models for those components is essential to ensure robust and efficient estimation in practice. In this subsection we justify that the random forest model is among the best in both propensity score and outcome models for feature selection under distribution shift. To demonstrate this choice, we conducted comprehensive sensitivity analysis over the propensity score model (logistic regression, random forest, XGBoost, and MLP) as well as the outcome model (kernel ridge regression, random forest, XGBoost, and MLP) in both the PermuCATE and DRPerm procedures to evaluate overall feature-selection efficiency in terms of F-1 and AUC under various perturbations. As demonstrated in Table 33 and 33, the random forest outcome model is consistently among the best methods for feature selection, better than MLP and kernel ridge by a large margin and more robust than XGBoost.

Similarly for the propensity score model, we compare feature-selection efficiency for different propensity score models—MLP, logistic regression, XGBoost, and random forest. Random forest also achieves enhanced efficiency (as demonstrated in Tables 31 and 32) compared with existing methods such as logistic regression and MLP, further supporting the practical recommendation that if one adapts the meta-learner for distance estimation, random forest is highly preferred in both the propensity score and outcome models. While we neither expect nor require the propensity score model to achieve strong predictive fit in practice, the random forest classifier preserves improved robustness along with the flexibility to capture more complex decision boundaries than logistic regression, which focuses on a linear decision boundary for distinguishing $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$. Although it contradicts common practice that logistic regression is preferred for the propensity score model in biomedical applications ([32], [72], [76]) and financial domains ([23], [65], [62]), we want

to emphasize this insight to every ML practitioner: for accurate feature selection via the meta-learner, random forest is definitely what you need, aligned with the regime for feature selection under distribution shift.

3.4 Random Forest's role in Feature Selection for Distribution Shift

We adopt random forest as both the propensity score model and the outcome model within the meta-learner framework for feature selection under concept drift and covariate shift. Relative to logistic regression, MLP, and XGBoost, random forest offers a key practical advantage: it effectively captures highly nonlinear and interacting decision boundaries without manual specification. More importantly, its robustness under a wide variety of challenging scenarios has long been demonstrated ([91], [56], and [55]), enabling robust fitting of the binary classifier. As established in [70], random forest is capable of adapting to sparsity efficiently, with convergence rates depending on the number of active features rather than the ambient dimension. This ensures reliable distance estimation between the existing and new data distributions, making the meta-learner procedure both efficient and robust for feature selection.

The adaptivity of random forests to the intrinsic low-dimensional structure, combined with their inherent robustness to noisy and high-dimensional covariates, naturally lends itself to improved out-of-distribution detection. By focusing on the active feature subspace and avoiding overfitting to irrelevant noise, the forest produces stable and reliable predictions that are sensitive to meaningful shifts in the covariate distribution. This adaptivity-robustness duality enables more reliable identification of distributional changes, which is precisely the foundation for effective feature selection under covariate shift and concept drift - in terms of the reliable OOD characterization across each of the dimension.

4 Performance Summarization

The proposed Feature Selection for Distribution Shift procedure has shown high competency in detecting complex, nonlinear concept drift, achieving very strong empirical performance and attaining state-of-the-art performance under nonlinear and arbitrary concept drift compared with C2ST, ARD (automatic relevance determination), Delta-Shapley value, knockoff, and penalized regression—as vividly demonstrated in the nonlinear concept drift and adversarial perturbation injection settings. Moreover, the DR (doubly robust) pseudo-outcome learner also shows highly competitive performance in both covariate shift and concept drift (Figures 6, 7, 8, and 9). We want to emphasize that, although very powerful and efficient for linear concept drift (Figures 8), the penalized regression procedure proposed in SGShift ([53]) is much less efficient under covariate shift and nonlinear concept drift, which aligns perfectly with the linearity assumption when it was proposed—illustrating its less favorable position for understanding complex shifts of conditional dependence, which further enhances the capability of our meta-learner-based hypothesis-testing methodology. In the meta-learner-based methods (PermuCATE and DR-Learner), we recommend random forest as the outcome model because it is relatively less sensitive to poor overlap; since our goal is distance estimation, not causal interpretation, this robustness makes it a good fit.

On the other hand, under covariate shift, the **MMD-LOCO** and **RF-domain** procedures dominate in terms of efficiency. Under very complex attacks on the classifier, the random forest procedure works slightly better (Table 18), whereas for simpler attacks or lower-dimensional data, the **MMD-LOCO** procedure is slightly better in terms of the tradeoff between computational efficiency and robustness. The degradation of **l1-misspec** and **l1-spec** under nonlinear, complex adversarial attacks demonstrates the significant limitation of the benchmark methodology ([53]), further enhancing the necessity for MMD/RF-based procedures. We want to emphasize that the low F1/AUC for the R-risk procedure is perfectly expected for datasets across modalities, as this procedure is intentionally designed to be invariant under covariate shift. Moreover, the Delta-Shapley value (absolute difference of Shapley values across each feature) procedure performs well under covariate shift rather than concept drift, aligning with its deviation of objective from characterizing distribution shift to characterizing the change in the contribution of predictive accuracy. Similarly to the meta-learner, the random forest and MMD-based procedures both possess exceptional robustness under poor overlap between the supports of $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$.

Apart from the feature-selection procedure, we want to emphasize that the highly competitive performance of C2ST (classifier two-sample test) via random forest is also very practical in production: as long as one random forest classifier is fit well, the feature-selection performance would be superior to most benchmark methods, highlighting the potential for random forest to shine in production environments. The MMD-based procedure is highly recommended given enough computational resources because of its highly robust and accurate feature-selection performance than the Random Forest Binary Classifieer Variable Importance as the Domain Classifier. More importantly, the density-estimation nature of random forest can further enhance the robustness and efficiency of the variable-importance procedure. Thus the random forest procedure would be highly preferred for understanding covariate shift at the feature level.

Tables 26 and 25 report results on low-SNR datasets with targeted perturbations. **FSDS (Feature Selection for Distribution Shift)** is the only method that remains competitive under *both* drift regimes, significantly outperforming the best baselines. Under covariate shift, FSDS's internal domain arms (RF-domain LOCO and MMD-LOCO) recover

the planted support almost perfectly, and FSDS inherits this advantage, whereas PO-risk alone stays near the chance level. Conversely, under concept drift, the ranking flips: PO-risk becomes the informative arm while MMD and RF-domain collapse, yet FSDS successfully tracks the stronger arm and consistently beats SHIFT and GRF. This confirms our design intent: FSDS is robust precisely because it *shares* RF/MMD components (for covariate shift) with the PO-risk component (for concept drift), rather than relying on a single mechanism, whereas every baseline is strong on at most one side. Beyond these low-SNR panels, we evaluate on thirteen semi-synthetic datasets created by injecting controlled shifts into real data. As detailed in the subsequent tables grouped by drift type (concept drift, nonlinear drift, adversarial covariate shift, and mixed shifts), the meta-learner-based procedures achieve the highest mean AUC/F1 in the majority of settings. The rare exceptions are expected by construction: ℓ_1 is competitive mainly under sparse linear concept drift, and certain datasets isolate pure covariate shift. Neither reflects the mixed, nonlinear, and unknown shifts typical of real data—precisely the regime where meta-learner procedures excel without pre-specifying the shift type.

This pattern is visible both in the aggregate summaries and in the per-dataset breakdowns, rather than being driven by a single dataset or perturbation type. As can be observed, the testing procedure based on meta-learners is highly competitive among all benchmark methods, especially when concept drift dominates covariate shift. On the other hand, the **Random Forest** based procedure has demonstrated high efficiency for characterizing covariate shift, regardless of the overlap. The SGShift (l1misspec and l1spec, [53]) based procedure works well under linear shift but struggles under nonlinear distribution shift, which aligns with its original objective for penalized regression. Among the testing procedures via meta-learners, the PermuCATE procedure is among the most efficient (especially when concept drift dominates covariate shift, as in Table 15), while the computationally efficient **GRF** (causal forest with split VIMP) possesses highly competitive performance compared with existing methods, making both procedures highly practical in real-world application scenarios.

To further evaluate the efficiency of the random forest classifier, we benchmark a variety of other datasets primarily for understanding the effect of covariate shift. Across other modalities of the datasets (FMoW Image dataset—[18], ChronoBerg—[33] semantic shift datasets), all of them are transformed via representation extraction procedures—Gemma3 ([67]) and Mongo-Leaf ([79]) for text data, and Video Transformer ([66]) for image data. We want to mention that for the ChronoBerg, FMoW ([18]), and Wilds Medical datasets, covariate shift is strictly dominated because labels are constructed via the order of observations—thus between-group covariate change dominates the change in labeling logic across different batches.

Specifically, five forms of adversarial perturbations are injected, and the procedure remains highly efficient in characterizing data change in continuously evolving data-generating processes. By construction, the ChronoBerg and Wilds datasets are constructed for covariate shift with pseudo-labels imposed with respect to time order (from index 1 to the number of observations in the dataset). As demonstrated in Tables 22 and 24, **LOCO-MMD** and **rf-domain** are among the most efficient, much more efficient than the knockoff and penalized regression versions—although in some circumstances the Delta-Shapley value based procedure also works efficiently, the objective mismatch is a significant drawback for interpretation.

From the time-profiling perspective, as demonstrated in Tables 28 and 29, the random forest classifier procedure followed by variable importance is among the most accurate and most computationally efficient for feature selection in the context of covariate shift, thanks to the very robust performance of random forest in high-dimensional covariates and efficient prediction in a wide variety of very challenging data-generating scenarios. While in the low-to-moderate number of covariates, LOCO-MMD does have its advantage, in the high-dimensional setting the random forest procedure plays its role, which is highly competitive with respect to NN-based methods including FSL-Net ([7]) and DataFix ([6]). Thus the very low training cost and deployment requirement, with near-real-time inference, equipped with the robustness of random forest, make it preferable in the production environment.

4.1 Additional Benchmark Comparison

We conduct additional benchmark comparisons for the most recent benchmark for the covariate shift - DataFix([6]) and the FSL-Net([7]) both in terms of the time complexity and the detection efficiency. The random forest based VIMP procedure is highly competitive in comparison to these two neural network based procedures. We want to mention that these are two parallel lines of research where their methods also employ good flexibilities but in practice, the random forest procedure does plays its role in feature selection for covariate shift. As demonstrated in the experimental results, the **FSL-Net** procedure works slightly better than the proposed methods but more computationally expensive and the **DataFix** procedure is less competitive than either the Random Forest based procedure or the MMD based procedure.

We also benchmark other feature-selection methods for covariate shift built upon tree-based models, including Boruta ([45]), ShadowVIMP ([58]), and VITA-permutation variable importance ([39]). These methods are competitive in some regimes, but they do not consistently provide additional benefit over the vanilla OOB-variable importance procedure based on random forest. Unlike the three alternatives—which involve non-trivial resampling procedures and multiple-testing corrections—RF-VIMP produces a direct importance ranking from the trained forest at virtually no extra cost. This simplicity not only speeds up daily re-evaluation but also makes the results easier to interpret and communicate across non-technical audiences, a practical advantage that often outweighs marginal gains in selection accuracy.

4.2 Summary

To summarize, the random forest classifier (RF-domain, high-dimensional covariates) and MMD-LOCO (lower-dimensional covariates) are best for feature selection under covariate shift. On the other hand, for feature selection under concept drift, the PermuCATE (concise feature selection) and causal forest split VIMP procedure (most parsimonious, relatively efficient) are recommended in practice; they achieve very strong empirical performance and attain state-of-the-art performance under nonlinear and arbitrary concept drift, outperforming existing benchmarks for any type of change more complex than linear concept drift. These are very valuable explorations, providing data scientists with a concise way to understand data change at the feature level. On the other hand, the RF-VIMP/MMD-based procedure is highly competitive with respect to the existing state of the art (FSL-Net) in terms of performance and is more computationally efficient than existing methods, demonstrating practical applicability. Additionally, careful sensitivity analysis has revealed the enhanced robustness of the classifier-based procedure.

The practical implications of these findings are compelling: several of our recommended methodologies offer practical advantages—from relatively light tuning and straightforward parallelization to exceptional robustness under poor overlap. We discuss their deployment in production in Section 7.

5 Variable Importance for Distribution Shift and its proxy to feature selection

In this section, we establish the duality between variable importance (VIMP) and feature selection within the proposed framework. We consider two complementary settings: a random forest based domain classifier for covariate shift, and a DR pseudo-outcome learner with PermuCATE and LOCO importance for concept drift. In both cases, the VIMP ranking serves as a coherent and interpretable proxy for feature selection, justifying our proposed Feature Selection for Distribution Shift (FSDS) procedure. Crucially, this ranking also acts as a reliable proxy for prediction accuracy, offering practical guidance for model diagnosis and feature pruning under distribution shift.

For characterizing covariate shift via random forest, the variable importance derived from a binary classifier trained to distinguish between $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ offers an efficient and interpretable measure of each feature's marginal contribution to the distributional change. This approach is rooted in the classic random forest framework of Breiman (2001), with extensive subsequent developments on the interpretation and calibration of variable-importance measures (e.g., [74] and [49]). In particular, [80] established the superior efficiency of out-of-bag (OOB) variable importance in both regression and classification settings, supporting its use as a reliable diagnostic tool for feature importance. More importantly, this procedure is highly efficient and agnostic to the underlying data-generating process. In our specific context—where the binary classifier distinguishes between the existing and new batches—the resulting variable importance can be directly interpreted as each feature's individual contribution to the observed covariate shift. This aligns with the classifier-based testing framework proposed and leveraged in [38], where separability between two batches serves as the basis for detecting and characterizing distributional changes in covariates X. We further mention that this notion of variable importance is indeed generalizable to other model types beyond random forest, such as MLP and XGBoost, where each feature's contribution collectively characterizes covariate shifts.

On the other hand, we leverage the pseudo-outcome risk equipped with the permutation variable importance and the LOCO(leave-one-covariate-out) procedure for estimating each individual feature's contribution to distribution shift with a specific focus on concept drift. The doubly robust pseudo-outcome learner is originally proposed to provide causal effect estimation where as long as one of the propensity score model(binary classifier to distinguish the two covariate spaces) and the outcome model(regress the response on the covariates) are correctly specified, the causal effect would be unbiased. In this work we leverage this pseudo-outcome risk as the objective function for estimating the distance between the two batches of data $(\mathbf{X}_{exist}, Y_{exist})$ and $(\mathbf{X}_{new}, Y_{new})$ within the meta-learner-as-distance-estimation framework we proposed.

The pseudo outcome is calculated as

$$\psi_i = \frac{T_i - \hat{e}(\mathbf{X}_i)}{\hat{e}(\mathbf{X}_i)(1-\hat{e}(\mathbf{X}_i))}(Y_i - \hat{\mu}_{T_i}(\mathbf{X}_i)) + (\hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i)) = (1) + (2).$$

Statistically, this object admits a direct interpretation: part (1) is the propensity-score-weighted residual, which measures how well the current model fits the data under the existing covariate distribution (introducing the shift in term 1). Consequently, a covariate shift in the new batch will alter the propensity score $e(X)$ (as reflected in its classification performance, the binary classifier distinguishing the two covariate spaces), thereby directly influencing the downstream PO-risk. The second term captures the change in the conditional difference between the two batches, which is precisely the estimand for the conditional mean difference $\tau(X) = E[Y|X, T = 1] - E[Y|X, T = 0]$. This term was originally designed to quantify the conditional mean structure—the first-order effect on the conditional expectation of φ (introducing the shift in term 2, the outcome model). Thus, the two components together provide a concise interpretation of variable importance for both concept drift and covariate shift, as each part contributes to a distinct aspect of the shift. We then estimate the doubly robust distance estimator $\tau_{DR}(\mathbf{X}) = E[\psi|\mathbf{X}]$ and compute the PO-risk as $R_{PO} = \sum_{i=1}^{n} \tau_{PO}(X)^2$,

which can efficiently handle both concept drift and covariate shift and serves as the distance estimator between the two batches of data $\{\mathbf{X}_{exist}, Y_{exist}\}$ and $\{\mathbf{X}_{new}, Y_{new}\}$. From this perspective, the first component serves primarily as a covariate-shift detector, while the second component is more directly aligned with concept drift. Together, these two components offer a concise and coherent interpretation of variable importance under both types of distributional change, with each part addressing a distinct source of shift. Under this framework, both permutation-based variable importance—defined as the difference in PO-risk before and after permuting a given feature—and the LOCO procedure—defined as the difference in PO-risk with and without the feature—effectively capture each feature's marginal contribution to the objective function (PO-risk) as a distance estimator. This naturally induces a notion of variable importance as the feature-specific contribution to the overall distribution shift between the two batches. We emphasize that the doubly robust property of this pseudo-outcome learner is established for treatment-effect estimation under counterfactual reasoning, not for our distance-estimation regime. The efficiency of our procedure is demonstrated directly from the construction of the pseudo-outcome (PO) risk, without causal interpretation.

Similarly, we leverage the R-risk ([60]) to characterize the distance between the conditional distributions of the response given predictors, $D(Y_{new}|\mathbf{X}_{new}, Y_{exist}|\mathbf{X}_{exist})$, where it is invariant under covariate shift justifiably from both a geometric aspect and a double/orthogonal machine learning ([16]) aspect. Thus it provides solid theoretical guarantees and statistical insights innate to distribution shift, primarily focusing on the different aspects of R-risk and PO-risk—focusing primarily on concept drift. The detailed justifications are deferred to the supplementary materials **??**.

Thus, combining all of the previous justifications, we are able to demonstrate that variable importance is capable of serving as a proxy for feature selection under both covariate shift and concept drift.

6 Feature Selection & Driver for Distribution Shift

A very important practical problem for feature selection under distribution shift lies in the fact that, given the selected top-K features $\mathbf{X}_1, ..., \mathbf{X}_K$, how can we give people concrete ideas of whether they are driving covariate shift, concept drift, or both, and can we attribute the distribution shift to individual features? Although empirical evidence and recommendations can be proposed given experimental results from benchmark methods, a concrete and concise procedure is highly desired. Our answer to this question is "impossible": under no other assumptions on correlation structures among features or the complexity of data-generating processes, it is impossible to attribute global distribution shift to individual feature levels, let alone disentangle covariate shift and concept drift. This corresponds to the identifiability issue regarding the impossibility of decomposing covariate shift and concept drift into separate sources and individual features. We carefully justify this via empirical studies.

6.1 Impossibility issues for the unique attribution for distribution shift

From a statistical standpoint, the decomposition of covariate shift and concept drift into individual feature levels, or even disentangling their effects, is inherently ill-posed. This challenge parallels the difficulty of attributing degradation in predictive performance—such as an increase in mean squared error (MSE)—to a single feature, since any marginal contribution is invariably conditional on the presence of all other covariates. Without imposing explicit or very strict structural assumptions on the underlying covariance or correlation matrices, achieving a unique and identifiable decomposition of the global distribution shift is fundamentally unattainable. We conduct a series of empirical data studies to illustrate this issue. Two batches of data are simulated through highly nonlinear data-generating processes with both concept drift and covariate shift, with distribution-shift measures MMD and HSIC (for covariate shift) and R-risk (for concept drift) measured for the two batches of data. Then random feature-inclusion paths are simulated: the order in which each feature is included is simulated with 20 repeats, and for each feature included in the data-generating model, the increase in MMD/HSIC and the decrease in R-risk are recorded. In this way the path of the difference in measures is recorded and constructed for each variable, and the path as well as the uncertainty of the risk-measure value are visualized. We note that in this process we make no assumptions regarding the structure of the dataset itself.

As demonstrated in Figures 12 and 13(a couple of different repeats are drawn on this plot), the feature insertion order is repeatedly permuted, and the corresponding incremental MMD and HSIC—both serving as proxies for covariate shift—are recorded at each step (with purple markers denoting the inclusion juncture of the genuine covariate-shift features). Crucially, the incremental paths exhibit considerable variability across permutations. Complementarily, the concept drift decomposition is evaluated via an R-risk-based procedure, with results for linear and nonlinear settings presented in Figures 14 and 15, respectively. In both cases, no consistent or uniform attribution path emerges across different inclusion orders - which gives people very vivid illustration that the decomposition of the global distribution shift(measured by MMD or R-risk) into different features is impossible as it depends heavily on the feature inclusion path and the increment of the distribution of the feature X_k is conditional on the existence of features $X_1, ..., X_{k-1}$.

These empirical observations align precisely with our theoretical intuition: the change in R-risk is strictly conditional on the current set of **existing variables**, just as incremental **MMD** is contingent upon the features already incorpo-

rated into the model. Consequently, any attempt to allocate distributional shifts to individual features remains inherently ambiguous. In summary, disentangling distribution shifts at the feature level is fundamentally impossible without imposing remarkably strong—and often empirically unverifiable—assumptions on the data-generating process itself. Given the theoretical non-identifiability, we propose a pragmatic production-level heuristic that integrates MMD/RF-based detectors and meta-learners for feature attribution: (i) unique to MMD/RF implies a covariate-shift driver; (ii) unique to a meta-learner-based procedure implies a concept-drift driver; and (iii) overlap implies joint driving. While this rule does not resolve the fundamental identifiability issue, it offers actionable statistical guidance for feature monitoring and prioritization in practice. Based on our analyses—including MSE error decomposition, MMD-based covariate-shift measurement, and R-risk concept-drift decomposition—we reveal a fundamental obstacle: the task of separating distribution shift from ordinary data changes is infeasible without extra assumptions. Despite extensive prior research in causal representation learning and online drift detection, achieving effective disentanglement in practice remains challenging. The core difficulty arises from the fact that the observed drop in predictive performance cannot be uniquely decomposed into independent contributions from concept drift and covariate shift without assumptions, which is impossible in practice.

Crucially, disentanglement remains intractable without imposing strong, often untestable assumptions on correlation structures and data-generating processes. Therefore, the guidance we offer should be viewed as a statistically grounded, actionable prioritization tool rather than a formal causal or attribution claim—the only concrete argument we can make is that the selected features are primary drivers of concept drift (DRPerm or PermuCATE) or covariate shift (RF-VIMP), but we truly will not know each feature's individual contribution to the distribution shift in the whole feature space, nor whether a feature is driving the other type of shift given that it contributes significantly to driving this type of shift. This is a fundamental limitation: without imposing a causal graph or running a randomized controlled trial, the "unique" attribution of distribution shift to individual features is infeasible unless very strong and concise dependence-graph assumptions and relationships among features are imposed.

On the other hand, while the proposed feature selection for distribution shift procedure can provide concise guidance for identifying root causes of distribution shift at the feature level, it cannot, by itself, indicate which upstream data sources should be explicitly investigated. It should therefore be complemented by marginal shift monitoring and missing-rate diagnostics to enable thorough upstream data-source-level root-cause analysis. Figures 17 and 16 show that RF-domain and MMD-LOCO are among the most efficient and robust approaches compared with the alternatives. In addition, the DRPerm procedure demonstrates strong capability in capturing distribution shifts, corroborating the insights we previously obtained. Collectively, these findings validate the comprehensiveness and practical utility of our proposed feature-selection framework, which integrates MMD-LOCO with a random forest classifier. We emphasize that, despite the wide range of existing methodologies for quantifying overlap, a unified proxy remains notably absent, which inevitably complicates systematic evaluation. Nevertheless, meaningful empirical guidance can still be extracted from these approaches. More importantly, the inherent robustness of random forests—and tree-based models in general—under poor overlap between batch feature spaces positions them as a concrete and dependable choice in practice.

6.2 Efficiency v.s. SNR

To further justify the efficiency for the **FSDS**, we investigate the relationship between the signal-to-noise ratio (SNR) and the efficiency of the FSDS procedure—measured by AUROC, precision, and recall—by incrementally injecting random noise across varying SNR levels. This analysis sets the stage for the subsequent section, where we demonstrate the algorithm's efficacy in highly challenging regimes, such as RNA-seq datasets characterized by extremely sparse signals and low SNR. These results substantiate the efficiency and robustness of the meta-learner-based feature selection methodology.

In the simulation setup comprising 6 batches of size 400, 25 replicates, and a fixed seed, the contribution region fraction is set to 0.25. Column 0 is the designated marked column, exhibiting a within-region mean lift of SNR standard deviations; the target batch introduces a shift of $\delta = 2$ on this region, while column 1 serves as a global distractor. We employ two evaluation metrics: (1) **accuracy**, defined as $|H \cap R|/|H|$ (with a baseline equal to the region fraction when H ignores R); and (2) **overlap**, defined as the Jaccard index $\text{Jaccard}(H, R)$. As the SNR increases, the marked-column **PO**-share and localization accuracy increase roughly in step, exhibiting clear proportional pattern.

As demonstrated in Table 1, under low SNR, noise dominates and PO efficiency is near baseline; as SNR rises, the marked-column PO-share and PO efficiency increase together once the mark outranks the distractor. TVD gains earlier at mid SNR but converges at high SNR and remains more noise-stable. In the two-region setup, efficiencies are computed from each block's own marked column.

6.3 Feature Selection Stability

We evaluate the stability and performance of feature selection under distribution shift via AUROC and confidence intervals of the Jaccard index, using synthetic data that plant covariate shift and concept drift. This synthetic suite is designed to isolate covariate shift: $X_0, X_1 \sim \mathcal{N}(0, I_p)$, the first K coordinates of X_1 are shifted by a magnitude of SNR, and the outcome Y is generated from the shifted block plus noise, so the planted change resides in the marginal distribution $P(X)$ rather than in the conditional distribution $P(Y \mid X)$. We set $n_0 = n_1 = 300$, $p = 30$, and $K = 10$,

Table 1: Efficiency of FSDS with respect to SNR. PO-risk and TVD denote the global pseudo-outcome risk and total-variation-distance values, respectively; "eff." is the localization accuracy (baseline equals the region fraction), and "Jac." is the Jaccard index between the selected set and the true marked region.

SNR	PO-risk	TVD	PO eff.	PO Jac.	TVD eff.	TVD Jac.
0.25	0.20	0.08	0.29	0.20	0.26	0.13
0.50	0.48	0.24	0.36	0.16	0.31	0.14
1.00	0.60	0.92	0.51	0.21	0.62	0.23
2.00	0.88	1.00	0.84	0.39	0.92	0.42
4.00	1.00	1.00	0.99	0.81	0.99	0.81

perform top-10 feature selection, use $B = 100$ bootstrap resamples per batch and $R = 10$ Monte Carlo (MC) repetitions, and define FSDS = (share(PO) + share(MMD))/2.

The evaluation metrics are essentially Top-10 precision/recall-type quantities. First, selection AUC is computed for each bootstrap, averaged over B bootstraps within each repetition, and reported as the mean with its 95% confidence interval (CI) across the R repetitions. Second, Jaccard stability is assessed by extracting the Top-10 feature set from each bootstrap, computing the pairwise Jaccard index $\frac{|H \cap R|}{|H|}$, averaging over all $i < j$ within each repetition, and reporting the 95% CI across the R mean Jaccard indices $\overline{J}$. Finally, "Jaccard versus truth" compares the full-sample Top-10 set against the planted true set S across the R draws, which directly corresponds to the precision@10/recall@10 comparison against the ground-truth support. The results are summarized in Table 2.

Table 2: Covariate shift setting: $p = 30$, $B = 100$, $R = 10$. Mean and 95% CI $[lo, hi]$ over MC repetitions.

SNR	PO-risk		FSDS	
	Jaccard C.I.	AUC C.I.	Jaccard C.I.	AUC C.I.
0.25	0.233 [0.216, 0.258]	0.500 [0.457, 0.531]	0.304 [0.261, 0.326]	0.707 [0.661, 0.745]
0.50	0.237 [0.220, 0.256]	0.508 [0.485, 0.539]	0.425 [0.397, 0.454]	0.858 [0.842, 0.873]
0.75	0.226 [0.213, 0.243]	0.507 [0.475, 0.528]	0.441 [0.410, 0.474]	0.886 [0.874, 0.902]
1.00	0.235 [0.213, 0.268]	0.512 [0.479, 0.561]	0.468 [0.427, 0.524]	0.900 [0.878, 0.925]
1.50	0.239 [0.216, 0.259]	0.513 [0.470, 0.556]	0.490 [0.427, 0.552]	0.907 [0.885, 0.932]
2.00	0.233 [0.224, 0.255]	0.499 [0.469, 0.547]	0.479 [0.428, 0.544]	0.902 [0.884, 0.921]
3.00	0.226 [0.216, 0.232]	0.516 [0.482, 0.540]	0.492 [0.459, 0.512]	0.912 [0.895, 0.923]

As shown in Table 2, the PO-risk AUC remains at chance level (CI covering 0.5), and its Jaccard index stays near the baseline. In contrast, the FSDS AUC rises with SNR, reaching the high 0.9s with a tight CI, while its Jaccard index roughly doubles that of PO-risk. Thus, under covariate shift, the MMD component effectively stabilizes feature selection, consistent with the five-panel analysis. We emphasize that this setup does not address concept drift, where $Y_1 \sim \text{Bernoulli}(\sigma(X_z\beta))$ with fixed X would likely reverse this ranking: PO-risk would improve, MMD/Jaccard would degrade, and FSDS would remain competitive primarily through the PO component. Furthermore, this synthetic setting excludes real-world panel noise and batch effects; specifically, X is drawn from an i.i.d. Gaussian distribution without feature truncation, missingness, or label imbalance. Finally, this evaluation does not represent the full FSDS framework, as the synthetic FSDS variant relies solely on the PO and MMD components and omits the random forest (RF) domain arm.

7 Product Consideration

Traditional monitoring methods across feature levels include Population Stability Index([88]), Kullback-Leibler Divergence([21]) for the predictive score distribution monitoring and the missing ratio statistics monitoring. Although all of those methods can only deal with individual dimension & marginal information, they do provide the complementary information on top of the proposed methods. Since only one random forest with default hyperparameters is needed, apart from serving it directly in the production system with near-real-time inference, this essentially provides every data scientist with a portable tool for diagnosing the covariate shift from the feature level.

7.1 Statistical Aspects for the production system

The computational cost would not be a bottleneck for this procedure to be embedded in the production ML system. Like the permutation test, the LOCO variable importance procedure can be straightforwardly parallelized, yielding

substantial computational savings—**JobLib** in Python and **multiprocess** in R serve as the standard backends. In practice, PermuCATE, particularly when implemented with a doubly robust pseudo-outcome learner (DR-Learner) under conditional outcome mechanisms, consistently ranks among the top-performing concept-drift feature-selection methods. Importantly, both feature-wise permutation and LOCO-based importance can be efficiently distributed with minimal engineering overhead, making the regular (every 2–3 minutes) testing procedure reasonable in deployed ML systems.

Thus in practice, when trying to characterize covariate shift, the **Random Forest Classifier (for high-dimensional covariates)** is preferred after looking at variable importance, serving as one of the most parsimonious yet efficient ways to characterize covariate shift at the feature level—thanks to its capability for modeling nonlinearity as well as exceptional robustness under many challenging scenarios. Users have the flexibility to choose among different notions of variable importance. The MMD-LOCO procedure, although slightly more computationally expensive, is still among the most efficient and trustworthy ways to characterize covariate shift at the feature level. On the other hand, to understand concept drift at the feature level, we recommend both the conditional permutation variable importance in the doubly robust pseudo-outcome learner (DR-learner) and the split-frequency variable importance in causal forest; they achieve very strong empirical performance and attain state-of-the-art performance under nonlinear and arbitrary concept drift, outperforming ARD and SGShift under complex shifts. The conditional permutation version of the R-risk is slightly less favorable yet still outperforms existing methods. We further emphasize that although there are many existing methodologies trying to estimate the distance between two batches of datasets, random forest keeps performing exceptionally well both in hypothesis testing, feature selection, and variable importance—highlighting its practical applicability in many real-world circumstances, expanding beyond traditional tabular datasets to many other modalities, including but not limited to image, text, audio, and video data. In line with what we observed in the **CFPerm** procedure, **PermuCATE** and **Split-Freq** variable importance outperform the **LOCO** version for detecting concept drift. However, to understand concept drift purely at the feature level, the **R-learner** is what we recommend regardless of sensitivity under higher degrees of model misspecification and ultra-complexity of data-generating processes. Our work makes a solid contribution toward understanding distribution shift (both covariate shift and concept drift) at the feature level—especially under concept drift, with tractable practical guidance.

We further justify that, for the domain-classifier procedure, it can be viewed as a class of models—starting from logistic regression, random forest, XGBoost, MLP, etc. We compare feature-selection efficiency for all those models; random forest is highly preferred over other variants that require extensive hyperparameter tuning, which introduces bias and reduces robustness for the feature-selection procedure. However, this does not mean the proposed R-risk/PO-risk-based procedure is perfect: first, the computational cost is much higher than the RF/MMD method (as demonstrated in Table 28, where meta-learner-based procedures are in the high range of computational cost even after parallelization); under circumstances where overlap between the support of $\mathbf{D}_{exist}$ and $\mathbf{D}_{new}$ is poor, the meta-learner will not only struggle to fit the propensity classifier and outcome model but also introduce significant bias in estimating the ITE accurately, thus undermining the efficiency of the proposed meta-learner-based methods for concisely characterizing distribution shift. Under poor overlap, both the MMD-LOCO and random forest procedures work well, especially when the support deviation between D_{exist} and D_{new} is large. The minimum sample size for effectively characterizing distribution shift via the hypothesis-testing procedure based on the meta-learner is around 200 per batch to enable robust and efficient estimation via the meta-learner testing procedure. On the other hand, the random forest variant would only require 100 observations per batch for efficient feature selection. From the monitoring and alerting aspect, we want to mention the following:

- Suppose H_0 in the **CFPerm** is not rejected, it provides strong statistical evidence that the data distribution has not meaningfully changed. Therefore, no full-retraining is required. This itself delivers tangible economic values via avoiding unnecessary computation & engineering overhead, where at the same time providing the deployment engineers the confidence to proceed with the current version of the models.
- Suppose H_0(both CFPerm or the Random-Forest Classifier procedures) is rejected and the top-K features are known to be low-risk(e.g. business-as-usual features) - then people should log and notify the data scientists to review them carefully throughout the ranked feature list based on the capacity of manual reviews. On the other hand, suppose H_0 is rejected and the top features belong to critical business metrics or the upstream data pipelines, then the immediate actions are required for the root-cause identification workflow. Under this circumstance, a proactive pre-warning threshold should be set up efficiently even before the significant distribution shift is observed.

From another aspect, we want to extend this process to a general-purposed class of procedures of building a binary classification model to distinguish the two batches of datasets $\mathbf{X}_{new}$ and $\mathbf{X}_{exist}$ - starting from the logistic regression, to random forest to gradient boosting to the MLP, we compare the overall efficiency and the robustness of the different versions of the models - especially comparing the MLP and XGBoost versions.

7.2 Potential Risks

The potential risk for the feature-selection procedure primarily focuses on the lack of overlap between covariates in $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$, under which circumstance the meta-learner-based procedure will struggle. When support shifts significantly

between different batches, the random forest classifier would be highly efficient, perhaps among the best-performing procedures compared with other versions such as the l_1 penalized regression procedures. Thus in practice, we propose the following general strategy for product best practice:

- Obvious covariate shift is observed, where there's almost no overlap between the distribution of the covariates in both X_{exist} and X_{new}, the random forest classifier is what we may need, however, the random forest VIMP can only tell concisely which feature shifts significantly conditional on the existence of all the other features, even including Y, which is fundamentally different from the concept drift detection use cases. For the relatively minor or gradual concept drift with covariate shift, the meta-learner method would work slightly better than the SGShift, Knockoff as well as the MMD for characterizing concept drift.
- In production or real-data analysis, the support shift can be quantified by the overlap score in [63]; the suggested threshold is around 0.3, so that if $OVL = \int \min\{f_1(e), f_0(e)\}\, de \leq 0.3$, the fallback procedure is triggered. Alternatively, o-values ([46]) or Bhattacharyya distances can also be leveraged to quantify the degree of overlap between $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$—the fallback procedure is also implemented in the submitted code.

On the other hand, continuously monitoring variable importance in production enables interpretable, feature-level characterizations of model behavior—specifically, the detection of both gradual and abrupt shifts in feature importance over time. This capability is particularly valuable in dynamic online environments where data streams will exhibit concept drift and models are updated incrementally. Several existing methods have been developed along this direction. iPFI([28]), iSAGE([59]) and Streaming SHAP([57], maintaining the running average for the shapley values) are among a few of them. We emphasize that while full configurations of these online interpretation methods fall outside the scope of our proposed statistical methodology, they would serve as a powerful complement in the production monitoring regime. Specifically, they can enable dashboard visualizations that provide operators with interpretable, real-time insights into which feature is driving significant distribution shift.

7.3 Deployment Checklist

In this subsection we discuss topics directly related to deployment in production ML systems. First, the random forest classifier procedure, with a high degree of robustness, efficiency, and computational efficiency, can serve both as a real-time feature-selection procedure for distribution shift and as a portable, efficient tool for checking distribution shift among batches of datasets. Second, this pipeline operates within the data-ingestion and model-inference cycles—whether micro-batch (every couple of minutes, enabling the alerting mechanism in a very short period of time), hourly, or daily. Rather than specifying a one-size-fits-all configuration, the system exposes a couple of factors, including the number of permutations B, degree of parallelization, and sample size (granularity) for each batch test. These allow deployment engineers to carefully tune runtime based on their specific cluster capacity and monitoring window. Third, the guiding principle is that the pipeline is required to complete its run before the next batch arrives in the inference engine; if this constraint cannot be met during peak load, the system automatically downgrades to the computationally cheaper random forest classifier VIMP fallback. This procedure ensures the monitoring loop never becomes a bottleneck while ensuring high-quality feature characterization regardless of resource constraints.

From cost-efficiency considerations, the primary economic value of feature selection for distribution shift lies in drastically shortening the latency between drift onset and actionable feature-level detection. Rather than waiting for requests from downstream business metrics to degrade before initiating a costly investigation, the pipeline proactively flags the exact variables driving the shift—enabling teams to diagnose root causes before the shift propagates into revenue-impacting outcomes. Upon output, the ranked feature list empowers deployment scientists to prioritize debugging efforts with surgical precision, narrowing a sprawling multi-hour or even multi-day investigation into a focused selection of top-K candidates. The rapid, almost real-time feature selection (especially for covariate shift) translates directly into improved **unit economics** across the following aspects. First, it minimizes compute waste by avoiding unnecessary full-model retraining. Second, it substantially reduces operational expenditure (OpEx) by shrinking engineering triage time. Third, it mitigates revenue loss by intercepting performance-degrading shifts before they cascade into production—intervening at the feature level rather than reacting to degraded business metrics after multiple days.

Validation of this pipeline is conducted through rigorous **offline back-testing** rather than online A/B testing, leveraging historical data with known distribution-shift events (recorded by logs). This procedure quantifies key performance metrics—including recall@K for truly shifted features, precision of the flagged set, and calibration of feature-specific p-values or the features with the highest variable importance in the random forest classifier procedure—under controlled and reproducible conditions. By establishing a robust baseline accuracy offline, engineers and ML scientists can confidently calibrate the decision threshold and validate the ML pipeline's diagnostic power before exposing it to live traffic, ensuring that production-level deployment is both safe and empirically grounded.

Overall, the feature-selection-for-distribution-shift procedure transforms ambiguous distributional changes into a concise, prioritized, and **comprehensive** action list, effectively shifting the operational cost curve from costly reactive firefighting to lean, proactive health management. The economic impact comes not merely from saving retraining efforts but also from embedding continuous and interpretable diagnostics into arbitrary ML systems—delivering measurable ROI at the model, batch, and incident levels while simultaneously reducing the cognitive burden on the engineering team.

7.4 ML System Level Relevance

We focus on two key aspects that connect distribution-shift procedures to ML system-level relevance. First, when should a given feature be removed from the deployed ML models? Second, when is an in-depth root-cause analysis required for downstream error diagnostics? To efficiently address these problems, the statistical methodology introduced in this work should be complemented by traditional statistical measures and system-level statistics. We assume that deployed ML models are already equipped with standard monitoring dashboards, including but not limited to missing-value statistics (e.g., the proportion of missing values per batch) and marginal shift monitoring procedures based on KL divergence [21] and the population stability index (PSI, [88]). Additionally, leveraging existing methods for online monitoring of variable importance in streaming settings is also required, including incremental permutation feature importance ([28]) and incremental SAGE ([59]). We then elaborate on each of these issues in detail.

Regarding the removal of corrupted features, we want to elaborate that the current statistical methodology should be complemented with the variable importance of deployed ML models, especially during back-testing stages. We recommend removing this feature when the following three conditions are met. First, the distribution of the feature itself oscillates a lot across different time periods, where PSI and KL-Divergence consistently exceed the critical threshold set by business relevance. Second, the feature has been actively contributing either to concept drift or covariate shift (staying among the top-K features for a good period of time via our **Feature Selection for Distribution Shift** procedure), contributing significantly to the distribution shift. Third, and foremost, the feature has no significant predictive power in the deployed model, either in the back-testing stage (in the past several days) or the testing stage. After satisfying these three conditions, we would claim that the feature should be removed immediately.

Second, to systematically determine whether and when a feature should be escalated to the downstream root-cause analysis(with respect to a certain time window, we denote it as the degradation window) and the deep-dive system, we propose a three-evidence criterion that must be jointly satisfied. This criterion is grounded in the observation that a single monitoring signal—such as a high PSI or a drop in variable importance—is often insufficient to justify a deep investigation, as it may arise from benign data fluctuations or model adaptations. Instead, we require a convergence of independent statistical signals that collectively point to a feature's anomalous behavior during a performance-degradation window, relative to a stable reference period. The three evidence types are designed to capture different facets of the feature's relationship with the model and the data: its changing role within the model, its net contribution to predictive performance, and the structural nature of its distribution shift across subpopulations.

To talk about this in detail, the first evidence requires a substantial rank shift in the variable importance across different time windows(the reference window and the degradation window) for these features - for example, a rank change of at least five positions or a relative VIMP change exceeding 50%. This indicates that the model's reliance on the feature has materially redistributed, signaling a potential instability. The second evidence is obtained by a leave-one-covariate-out (LOCO) evaluation on the degradation window: removing the feature should yield a statistically significant improvement in performance, such as an AUC gain above 2% with a 90% or 95% confidence interval that would not include zero (these values are just for illustration and would be subject to discussion with business teams). It indicates that removing the feature improves performance, suggesting that during the degradation period the feature has become a net source of noise or misleading signal rather than a useful predictor. The third evidence examines the feature's marginal distribution shift across subpopulations(although the meta-learner testing as feature selection procedure can also induce subgroup level insights, it would be out of our scope); we require that the shift be concentrated in a specific subgroup, as measured by a subgroup-level PSI(Population stability index) that exceeds the global PSI by more than 0.15, with that subgroup accounting for at least 5% of the total sample. Only when all three conditions are concurrently met does the system generate a 'recommend for the downstream deep-dive' signal. This joint criterion provides a statistically grounded and evidence-based trigger for further investigation, while explicitly avoiding any causal claim—the evidence is intended to guide, not replace, human diagnostic efforts, and serves as a prioritization tool for allocating limited analytical resources. We want to emphasize that, among all of these procedures, the outputs are only for reference; there is no deterministic output. Results should be blended with business context and relevance for the decision-making process. **We focus on the evidence—not the decision—and frame all outputs as recommendations to guide, rather than replace, human diagnostic efforts.**

We also acknowledge that ML monitoring mechanisms are far more comprehensive than the aspects covered here. They include, but are not limited to, deciding whether a new feature should be incorporated into currently deployed models and how frequently the model should be retrained. The former can leverage existing methods such as the adversarial random forest ([10]), while the latter is often heuristic-based rather than purely data-driven. Streaming VIMP and incremental feature acquisition and selection ([86]) represent another important class of statistical and ML problems. Although outside the scope of this work, we note that these techniques would be highly desirable to ensure that scientists and engineers can extract interpretable, concise, real-time insights for smoother and more tractable decision-making.

8 Summary

We present a comprehensive evaluation of feature-selection strategies for distribution shifts, encompassing both concept drift and covariate shift—a critical yet often overlooked procedure in machine learning system design. For concept drift detection, the meta-learner approach achieves very strong empirical performance and attains state-of-the-art performance under nonlinear and arbitrary concept drift. For covariate shift, we adopt a random forest-based variable importance procedure on out-of-fold samples, which achieves highly competitive performance relative to the existing state of the art (FSL-Net [**barrabes2025feature**]) while exhibiting slightly better computational efficiency. We demonstrate this through both simulations and extensive real-data applications equipped with a variety of adversarial perturbations. The variable-importance-based procedure has been rigorously justified for efficiently characterizing feature-level contributions to distribution shift. Most importantly, the strong performance of random forest—both as the outcome and propensity models within the meta-learner, and as the binary classifier trained for covariate shift with VIMP-based feature selection—is solidly justified, demonstrating its strong practical value. Integrated with standard ML monitoring best practices, the proposed framework can be readily deployed in production systems, delivering interpretable top-K feature rankings, actionable recommendations for root-cause analysis, and near-real-time business insights into variable changes—as demonstrated across diverse real-world datasets. In particular, the encouraging results on RNA-seq datasets demonstrate strong practical value in a wide range of real-world datasets for helping data science practitioners. Empirically, our results reaffirm the fundamental impossibility of unambiguously disentangling overall distribution shifts into feature-level contributions of covariate shift versus concept drift.

Looking ahead, several directions remain open. First, extending this feature-selection framework to continuous batch settings with adaptive updating strategies—potentially drawing from model-selection problems—could offer more flexible solutions tailored to streaming industrial demands. Second, incorporating multimodal information with modality-wise contributions of pseudo-outcome risk would provide a direct interpretation of how each modality contributes to overall out-of-distribution (OOD) shift. Third, developing a differentially private version of the one-shot aggregation, where the uploaded VIMP vectors are perturbed under a formal privacy budget while preserving the Matrix Bernstein concentration, would further strengthen the practical deployability of the framework in sensitive domains. We leave these as promising avenues for future research. The Python software and implementations are available at `https://github.com/ruanhq-StatsML/FSDS-FeatureSelection_for_DistributionShift`.

References

[1] Mélisande Albert et al. "Adaptive test of independence based on HSIC measures". In: *The Annals of Statistics* 50.2 (2022), pp. 858–879. DOI: 10.1214/21-AOS2129. URL: https://doi.org.

[2] Haider Ali, Peter Yurkovich, and Jamous Bitrick. "MI-RFGSM: Efficient and Robust Adversarial Attacks Against Deep Reinforcement Learning". In: (2022).

[3] Alejandro Almodóvar, Juan Parras, and Santiago Zazo. "Propensity Weighted Federated Learning for Treatment Effect Estimation in Distributed Imbalanced Environments". In: *Computers in Biology and Medicine* 178 (2024), p. 108779. DOI: 10.1016/j.compbiomed.2024.108779.

[4] Maksym Andriushchenko and Nicolas Flammarion. "Understanding and improving fast adversarial training". In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 16048–16059.

[5] Susan Athey, Julie Tibshirani, and Stefan Wager. "Generalized random forests". In: (2019).

[6] Míriam Barrabés et al. "Adversarial learning for feature shift detection and correction". In: *Advances in Neural Information Processing Systems* 36 (2023), pp. 57597–57638.

[7] Míriam Barrabés et al. "Feature Shift Localization Network". In: *arXiv preprint arXiv:2506.09101* (2025).

[8] Tom Bewley et al. "ShapShift: Explaining Model Prediction Shifts with Subgroup Conditional Shapley Values". In: *arXiv preprint arXiv:2604.11200* (2026).

[9] Bernd Bischl et al. "OpenML: Insights from 10 years and more than a thousand papers". In: *Patterns* 6.7 (2025), p. 101317. DOI: 10.1016/j.patter.2025.101317.

[10] Kristin Blesch et al. "Conditional feature importance with generative modeling using adversarial random forests". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 39. 15. 2025, pp. 15596–15604.

[11] Valerio Bonsignori et al. "Interpreting Federated Learning by Aggregating SHAP Explanations". In: *IEEE Access* 14 (2026), pp. 80280–80297. DOI: 10.1109/ACCESS.2026.3696009.

[12] Leo Breiman. "Random forests". In: *Machine learning* 45 (2001), pp. 5–32.

[13] Sebastian Caldas et al. "Leaf: A benchmark for federated settings". In: *arXiv preprint arXiv:1812.01097* (2018).

[14] Tianqi Chen and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. KDD '16. ACM, Aug. 2016, pp. 785–794. DOI: 10.1145/2939672.2939785. URL: http://dx.doi.org/10.1145/2939672.2939785.

[15] Xue-wen Chen and Jong Cheol Jeong. "Enhanced recursive feature elimination". In: *Sixth international conference on machine learning and applications (ICMLA 2007)*. IEEE. 2007, pp. 429–435.

[16] Victor Chernozhukov et al. *Double/debiased machine learning for treatment and structural parameters*. 2018.

[17] Leo H. Chiang, Evan L. Russell, and Richard D. Braatz. *Fault Detection and Diagnosis in Industrial Systems*. Tennessee Eastman fault files `d00_te.dat` and `d01_te.dat`. London: Springer, 2001. DOI: `10.1007/978-1-4471-0347-9`.

[18] Gordon Christie et al. "Functional map of the world". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2018, pp. 6172–6180.

[19] Oscar Clivio et al. "Deconfounding Scores: Representation Learning for Causal Effect Estimation with Weak Overlap". In: *arXiv preprint arXiv:2604.00811* (2026).

[20] John Clore et al. *Diabetes 130-US Hospitals for Years 1999–2008*. UCI Machine Learning Repository. File `diabetic_data.cs` 2014. DOI: `10.24432/C5230J`. URL: `https://doi.org/10.24432/C5230J`.

[21] Jiequan Cui et al. "Generalized kullback-leibler divergence loss". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026).

[22] Alexander D'Amour et al. "Overlap in observational studies with high-dimensional covariates". In: *Journal of Econometrics* 221.2 (2021), pp. 644–654.

[23] Lei Ding et al. "Risky borrowers or risky mortgages disaggregating effects using propensity score models". In: *Journal of Real Estate Research* 33.2 (2011), pp. 245–278.

[24] J. J. Downs and E. F. Vogel. "A Plant-Wide Industrial Process Control Problem". In: *Computers & Chemical Engineering* 17.3 (1993), pp. 245–255. DOI: `10.1016/0098-1354(93)80018-I`.

[25] Pietro Ducange et al. "Federated SHAP: Privacy-Preserving and Consistent Post-Hoc Explainability in Federated Learning". In: *Machine Learning* 115.2 (2026), p. 24. DOI: `10.1007/s10994-025-06956-1`.

[26] François Fleuret. "Fast binary feature selection with conditional mutual information". In: *Journal of Machine learning research* 5.Nov (2004), pp. 1531–1555.

[27] Rui Fu et al. "FEAST: A Communication-Efficient Federated Feature Selection Framework for Relational Data". In: *Proceedings of the ACM on Management of Data* 1.1 (2023), 107:1–107:28. DOI: `10.1145/3588961`.

[28] Fabian Fumagalli et al. "Incremental permutation feature importance (iPFI): towards online explanations on data streams". In: *Machine Learning* 112.12 (2023), pp. 4863–4903.

[29] Josh Gardner, Zoran Popovic, and Ludwig Schmidt. "Benchmarking distribution shift in tabular data with tableshift". In: *Advances in Neural Information Processing Systems* 36 (2023), pp. 53385–53432.

[30] Isabelle Guyon et al. "Gene selection for cancer classification using support vector machines". In: *Machine Learning* 46.1-3 (2002), pp. 389–422.

[31] Hiroyuki Hanada et al. "Safe Distributionally Robust Feature Selection under Covariate Shift". In: *arXiv preprint arXiv:2603.16062* (2026).

[32] Jason S Haukoos and Roger J Lewis. "The propensity score". In: *Jama* 314.15 (2015), pp. 1637–1638.

[33] Niharika Hegde et al. "CHRONOBERG: Capturing Language Evolution and Temporal Awareness in Foundation Models". In: *arXiv preprint arXiv:2509.22360* (2025).

[34] Jennifer L. Hill. "Bayesian Nonparametric Modeling for Causal Inference". In: *Journal of Computational and Graphical Statistics* 20.1 (2011), pp. 217–240. DOI: `10.1198/jcgs.2010.08162`.

[35] Xiaoyu Hu and Jing Lei. "A distribution-free test of covariate shift using conformal prediction". In: *arXiv e-prints* (2020), arXiv–2010.

[36] Kosuke Imai and Marc Ratkovic. "Estimating treatment effect heterogeneity in randomized program evaluation". In: (2013).

[37] Sooyong Jang et al. "Sequential covariate shift detection using classifier two-sample tests". In: *International conference on machine learning*. PMLR. 2022, pp. 9845–9880.

[38] Sooyong Jang et al. "Sequential covariate shift detection using classifier two-sample tests". In: *International Conference on Machine Learning*. PMLR. 2022, pp. 9845–9880.

[39] Silke Janitza, Ender Celik, and Anne-Laure Boulesteix. "A computationally fast variable importance test for random forests for high-dimensional data". In: *Advances in Data Analysis and Classification* 12.4 (2018), pp. 885–915.

[40] Alistair E. W. Johnson et al. *MIMIC-IV, a freely accessible electronic health record dataset*. Scientific Data. PhysioNet; DOI: 10.1038/s41597-022-01899-x. 2023.

[41] Edward H Kennedy et al. "Optimal doubly robust estimation of heterogeneous causal effects". In: *arXiv preprint arXiv:2004.14497* 5 (2020).

[42] Rémi Khellaf, Aurélien Bellet, and Julie Josse. "Federated Causal Inference: Multi-Study ATE Estimation beyond Meta-Analysis". In: *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*. Vol. 258. Proceedings of Machine Learning Research. PMLR, 2025, pp. 3448–3456.

[43] Rémi Khellaf et al. “Principled Federated Random Forests for Heterogeneous Data”. In: *arXiv preprint arXiv:2602.03258* (2026). arXiv: 2602.03258.

[44] Jonas M Kübler et al. “Automl two-sample test”. In: *Advances in Neural Information Processing Systems* 35 (2022), pp. 15929–15941.

[45] Miron B Kursa and Witold R Rudnicki. “Feature selection with the Boruta package”. In: *Journal of statistical software* 36 (2010), pp. 1–13.

[46] Lihua Lei et al. “Distribution-free assessment of population overlap in observational studies”. In: *Preprint Available at https://lihualei71. github. io/ovalue. pdf* (2021).

[47] Anran Li et al. “FedSDG-FS: Efficient and Secure Feature Selection for Vertical Federated Learning”. In: *arXiv preprint arXiv:2302.10417* (2023). arXiv: 2302.10417.

[48] Jundong Li et al. “Feature selection: A data perspective”. In: *ACM computing surveys (CSUR)* 50.6 (2017), pp. 1–45.

[49] Xiao Li et al. “A Debiased MDI Feature Importance Measure for Random Forests”. In: *arXiv preprint arXiv:1906.10845* (2019).

[50] Jiadong Lin et al. “Nesterov accelerated gradient and scale invariance for adversarial attacks”. In: *arXiv preprint arXiv:1908.06281* (2019).

[51] Jiashuo Liu et al. “On the need for a language describing distribution shifts: Illustrations on tabular datasets”. In: *Advances in Neural Information Processing Systems* 36 (2023), pp. 51371–51408.

[52] Ruiqi Lyu, Alistair Turcan, and Bryan Wilder. “Explaining Concept Shift with Interpretable Feature Attribution”. In: *arXiv preprint arXiv:2505.20634* (2025).

[53] Ruiqi Lyu, Alistair Turcan, and Bryan Wilder. “Explaining Concept Shift with Interpretable Feature Attribution”. In: *arXiv preprint arXiv:2505.20634* (2025).

[54] Michael McCann and Adrian Johnston. *SECOM Data Set*. UCI Machine Learning Repository. 2008. URL: https://archive.ics.uci.edu/dataset/245/secom.

[55] Lucas Mentch and Siyu Zhou. “Getting better from worse: augmented bagging and a cautionary tale of variable importance”. In: *arXiv preprint arXiv:2003.03629* (2020).

[56] Lucas Mentch and Siyu Zhou. “Randomization as regularization: A degrees of freedom explanation for random forest success”. In: *Journal of Machine Learning Research* 21.171 (2020), pp. 1–36.

[57] Maciej Mozolewski et al. “Towards evaluation of explainable artificial intelligence in streaming data”. In: *World conference on explainable artificial intelligence*. Springer. 2024, pp. 145–168.

[58] Tim Müller et al. “ShadowVIMP: permutation-based multiple testing-controlled variable selection”. In: *BMC bioinformatics* (2026).

[59] Maximilian Muschalik et al. “isage: An incremental version of SAGE for online explanation on data streams”. In: *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer. 2023, pp. 428–445.

[60] Xinkun Nie and Stefan Wager. “Quasi-oracle estimation of heterogeneous treatment effects”. In: *Biometrika* 108.2 (2021), pp. 299–319.

[61] Joseph Paillard et al. “Measuring Variable Importance in Individual Treatment Effect Estimation with High Dimensional Data”. In: *arXiv preprint arXiv:2408.13002* (2024).

[62] Elisabeta Pana, Sascha Vitzthum, and David Willis. “The impact of internet-based services on credit unions: a propensity score matching approach”. In: *Review of Quantitative Finance and Accounting* 44.2 (2015), pp. 329–352.

[63] Massimiliano Pastore and Antonio Calcagnı. “Measuring distribution similarities between samples: A distribution-free overlapping index”. In: *Frontiers in psychology* 10 (2019), p. 1089.

[64] Hanchuan captives Peng, Fuhui Long, and Chris Ding. “Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 27.8 (2005), pp. 1226–1238.

[65] Jennifer Lewis Priestley and Eric VonDohlen. “Propensity score matching: a tool for consumer risk modeling and portfolio underwriting”. In: *Journal of Applied Statistics* 51.12 (2024), pp. 2481–2488.

[66] Alec Radford et al. “Learning transferable visual models from natural language supervision”. In: *International conference on machine learning*. PmLR. 2021, pp. 8748–8763.

[67] Henrique* Schechter Vera et al. “EmbeddingGemma: Powerful and Lightweight Text Representations”. In: (2025). URL: https://arxiv.org/abs/2509.20354.

[68] Antonin Schrab et al. “MMD aggregated two-sample test”. In: *Journal of Machine Learning Research* 24.194 (2023), pp. 1–81.

[69] Leo Schwinn et al. “Exploring misclassifications of robust neural networks to enhance adversarial attacks”. In: *Applied intelligence* 53.17 (2023), pp. 19843–19859.

[70] Erwan Scornet, Gérard Biau, and Jean-Philippe Vert. "Consistency of random forests". In: (2015).

[71] Carlos Sebastián and Carlos E González-Guillén. "A feature selection method based on Shapley values robust for concept shift in regression". In: *Neural Computing and Applications* 36.23 (2024), pp. 14575–14597.

[72] Steven J Staffa and David Zurakowski. "Five steps to successfully implement and evaluate propensity score matching in clinical research studies". In: *Anesthesia & Analgesia* 127.4 (2018), pp. 1066–1073.

[73] Beata Strack et al. *Diabetes 130-US hospitals for years 1999–2008 Data Set*. UCI Machine Learning Repository; hospital EHR proxy for MIMIC-style tasks. 2014. URL: https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008.

[74] Carolin Strobl et al. "Bias in random forest variable importance measures: Illustrations, sources and a solution". In: *BMC bioinformatics* 8 (2007), pp. 1–21.

[75] Hind Taud and Jean-Franccois Mas. "Multilayer perceptron (MLP)". In: *Geomatic approaches for modeling land change scenarios*. Springer, 2017, pp. 451–455.

[76] Laine E Thomas, Fan Li, and Michael J Pencina. "Overlap weighting: a propensity score method that mimics attributes of a randomized clinical trial". In: *Jama* 323.23 (2020), pp. 2417–2418.

[77] UCI Machine Learning Repository. *Gene expression cancer RNA-Seq Data Set*. TCGA-PANCAN-HiSeq-801x20531. 2016. URL: https://archive.ics.uci.edu/dataset/401/gene+expression+cancer+rna+seq.

[78] Thanh Vinh Vo et al. "Bayesian Federated Estimation of Causal Effects from Observational Data". In: *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*. Vol. 180. Proceedings of Machine Learning Research. PMLR, 2022, pp. 2024–2034.

[79] Robin Vujanic and Thomas Rueckstiess. *LEAF: Knowledge Distillation of Text Embedding Models with Teacher-Aligned Representations*. 2025. arXiv: 2509.12539 [cs.IR]. URL: https://arxiv.org/abs/2509.12539.

[80] Meredith L Wallace et al. "Use and misuse of random forest variable importance metrics in medicine: demonstrations through incident stroke prediction". In: *BMC Medical Research Methodology* 23.1 (2023), p. 144. DOI: 10.1186/s12874-023-01965-x.

[81] Guan Wang. "Interpret Federated Learning with Shapley Values". In: *arXiv preprint arXiv:1905.04519* (2019). arXiv: 1905.04519.

[82] Xiaosen Wang and Kun He. "Enhancing the transferability of adversarial attacks through variance tuning". In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2021, pp. 1924–1933.

[83] John N. Weinstein et al. "The Cancer Genome Atlas Pan-Cancer analysis project". In: *Nature Genetics* 45.10 (2013), pp. 1113–1120. DOI: 10.1038/ng.2764.

[84] David Wipf and Srikantan Nagarajan. "A new view of automatic relevance determination". In: *Advances in neural information processing systems* 20 (2007).

[85] Ruoxuan Xiong et al. "Federated Causal Inference in Heterogeneous Observational Data". In: *Statistics in Medicine* 42.24 (2023), pp. 4418–4439. DOI: 10.1002/sim.9868.

[86] Dianlong You et al. "Online causal feature selection for streaming features". In: *IEEE Transactions on Neural Networks and Learning Systems* 34.3 (2021), pp. 1563–1577.

[87] Ming Yuan and Yi Lin. "Model selection and estimation in regression with grouped variables". In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 68.1 (2006), pp. 49–67.

[88] Bilal Yurdakul. *Statistical properties of population stability index*. Western Michigan University, 2018.

[89] Michael J Zellinger and Peter Bühlmann. "Natural Language-Based Synthetic Data Generation for Cluster Analysis". In: *Journal of Classification* 42.3 (2025), pp. 517–543.

[90] Grace X. Y. Zheng et al. "Massively parallel digital transcriptional profiling of single cells". In: *Nature Communications* 8 (2017), p. 14049. DOI: 10.1038/ncomms14049.

[91] Siyu Zhou and Lucas Mentch. "Trees, forests, chickens, and eggs: when and why to prune trees in a random forest". In: *Statistical Analysis and Data Mining: The ASA Data Science Journal* 16.1 (2023), pp. 45–64.

9 Supplementary Materials: Implementation Details

We provide all the experimental results across diverse types and modalities of the datasets, ranging from linear & nonlinear synthetic data to tabular synthetic datasets and the chronoberg text corpus dataset as well as the FMoW dataset. Additionally sensitivity analysis in terms of both component models in meta-learners and the overlap are also provided via justifiable empirical studies. The shadowVIMP and VitaPIMP procedures are adapted from **R** through the rpy2 package with python benchmark for evaluation. The rf-domain, delta-shap, MMD-LOCO, DRPerm, PermuCATE, ARD methods are implemented in Python, the adversarial perturbations procedures on the domain regressor trained is adapted from that on the image dataset. The FSL-Net, DataFix, knockoff, l1-spec and l1-misspec are leveraged from the existing implementations.

9.1 Detailed Settings for the FSDS

We present some of the detailed settings for the **FSDS/CFPerm** procedures in production.

1. The default hyperparameter setting for random forest on a dataset with n observations and p features is: for continuous response, $mtry = \frac{p}{3}$; for discrete response, $mtry = \sqrt{p}$. The minimal node size is set as $node_size = \frac{\sqrt{n}}{2}$, and the number of trees in the outcome model is set as $n_{trees} = 150$. For logistic regression, the **lbfgs** procedure is adapted for optimization, and the number of iterations is set as 2000 by default.

2. The cross-feature quantile (95%) and within-feature quantile (99%) are pre-set and not recommended to be modified. Regarding the notion of variable importance, we suggest **PermuCATE** (conditional permutation variable importance, [61], more accurate with lower uncertainty than LOCO) for more accurate calculation, and the split-frequency version by causal forest for parsimonious output generation (with significantly faster computational speed).

3. From the statistical aspect, we emphasize again that the assumption for the causal inference procedure is no longer satisfied in terms of the strong ignorability condition, especially under the circumstance where H_0 is rejected.

4. In production, there is a fair degree of flexibility in choosing among outcome and propensity score models. However, for tabular data or embedding-representation data, the random forest outcome model and logistic regression propensity score model are recommended for the sake of robustness and system reliability.

5. One additional aspect worth emphasizing concerns sample-size requirements for each batch in the testing procedure in **MT-as-FS**. In particular, the new batch size n_{new} should not fall below a certain minimum threshold (around 200 for leveraging random forest as the outcome model with cross-fitting); otherwise, the lack of adequate fit in both outcome and propensity score models may introduce non-negligible bias into the estimated variable importance. We note that in a production environment this constraint is reasonable because the existing batch is typically much larger than the new batch by default (e.g., accumulated historical data).

6. Although additional computation is introduced, the number of permutations should not fall below 100 to enable a valid data-driven threshold for the testing procedure. Given ample computational resources, the number of permutations in **CFPerm** should be set to 200 or 500 by default to ensure false-alarm control as well as robustness of the feature-selection stage.

10 Supplementary Materials: Meta-Learner Testing as Feature Selection

This section briefly discusses the meta-learner testing procedure that employs the permute-then-refit framework to assess variable importance. This procedure serves as a complement to the main contribution of our work, which focuses on feature selection for distribution shift. From an overarching perspective, this supplementary procedure is more stringent and concise, yielding higher confidence in the selected features, while with lower sensitivity compared to our proposed Feature Selection for Distribution Shift approach.

10.1 Efficiency Justification

We justify the efficiency of the proposed **Meta-Learner Testing as Feature Selection Procedure** under both concept drift and covariate shift. As demonstrated in Figures 1 and 2 (the 'X' and dot are those points with rejection), the features with concept drift $\mathbf{X}_4 - \mathbf{X}_7$ are ranked among the top identified features with the smallest feature-wise p-value as the degree of concept drift increases. Moreover, as demonstrated in Figures 4 and 3, both **PermuCATE** and **LOCO** have also demonstrated their efficiency in characterizing mean shift and variance shift, with conditional permutation VIMP slightly more efficient.

The crucial metrics for evaluating the efficiency for the **Meta Learner-Testing-as-Feature-Selection** including the following given ground-truth shifted feature set $\mathcal{S}_{gt}$

- The kendall's correlation of the ranking of the identified feature set $\mathcal{S}$ and ground-truth shifted feature set $\mathcal{S}_{gt}$.
- The recall for the features identified : $\frac{|\mathcal{S} \cap \mathcal{S}_{gt}|}{|\mathcal{S}|}$ and the precision@k: $\frac{|\mathcal{S} \cap \mathcal{S}_{gt}|}{|\mathcal{S}_{gt,k}|}$

We want to mention that although in real-world applications no ground-truth indices of features with significant contribution to distribution shift are labeled, the statistical and numerical results in our **CFPerm** methodology have showcased the efficiency of the procedure under different types of distribution shift justifiably.

Thus, the primary utility of **Meta-Learner Testing-as-Feature-Selection** procedure lies in its feature-selection efficiency — and that's precisely the dimension that truly matters for production. Unlike the R-RiskPerm (via leveraging the R-risk objective function, only tells you whether a shift has occurred from the batched dataset level), CFPerm enables a ranked and **interpretable** set of features that are primarily driving the shifts. This feature-level output directly translates into actionable diagnostics: engineers can trace the identified features back to upstream data pipelines, schema changes, or operational anomalies, dramatically reducing the time spent on root-cause investigation in the production

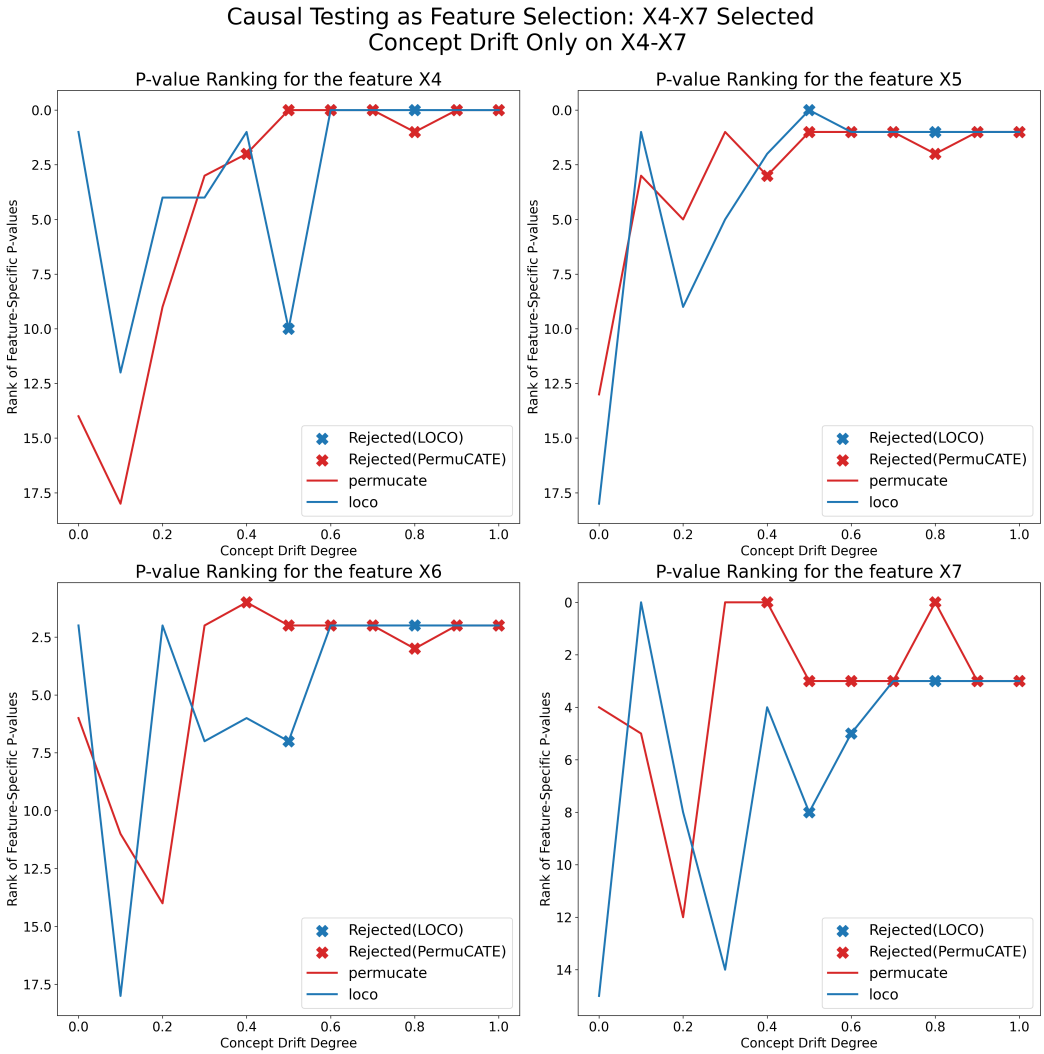


Figure 1: Meta-Learner Testing as Feature Selection: Concept Drift 1

environment.

We want to further mention that the **Meta-Learner Testing-as-Feature-Selection** procedure further outputs the ranking of variable importance for each considered notion, including but not limited to PermuCATE, LOCO, and GRF. These rankings can be viewed as an alternative notion of out-of-distribution (OOD) feature importance. Specifically, the ranking of feature-specific p-values characterizes each feature's relative contribution to the distribution shift between the two batches of data. The dashboard can be built in production environment for ML engineers to monitor each metric/feature's contribution to the change of the Topline Business Metrics.

10.2 System Actions & Operations

The primary value of the CFPerm detection framework lies not in providing a binary "retrain or not" signal, but in delivering a structured, feature-level diagnostic that directly informs downstream engineering and analytical workflows. The **Meta-Learner Testing-as-Feature-Selection** procedure's primary value is not limited to providing a binary 'retrain or not' signal, but in delivering a structured, feature-level diagnostics that directly informs downstream engineering and analytical workflows.

1. **Guiding downstream root-cause analysis and subgroup investigation.** The features flagged by the significance test serve not only as proxies for distribution-shift contributors, but also as natural stratification axes for deeper diagnostic exploration. Distribution shift localization can be achieved by partitioning the population along the ranges of these selected features. For example, if a categorical feature is flagged, comparing the distribution shift across its different levels provides a principled and practical basis for differential analysis. This transforms the testing result into a directly actionable input for case-study investigation and root-cause identification.

2. The features selected given the significance-testing results will not only serve as a proxy for those contributing to distribution shift but also guide downstream subgroup analysis or in-depth case studies for root-cause analysis of data change. Distribution-shift localization can be achieved by stratifying the population with respect to grids expanded by the ranges of these selected features. Specifically, if a categorical feature is selected, the subgroups

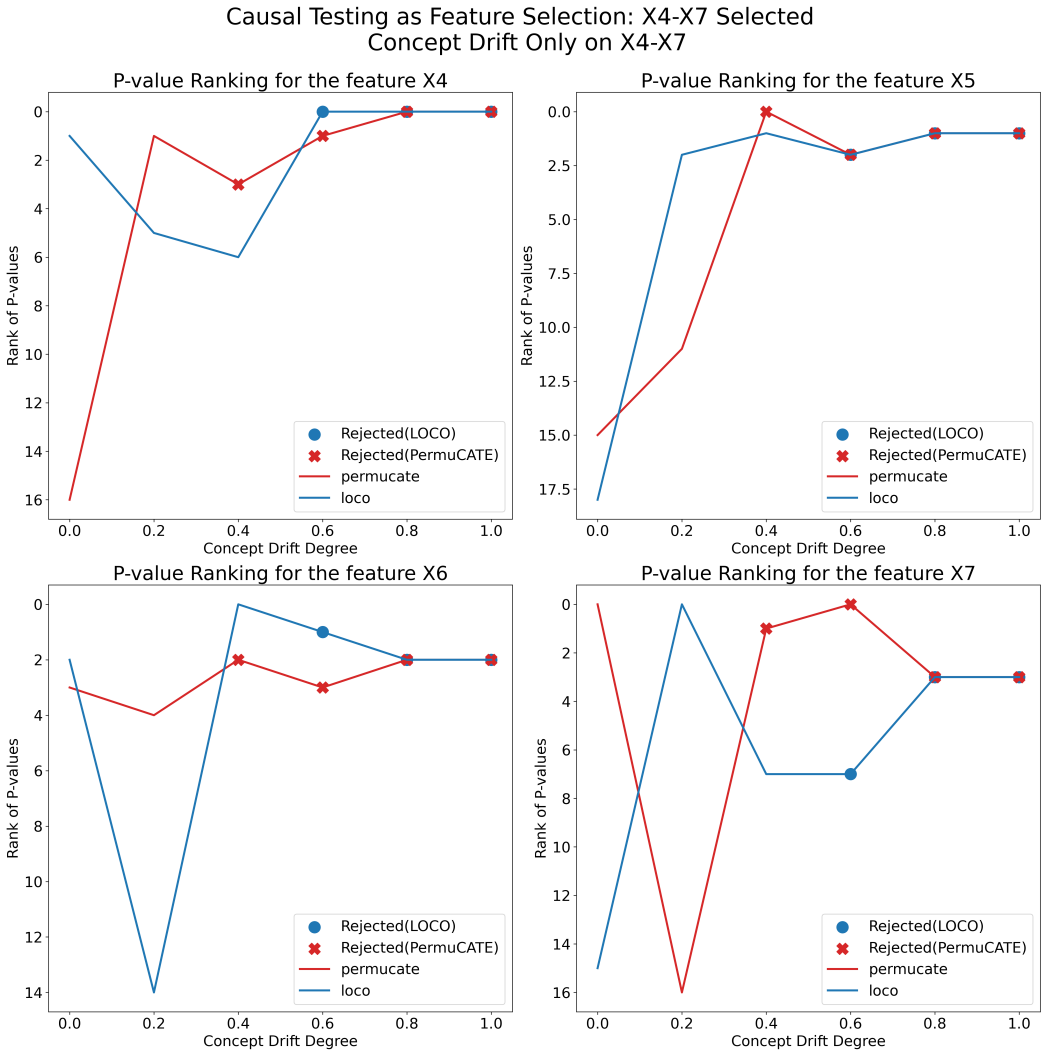


Figure 2: Meta-Learner Testing as Feature Selection: Concept Drift 2

characterized by the different levels of that feature will enhance differential analysis of distribution shift among subgroups. This provides a practical and statistically principled diagnostic tool for downstream case studies and root-cause analysis.

3. The variable importance/feature-specific p-value identified via meta-learners can serve as a notion of **Online** or **OOD** variable importance that characterizes their contribution to distribution shift; with proper interpretation, it implicitly gives deployment engineers concise information about the relative contribution of each metric/feature to data change.

4. Most importantly, from the system level, the top-K features selected can serve as an early alert for model performance/feature-quality degradation with no additional computational cost on top of the hypothesis-testing procedure. From a system-engineering perspective, this enables a proactive reliability paradigm: the monitoring stack can be configured to trigger a 'yellow' alert (i.e., a pre-warning) simply when VIMP for a certain feature exceeds a predefined threshold, without waiting for business metrics to degrade in real-world scenarios.

The entire diagnostic can be run offline, decoupled from real-time inference, so it adds zero latency to serving; within the monitoring window (e.g., hourly or daily), the parallelized pipeline comfortably completes well before the next batch arrives. Scalability is further ensured by highly scalable $O(p)$ post-fit operations (threshold comparison and ranking) without requiring additional model fitting, which remain negligible regardless of p. The data-driven threshold efficiently avoids external data dependencies, which allows the system to handle fluctuating batch sizes without re-engineering.

10.3 Code Availability

The code for implementing the procedure is available in `https://github.com/ruanhq-StatsML/CFPerm` in both **Python** and **R** with both parallelized and serial versions available. The google colab for reproducing the visualization can be found in `https://colab.research.google.com/drive/1EZ2CcPcETwegopk30hpPrDyRUF6a5lNM`.

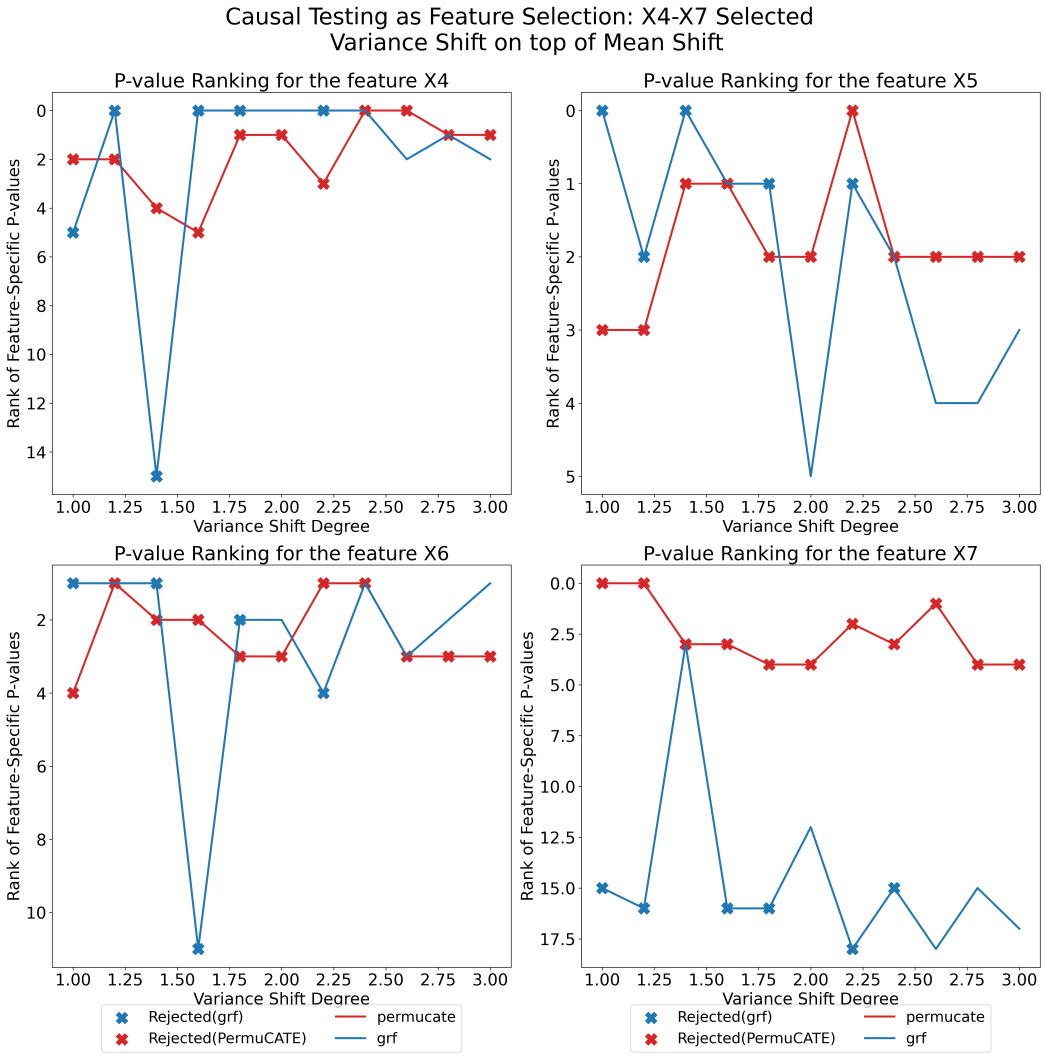


Figure 3: Meta-Learner Testing as Feature Selection: Variance Shift on top of Mean Shift

10.4 Computational Aspect and Efficiency

From the computational efficiency aspect, we want to emphasize that the feature selection procedure itself would enable near-real-time inference with almost no latency especially for the feature ranking indices as well as the rank correlations. Thus in production, the latency itself won't be an issue. The CFPerm procedure is straightforward to parallelize in both R and Python, as each refitting step in the permutation test can be distributed across independent computational nodes. In Python, this is implemented via the `joblib` library; in R, via the `foreach` package. As shown in Figure **??**, parallelization reduces computational cost by approximately 60%–70% compared with the serial implementation on the 2-core **CPU**, achieving computation savings comparable to the DRPerm procedure (based on the pseudo-outcome learner). Like the DRPerm procedure, the CFPerm test can be readily deployed in production environments by leveraging multiple computational nodes in a cluster for configuration of the batch testing procedure every a few minutes justifiably.

Another potential risk for this attribution procedure primarily focuses on the lack of overlap between covariates in $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$, under which circumstance the causal-inference-based procedure will struggle. When support shifts significantly between different batches, the random forest classifier would be highly efficient, perhaps among the best-performing procedures compared with other versions such as the l_1 penalized regression procedures. Thus in practice, we propose the following general strategy for product best practice:

- Obvious covariate shift is observed, where there is almost no overlap between the distributions of covariates in both $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$; the random forest classifier is what we may need. However, random forest VIMP can only tell concisely which feature shifts significantly conditional on the existence of all other features, even including Y, which is fundamentally different from concept-drift detection use cases.
- For latent or gradual concept drift with covariate shift, the meta-learner-based method would work slightly better than SGShift, Knockoff, and MMD for attributing concept drift.
- Under circumstances where both covariate shift and concept drift are significant, the combination of CFPerm and the RF-Domain classifier is recommended.
- In production or real-data analysis, the support shift can be quantified by the overlap score in [63].

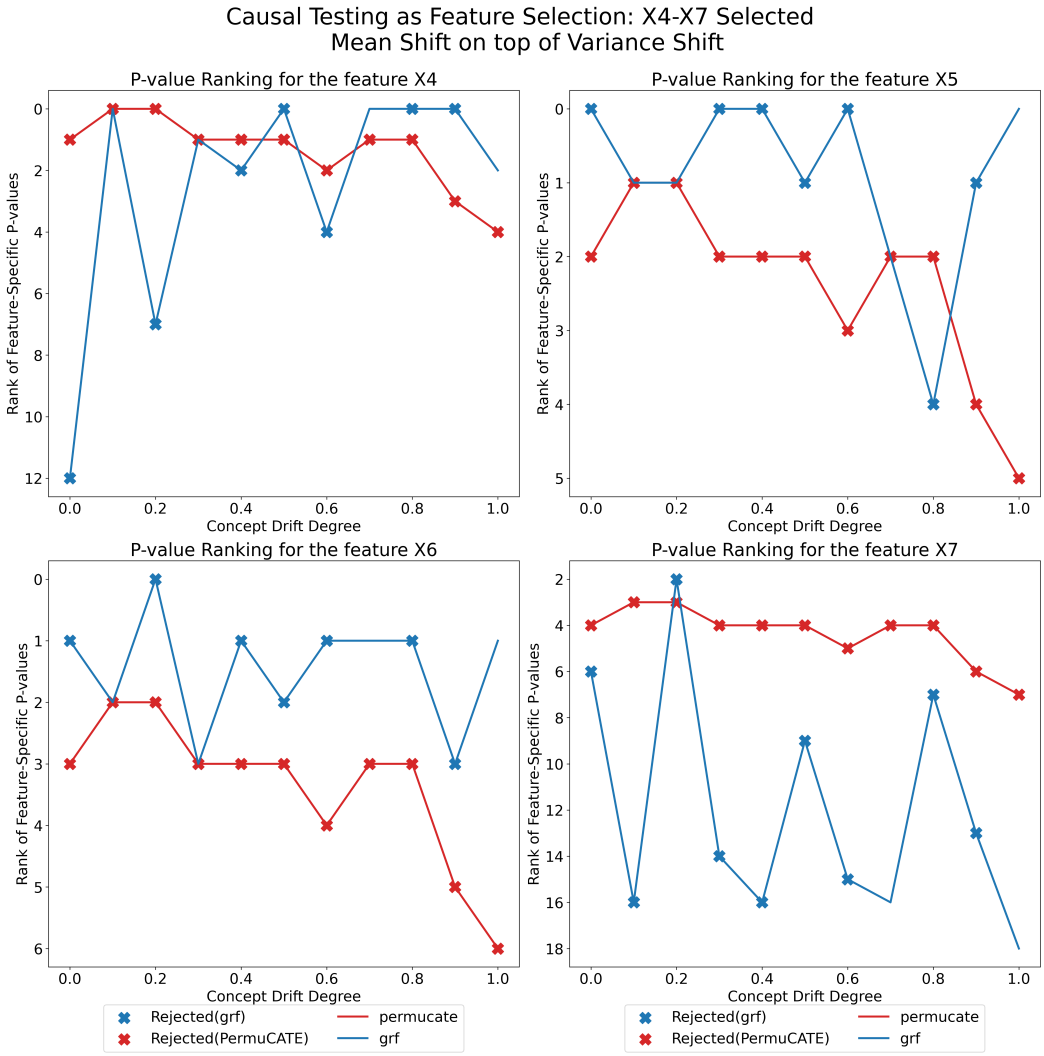


Figure 4: Meta-Learner Testing as Feature Selection: Mean Shift on top of Variance Shift

11 Aggregate the distribution shift variable importance through - Federated Aggregation

In federated learning, ML outputs are typically aggregated across sites in subsequent stages, with the aim of preserving as much information as possible. We therefore examine whether our FSDS procedure, together with out-of-distribution variable importance, can be integrated into this aggregation pipeline—specifically, to help reduce information loss when results are combined across sites. This aggregation procedure has a significant relationship with the federated learning regime. Federated aggregation typically keeps raw rows local while moving only scores or nuisance fits. For variable importance, [81] computes Shapley values for host features in vertical FL, while [25] constructs row-free SHAP backgrounds via federated clustering and GANs. [11] aggregates client-specific SHAP vectors using accuracy and faithfulness weights instead of a flat mean. For vertical feature selection, [27] ranks features via conditional mutual information on statistical summaries, and [47] refines this with Gini initialization and encrypted stochastic gates. To mitigate shift, [43] pools split statistics to grow a federated forest whose impurity decrease approximates a pooled forest under covariate and outcome shift. For causal effects, [85] ships propensity summaries to construct IPW/AIPW estimators, and [78] learns a posterior over potential outcomes via Gaussian processes across sites. [42] compares meta-analysis and plug-in estimators, whereas [3] uses propensity-weighted FedAvg to prevent sites with scarce treated rows from dominating. Most recently, [43] constructs global propensity scores via membership weights, yielding Fed-IPW and Fed-AIPW that explicitly leverage cross-site treatment heterogeneity to improve overlap.

We leverage the **FSDS** procedure applying on this federated aggregation task. We assess rank correlations of variable importance from MMD-LOCO, a random forest binary classifier, and pseudo-outcome risk, in order to evaluate how well the aggregated variable importance can be recovered. Because values and variable importances are scattered across isolated and distributed sites, some information loss is unavoidable. We generated synthetic data with different types of distribution shift across subsets (simulating different sites/regions) to examine the consistency of variable importance rankings under distribution shift and for the predictive model using permutation variable importance. We note that for variable importance—both its ranking and its uncertainty, as with permutation variable importance or **Permu-**

CATE (for the pseudo-outcome learner or other meta-learner)—we use inverse variance weighting (IVW) to account for uncertainty and magnitude at each site during aggregation. Otherwise, we use sample-size weighting and the median to aggregate VIMP magnitude and compute the Spearman rank correlation between site-wise variable importance and global variable importance. The robustness analyses and real-data applications demonstrate the efficiency and robustness of our approach, supported by the statistical justifications provided herein.

11.1 Pressure Tests

A crucial aspect of evaluating efficiency is assessing robustness under varying degrees of data-quality corruption. We consider a scenario with 20 sites, where the variable importance for distribution shift is computed using various notions (e.g., PO-risk, MMD, and RF domain classifier) and subsequently aggregated. We conduct robustness checks by systematically varying both the number of corrupted data centers and the rate of SNR degradation. The aggregated metrics are then compared against the ground-truth feature-shift indices of the entire dataset to verify whether our federated algorithm can still preserve the VIMP ranking. More importantly, we explicitly bypass overly simplified benchmarks like IHDP (with fewer than 500 observations in total), whose mild data-generating process (DGP) design trivially saturates existing methods and fails to expose the breakdown points of aggregation under extreme corruption. Instead, we construct a high-dimensional synthetic setting to rigorously evaluate the robustness of our procedure in recovering weak shift signals from a vast number of features. We emphasize that in the absence of ground-truth indices for the shifted features, the relative comparison across different methods becomes more informative than their absolute magnitudes. Unless otherwise specified, local VIMP is defined as the FSDS share of degree, RF feature importance, and $|\Delta\text{corr}|$; a dash in the tables indicates that the threshold is not crossed on the reported grid. IVW refers to the inverse-variance weight used in the sample-size simulation and in the repliclust / IHDP overlap runs. The high-dimensional active-fraction grid below reports mean share, mean rank, and median share; IVW on the product $n_{\text{bad}} \times \text{SNR} \times \alpha$ is deliberately excluded from these tables, so that the corruption logic remains transparent and unconfounded by weight choice.

The empirical results further demonstrate exceptional robustness. As detailed in Tables 3, 4, 39, and Figure 5, both AUC and $P@K$ are remarkably well-preserved in high-dimensional settings, even when nearly half of the data centers are corrupted or under severely degraded SNR. Specifically, mean share and IVW maintain near-perfect recall under extreme corruption. In contrast, the median share proves highly fragile under partial overlap and harsh SNR regimes. While IVW is susceptible to weight inflation from corrupted sites with underestimated variance, its variance-aware weighting mechanism generally yields slightly superior robustness compared to both mean and median aggregation. Furthermore, the procedure is fairly robust to sample-size imbalance across different data centers (Table 6). Notably, even under severe imbalance conditions (e.g., Dirichlet concentration parameter 10), both the AUC and precision remain well preserved. Overall, the proposed aggregation protocol exhibits strong resilience against severe federated heterogeneity.

Finally, we extend the evaluation beyond the high-dimensional corruption scale to the heterogeneous shift. Tables 37, 13, and 36 present the aggregated VIMP rankings across broader two-batch benchmarks under heterogeneous site-level shifts, confirming that the framework provides a highly efficient proxy for the aggregated version of VIMP. In the synthetic setting detailed in Table 13, the covariates are of dimension $d = 10$, with the signal residing on the first five coordinates $m(X) = X_{1:5}^{\top}\beta$ where $\beta = \mathbf{1}_5$, plus $N(0, 1)$ noise. The total sample size is $n = 2400$, and client sizes are either equal or drawn from a Dirichlet allocation. Three of the six clients carry the stated shift while the other three remain clean, following standard covariate-shift and concept-drift designs (e.g., a mean shift on the first four coordinates, a Gaussian covariance shift, a two-component mixture, and concept drifts of linear, nonlinear, cubic, interaction, and threshold types). The results in Table 13 are averaged over two seeds and both size allocations.

11.2 Real Data Applications

To further validate the efficiency of our aggregation strategy across diverse two-batch benchmarks, we conduct extensive evaluations on real-world datasets. Since ground-truth indices for shifted features are typically unavailable in practice, we manually inject synthetic linear and nonlinear covariate shifts and concept drifts into half of the features to establish the ground truth. As detailed in Tables 7 and 8, across challenging datasets such as SECOM [54], Diabetes130 [20], TEP [17], and TCGA [83], the aggregated procedure remains highly robust even when nearly half of the data centers are corrupted. Furthermore, both the PO-risk (pseudo-outcome risk) and LOCO-MMD components demonstrate exceptional performance under these settings.

Beyond these tabular and biomedical benchmarks, we extend our evaluation to image and text modalities using the FEMNIST [13] and WhyShift [51] datasets to conduct the ranking consistency evaluation for the variable importance aggregation. For the FEMNIST dataset, we split it into individual sets by pairs of different labels (manually written digits), and for the WhyShift dataset, we randomly shuffle them into 10 folds and conduct the comparison within each fold. The Spearman rank correlation is calculated between the aggregated variable importance ranking and the whole-fold variable importance rankings.

Specifically for the FEMNIST image-based two-batch settings, our methodology returns three vectors for a pair of batches: the random-forest domain-classifier importance, the leave-one-coordinate RBF MMD, and the increase in

pseudo-outcome risk under conditional permutation. For character-pair partitions, a 30-component PCA is fitted on the pooled pair, and each batch is partitioned into $K \in \{5, 10, 20, 30\}$ blocks. Table 11 reports the mean Spearman correlation over four pairs from the five most common characters between the sample-size weighted merge and the pooled vector, whereas Table 12 shows the consistency of ranks for covariate shift attribution after aggregating multiple data sources.

Using the same digit images as a direct prediction problem—rather than through the OOD wrapper—provides a sharper two-batch validation check. A random forest classifies digits 0–9 from 30 pooled PCA scores on 200,000 images, where importance is defined as the drop in accuracy when a component is permuted. The rows are cut into 20 blocks of 10,000. Because the blocks have equal size, the uniform average and the sample-size weighted average coincide. Their Spearman correlation with the pooled importance is 0.996 (Pearson 0.996), and inverse-variance weighting yields a Spearman correlation of 0.993. Notably, a single block, prior to merging, already correlates at 0.975, and the first four components are $2, 3, 6, 9$ in both the pooled vector and the merge. This demonstrates that the procedure efficiently aggregates variable importance across subsets, and that information loss from inter-site interactions does not substantially degrade the combined VIMP ranking.

Interestingly, Pseudo-outcome (PO) risk does not follow this pattern in the image-based two-batch setting. On the 200,000 digit images with a balanced random treatment arm and the digit as the outcome, the pooled PO-risk vector has a mean absolute entry of 0.0007. Inverse-variance and sample-size merges correlate with it at a Spearman rank of -0.02 and -0.01, respectively. Since the treatment arm is independent of the image, the contrast estimated by the wrapper contains no residual signal after the outcome regression and propensity score adjustment.

11.3 Statistical Justification

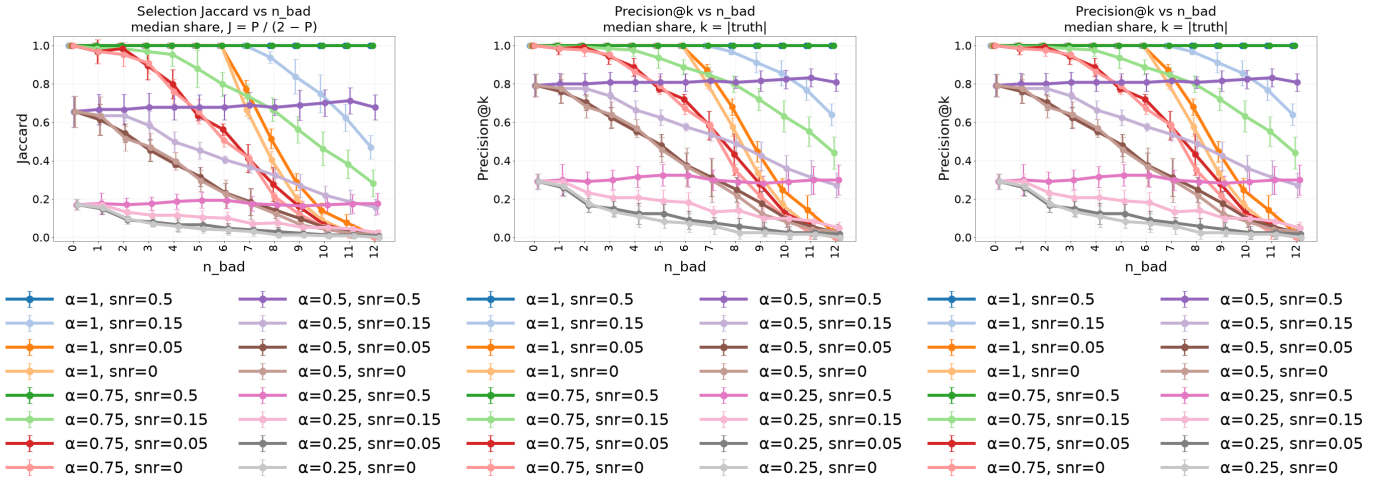
Here we denote the variable importance for site k and feature j as $V_{j,k}$, where $j \in \{1, \ldots, p\}$ and $k \in \{1, \ldots, K\}$. We note that the dependence across the variable importances is unknown within each individual site (data center), but for statistical justification, we assume that the VIMPs are independent across different sites. (This is distinct from multimodal attribution, where multiple data loaders are digested to attribute to individual feature subsets.) To establish the concentration of the variable importance aggregation, our statistical guarantee requires only a minimal condition: the local VIMP estimators are consistent in both mean and variance. That is, for each site k and feature j, $\hat{V}_{j,k} \xrightarrow{p} V^*_{j,k}$ and $\hat{\sigma}^2_{j,k} \xrightarrow{p} \sigma^2_{j,k}$. By independently normalizing each local vector such that $\mathbf{V}_k \in [0,1]^p$ and $\sum_{j=1}^p V_{k,j} = 1$, we ensure the spectral norm is strictly bounded: $\|\mathbf{V}_k\|_2 \leq \|\mathbf{V}_k\|_1 = 1$. Consequently, the aggregated vector $\bar{\mathbf{V}}$ converges to its expectation. Crucially, the Matrix Bernstein inequality upgrades this pointwise consistency into a *uniform* concentration bound across all p features, where the bound remains entirely agnostic to their unknown cross-feature correlations.

One aspect that requires elaboration is the aggregation scheme. Regardless of whether the aggregator is a median or a mean, as long as a consistent variance estimator exists (e.g., via bootstrap, jackknife, or bag-of-little-bootstrap, potentially depending on the number of sites K), the aggregated VIMP satisfies the Matrix Bernstein bound. Specifically for the Inverse-Variance Weighting (IVW) procedure, it additionally requires the variance estimates to be consistent and strictly bounded below. Given this minimal condition, the Matrix Bernstein bound holds for the IVW aggregation as well.

The Matrix Bernstein bound fundamentally bypasses the need to estimate the unknown $p \times p$ covariance matrix. The ambient dimension p enters only through $\log(2p/\delta)$, which establishes a joint confidence ellipsoid avoiding the Family-wise Error Rate (FWER). The net effect is a statistically rigorous aggregation that is equally efficient computationally and operationally. In terms of communication, each site only needs to upload a single p-dimensional normalized VIMP vector exactly *once*. The total communication cost is strictly $O(Kp)$ floats. Compared to federated learning protocols that iteratively synchronize high-dimensional model parameters, our one-shot aggregation reduces the communication overhead by a factor of $1/K$, requiring no transmission of raw data or iterative model updates. Thus, the framework achieves finite-sample valid uncertainty quantification at a minimal communication cost.

Theorem (Matrix Bernstein Bound for aggregated VIMP). Let the local VIMP vectors be normalized such that $\mathbf{V}_k \in [0,1]^p$ and $\sum_{j=1}^p V_{k,j} = 1$ for all $k \in \{1, \ldots, K\}$. Define the aggregated VIMP as $\bar{\mathbf{V}} = \frac{1}{K}\sum_{k=1}^K \mathbf{V}_k$. For any $\delta \in (0,1)$, with probability at least $1 - \delta$, the spectral-norm deviation of $\bar{\mathbf{V}}$ from its expectation satisfies

$$\left\|\bar{\mathbf{V}} - \mathbb{E}[\bar{\mathbf{V}}]\right\|_2 \leq \sqrt{\frac{2\log(2p/\delta)}{K}} + \frac{2\log(2p/\delta)}{3K}. \tag{1}$$



(a) (b) (c)

Figure 5: Sensitivity Analysis for the corrupted data-centers/sites on the efficiency of the federated aggregation procedure

11.4 Summary

In summary, the VIMP rankings for covariate-shift attribution and predictive accuracy are reasonably effective under aggregation, whereas ranking consistency tends to be weaker for PO-risk due to information loss across blocks and sites. While we do not claim this as a comprehensive federated learning optimization framework, it delivers two highly practical benefits. First, it enables feature-level attribution of distribution shift across isolated sites, providing interpretable insights into which features are driving the data change. Second, by only requiring a single upload of local VIMP vectors, it significantly reduces communication costs compared to standard iterative federated learning. Thus, it serves as a direct interpretation and extension of the FSDS procedure to domains such as federated learning and multimodal learning, where understanding feature-level data changes is crucial.

Looking ahead, several directions remain open. First, extending the framework to continuous batch settings with adaptive updating strategies could offer more flexible solutions tailored to streaming industrial demands. Second, incorporating modality-wise contributions of pseudo-outcome risk would provide a direct interpretation of how each modality contributes to overall OOD shift. Third, developing a differentially private version of the one-shot aggregation, where the uploaded VIMP vectors are perturbed under a formal privacy budget while preserving the Matrix Bernstein concentration, would further strengthen the practical deployability of the framework in sensitive domains. We leave these as promising avenues for future research.

12 Refinements relative to the Matrix Bernstein baseline

Claim. Under Assumptions (A1)–(A4), the one-shot aggregate $\bar{V}$ obeys the scalar and coordinate-wise bounds (2)–(3). Under the additional gap condition (A5), the empirical Top-K set of $\bar{V}$ coincides with that of $\mathbb{E}\bar{V}$ on the event $\{\|\bar{V} - \mathbb{E}\bar{V}\|_\infty < \text{margin}/2\}$. Under (A6), the leading range factor in (3) may be replaced by an empirical variance proxy. The assumptions are satisfied by any simplex-normalized upload; they do not require a low-rank or sparse model for $\Sigma = \mathrm{Cov}(V_k)$. The resulting inequalities are not asserted to dominate the spectral Matrix Bernstein bound for every Σ, nor to be minimax optimal; they are sufficient for coordinate-wise control and for a standard Top-K corollary.

Assumptions.

(A1) $V_k \in [0,1]^p$ with $\mathbf{1}^\top V_k = 1$ for $k = 1, \ldots, K$; the sites are independent; $\bar{V} = \sum_k w_k V_k$ with $w_k \geq 0$ and $\sum_k w_k = 1$.

(A2) $\hat{R} := \max_k \|V_k\|_2$ and $R_\star \geq \hat{R}$ (e.g. $R_\star = 1$).

(A3) K_{eff} denotes the effective sample size induced by (w_k) ($K_{\text{eff}} = K$ when $w_k = 1/K$).

(A4) A fixed contrast $a \in \mathbb{R}^p$ with $\|a\|_2 \leq 1$; $v_a := \mathrm{Var}(a^\top V_1)$ under identical site laws, or the analogous weighted variance otherwise.

(A5) *(Top-K corollary.)* The population shares separate a designated index set of size K from its complement by margin > 0.

(A6) *(Empirical Bernstein / IVW.)* Consistent estimates $\hat{\sigma}_k^2 \geq \underline{\sigma}^2 > 0$ for the coordinates or contrast in use.

Dependence among coordinates within a site is unrestricted, and Σ is not estimated. No condition is imposed on $\mathrm{rank}(\Sigma)$. In particular, (A1)–(A2) hold for every probability-simplex vector, including any FSDS share upload.

Spectral baseline. Under (A1)–(A3) and a bound R on $\|w_k(V_k - \mathbb{E}V_k)\|_2$, Matrix Bernstein [**tropp2012user**, **mackey2014matrix**] yields, with probability at least $1-\delta$,

$$\|\bar{V} - \mathbb{E}\bar{V}\|_2 \le C\Big(R\sqrt{\tfrac{\log(2p/\delta)}{K_{\text{eff}}}} + R\tfrac{\log(2p/\delta)}{K_{\text{eff}}}\Big).$$

The sole modification used below is the substitution $R \leftarrow R_\star$ from (A2).

Scalar and ℓ_∞ bounds. Under (A1)–(A4), with probability at least $1-\delta$,

$$\big|a^\top(\bar{V} - \mathbb{E}\bar{V})\big| \le C\Big(\sqrt{\tfrac{v_a \log(2/\delta)}{K_{\text{eff}}}} + R_\star\tfrac{\log(2/\delta)}{K_{\text{eff}}}\Big), \tag{2}$$

and, by a union bound over coordinates,

$$\|\bar{V} - \mathbb{E}\bar{V}\|_\infty \le C\Big(R_\star\sqrt{\tfrac{\log(2p/\delta)}{K_{\text{eff}}}} + R_\star\tfrac{\log(2p/\delta)}{K_{\text{eff}}}\Big). \tag{3}$$

Both displays are of Bernstein type. Equation (3) supplies the deviation that enters the Top-K event; it is not presented as a strict improvement of the spectral-norm inequality for all covariance structures.

Top-K corollary. Under (A1)–(A5), the event $\{\|\bar{V} - \mathbb{E}\bar{V}\|_\infty < \text{margin}/2\}$ implies equality of the empirical and population Top-K sets. Together with (3), a sufficient condition is

$$C\Big(R_\star\sqrt{\tfrac{\log(2p/\delta)}{K_{\text{eff}}}} + R_\star\tfrac{\log(2p/\delta)}{K_{\text{eff}}}\Big) < \tfrac{1}{2}\,\text{margin}.$$

If the margin lies below this threshold, Top-K instability is attributable to insufficient separation of population shares rather than to the choice among mean, median, or IVW aggregation. Concordance of the full Spearman ranking is not claimed.

Empirical Bernstein. Under (A1)–(A4) and (A6), empirical Bernstein [**wainwright2019high**, **vershynin2018high**] replaces the leading factor $R_\star$ in (3) by a consistent variance proxy, up to a remainder of order $R_\star \log(2p/\delta)/K_{\text{eff}}$. The same variance input is required for IVW [**dersim**]. If the lower bound in (A6) fails for some k, truncated weights or equal weights restore (3).

Table 3: High-Dimensional Robustness under Varying Local Support Sizes (α). We report the Spearman rank correlation ρ, aggregated AUC, and Precision@K ($P@K$) as the number of corrupted sites (n_{bad}) increases from 0 to 12.

$n_{\text{bad}}/12$	$\alpha = 1$			$\alpha = 0.75$			$\alpha = 0.5$			$\alpha = 0.25$		
	ρ	AUC	$P@K$	ρ	AUC	$P@K$	ρ	AUC	$P@K$	ρ	AUC	$P@K$
0	1.000	1.000	1.00	1.000	1.000	1.00	1.000	1.000	1.00	1.000	1.000	1.00
1	0.938	1.000	1.00	0.942	1.000	1.00	0.954	1.000	1.00	0.973	0.991	0.98
2	0.890	1.000	1.00	0.898	1.000	1.00	0.915	1.000	1.00	0.949	0.990	0.98
3	0.852	1.000	1.00	0.858	1.000	1.00	0.885	1.000	1.00	0.929	0.984	0.97
4	0.818	1.000	1.00	0.827	1.000	1.00	0.857	1.000	1.00	0.909	0.979	0.95
5	0.791	1.000	1.00	0.795	1.000	1.00	0.833	1.000	0.99	0.888	0.968	0.92
6	0.767	1.000	1.00	0.770	1.000	1.00	0.811	1.000	0.99	0.874	0.953	0.86
7	0.746	1.000	1.00	0.748	1.000	1.00	0.795	0.999	0.97	0.856	0.930	0.79
8	0.730	1.000	1.00	0.729	1.000	1.00	0.777	0.985	0.94	0.837	0.905	0.72
9	0.713	1.000	1.00	0.716	0.999	0.98	0.760	0.980	0.90	0.819	0.880	0.65
10	0.699	1.000	1.00	0.701	0.997	0.94	0.741	0.962	0.80	0.801	0.857	0.52
11	0.686	1.000	1.00	0.688	0.985	0.86	0.724	0.936	0.64	0.785	0.799	0.37
12	0.675	0.988	0.75	0.672	0.945	0.60	0.706	0.890	0.37	0.768	0.744	0.19

Table 4: Shared Support: Dimensional Scaling and SNR Swap. The top panel reports the tipping points (minimum n_{bad}) at which ρ falls below specific thresholds. The bottom panel reports AUC and $P@K$ under varying SNR regimes.

p	K	Tipping Points (n_{bad})			$n_{\text{bad}} = 10$
		$\rho < 0.95$	$\rho < 0.90$	$\rho < 0.80$	$((\rho, \text{AUC}, P@K)$
500	20	1	3	7	(0.718, 0.987, 0.81)
1000	25	1	2	4	(0.655, 0.981, 0.69)
2000	50	1	1	3	(0.538, 0.959, 0.57)

SNR Swap at Fixed p (Clean SNR = 1.5, AUC = 1.0)

SNR_{low}	p	$n_{\text{bad}} = 5$		$n_{\text{bad}} = 10$	
		AUC	$P@K$	AUC	$P@K$
0.5	1000	1.000	1.00	1.000	1.00
0.5	2000	1.000	1.00	1.000	1.00
0.25	1000	1.000	1.00	1.000	1.00
0.25	2000	1.000	1.00	1.000	0.97
0.05	1000	1.000	1.00	0.628	0.10
0.05	2000	1.000	1.00	0.605	0.06
0	1000	1.000	1.00	0.487	0.02
0	2000	1.000	1.00	0.497	0.04

Table 5: Contamination Tipping Points: Mean Share vs. Median Share. Entries denote the minimum n_{bad} at which the metric falls below the threshold. A dash (—) indicates the threshold is not crossed on the reported grid.

Mode	SNR_{low}	Mean Share (n_{bad} for $\downarrow$)			Median Share (n_{bad} for $\downarrow$)		
		$\rho < 0.90$	AUC< 0.95	$J@K < 0.7$	$\rho < 0.90$	AUC< 0.95	$J@K < 0.7$
low_snr	0.5	—	—	—	—	—	—
low_snr	0.25	—	—	—	3	—	—
low_snr	0.1	10	10	10	3	9	9
low_snr	0	7	9	9	3	8	7
null_snr	—	5	9	9	3	8	7
wrong_plant	—	1	7	6	4	6	6
y_scramble	—	4	—	—	2	—	—

Table 6: Sample-Size Imbalance (Dirichlet α) and Mixed Sites. We report the Spearman ρ of the observed aggregate against the clean aggregate. IVW denotes inverse-variance weighting. Rows $\alpha \in \{0.05, 20\}$ use the same setting as $\alpha \in \{0.05, 0.1, 1, 10, 20\}$: $B = 10$, $p = 80$, $K = 10$, $N = 4000$, five replicates.

α	Regime	Uniform	n-weighted	IVW	Median
0.05	Covariate	1.000	1.000	0.788	1.000
0.05	Concept	0.996	0.913	0.876	0.975
0.1	Covariate	1.000	1.000	0.754	1.000
0.1	Concept	0.996	0.921	0.888	0.985
1	Covariate	1.000	1.000	0.883	1.000
1	Concept	0.991	0.952	0.738	0.964
10	Covariate	1.000	1.000	0.971	1.000
10	Concept	0.984	0.965	0.610	0.961
20	Covariate	1.000	1.000	0.977	1.000
20	Concept	0.985	0.976	0.749	0.950
Mixed 5 CS + 5 CD		0.300	0.300	—	0.295
Mixed (vs All-CD)		0.334	0.334	—	0.372

Table 7: Federated Alignment on Challenging Real-World Datasets under Concept Drift. We report AUC, Precision@K ($P@K$), and Spearman ρ under varying numbers of corrupted sites (n_{bad}).

Dataset	n_{bad}	Mean Share			Median Share		
		AUC	$P@K$	ρ	AUC	$P@K$	ρ
SECOM	0	1.000	1.00	1.000	1.000	1.00	1.000
	5	1.000	0.98	0.894	0.995	0.92	0.917
	10	0.708	0.40	0.657	0.683	0.34	0.758
TCGA	0	1.000	1.00	1.000	1.000	1.00	1.000
	5	1.000	1.00	0.869	0.999	0.98	0.877
	10	0.977	0.84	0.771	0.974	0.82	0.758
Diabetes	0	1.000	1.00	1.000	1.000	1.00	1.000
	5	1.000	1.00	0.887	1.000	1.00	0.796
	10	0.920	0.75	0.814	0.940	0.85	0.635
TEP	0	1.000	1.00	1.000	1.000	1.00	1.000
	5	0.996	0.88	0.868	0.970	0.68	0.728
	10	0.580	0.12	0.582	0.586	0.20	0.581

Table 8: Pooled FSDS Benchmark on Challenging Datasets. We compare FSDS against baselines under linear and complex concept drift (random and low-$|\beta|$ halves). The best-performing FSDS results are highlighted in bold.

Dataset	Drift Type	AUC					$P@K$				
		PO	PermuCATE	FSDS	RF	LOCO-MMD	PO	PermuCATE	FSDS	RF	LOCO-M
SECOM	Linear	0.505	0.465	**0.827**	0.937	0.643	0.35	0.30	**0.65**	0.80	
	Complex (Rand)	0.302	0.366	0.419	0.506	0.500	0.30	0.40	0.45	0.55	
	Complex (Low-β)	0.365	0.603	0.292	0.465	0.500	0.40	0.55	0.35	0.45	
TCGA-PANCAN	Linear	0.458	0.452	**0.890**	0.967	0.872	0.25	0.35	**0.75**	0.90	
	Complex (Rand)	0.693	0.445	0.540	0.573	0.542	0.65	0.45	0.55	0.55	
	Complex (Low-β)	0.667	0.297	0.551	0.603	0.542	0.55	0.40	0.60	0.60	
PBMC3k	Linear	0.495	0.623	**0.858**	0.967	1.000	0.25	0.40	**0.55**	0.90	
	Complex (Rand)	0.430	0.772	0.430	0.460	0.432	0.50	0.70	0.45	0.45	
	Complex (Low-β)	0.430	0.287	0.352	0.448	0.430	0.40	0.35	0.40	0.45	
Diabetes130	Complex (Rand)	0.691	0.682	0.627	0.600	0.518	0.60	0.60	0.60	0.60	
	Complex (Low-β)	0.373	0.600	0.345	0.600	0.200	0.30	0.60	0.40	0.60	
TEP	Complex (Rand)	0.512	0.598	0.529	0.600	0.542	0.50	0.50	0.45	0.60	
	Complex (Low-β)	0.435	0.675	0.497	0.422	0.372	0.40	0.65	0.55	0.50	

Table 9: Comparison of feature selection methods on SECOM and Diabetes130 datasets under Covariate (Cov) and Concept (Con) shifts, evaluated by AUC and Precision@10 ($P@10$). Two split strategies are reported: random split and `max_feature` split. Best FSDS results are in bold.

Method	SECOM (random)				Diabetes130 (random)				SECOM (`max_feature`)				Diabetes130 (`max_feature`)			
	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$
PO	0.452	0.400	0.619	0.600	0.308	0.633	0.892	0.867	0.487	0.267	0.579	0.600	0.492	0.733	0.758	0.767
FSDS	**0.716**	**0.433**	**0.598**	**0.400**	**0.792**	**0.867**	**0.742**	**0.800**	**0.650**	**0.267**	**0.442**	**0.200**	**0.483**	**0.733**	**0.713**	**0.767**
deg	0.950	0.833	0.552	0.333	0.992	0.967	0.467	0.733	0.916	0.767	0.429	0.167	0.850	0.900	0.583	0.733
RF	1.000	1.000	0.549	0.300	1.000	1.000	0.558	0.800	0.920	0.733	0.459	0.200	0.917	0.967	0.617	0.767
RF-LOCO	0.622	0.367	0.447	0.267	0.758	0.767	0.475	0.733	0.631	0.367	0.418	0.200	0.575	0.767	0.433	0.667
LOCO-MMD	0.987	0.967	0.496	0.233	0.983	0.967	0.475	0.733	0.897	0.800	0.517	0.233	0.700	0.767	0.408	0.700
HSIC-LOCO	0.402	0.167	0.577	0.367	0.575	0.767	0.475	0.767	0.505	0.367	0.612	0.367	0.558	0.767	0.475	0.700
GRF	0.543	0.267	0.774	0.633	0.442	0.700	0.692	0.833	0.526	0.300	0.667	0.433	0.433	0.700	0.592	0.733
ℓ_1 mis	0.580	0.267	0.803	0.667	0.458	0.700	0.725	0.767	0.718	0.500	0.697	0.500	0.533	0.767	0.700	0.767
ℓ_1 spec	0.587	0.367	0.796	0.633	0.583	0.767	0.550	0.700	0.657	0.467	0.749	0.567	0.550	0.733	0.683	0.767
ΔSHAP	0.587	0.267	0.778	0.633	0.475	0.733	0.750	0.800	0.478	0.233	0.744	0.600	0.333	0.633	0.792	0.800
knockoff	0.578	0.367	0.796	0.600	0.500	0.733	0.996	0.967	0.626	0.367	0.749	0.633	0.329	0.667	0.883	0.900
PermuCATE*	0.523	0.350	0.748	0.600	0.492	0.733	0.808	0.800	0.539	0.367	0.709	0.567	0.450	0.667	0.717	0.767

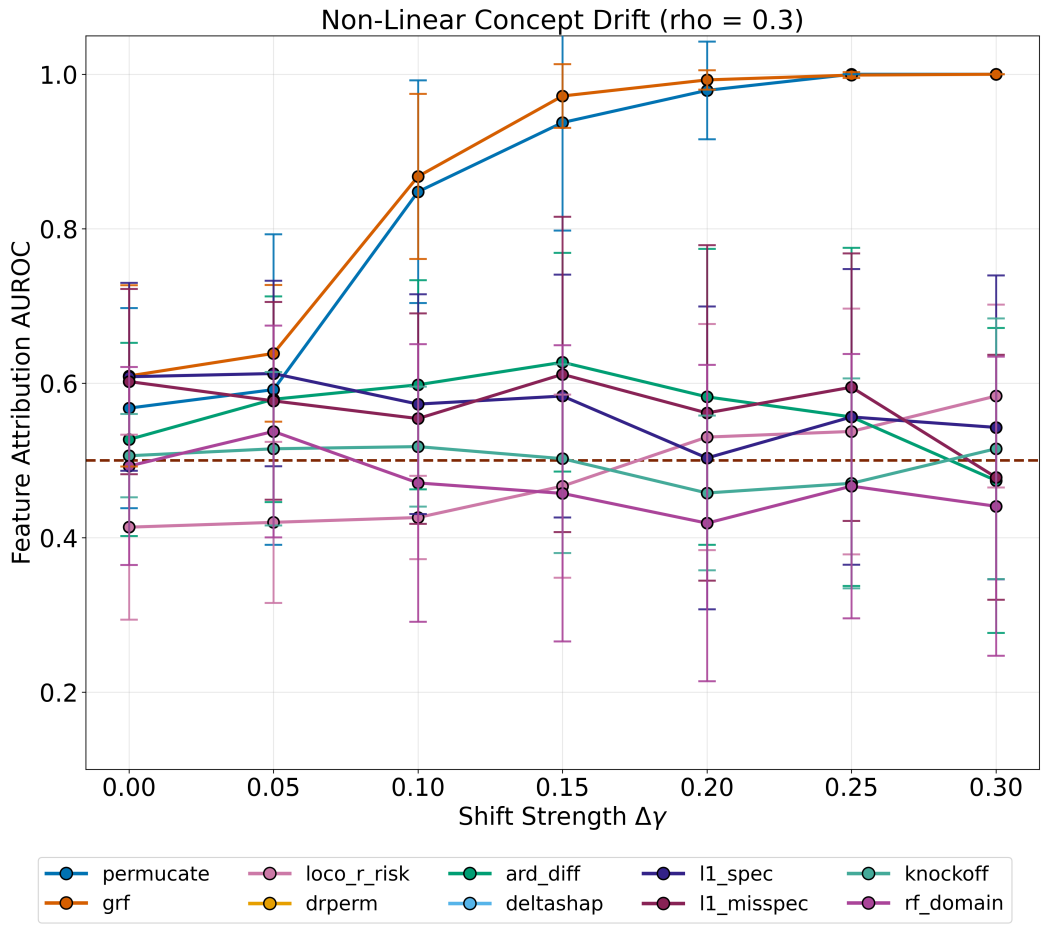
* PermuCATE on SECOM random covariate is the mean of 2 reps; rep 0 had a one-class label batch and the fit was skipped.

Table 10: ShiftDGP. Spearman correlation of the merged permutation importance with the pooled importance.

Shift	(δ_X, δ_C)	Uniform	n-weighted	IVW	Median
None	$(0, 0)$	0.891	0.900	0.903	0.885
Mean	$(0.5, 0)$	0.815	0.794	0.791	0.815
Mean	$(1.5, 0)$	0.839	0.827	0.833	0.861
Covariance	$(0.6, 0)$	0.842	0.836	0.833	0.845
Mixture	$(0.8, 0)$	0.809	0.845	0.848	0.848
Linear drift	$(0, 1)$	0.903	0.933	0.921	0.897
$\sin(X_0 X_1)$	$(0, 1)$	0.867	0.894	0.888	0.873
Cubic	$(0, 0.4)$	0.842	0.845	0.852	0.888
$X_0 X_2 X_3$	$(0, 1)$	0.803	0.830	0.855	0.830
Threshold	$(0, 1)$	0.867	0.867	0.873	0.873
Mean + sine	$(1, 1)$	0.836	0.830	0.821	0.848
Mean + product	$(1, 1)$	0.836	0.848	0.845	0.909

Table 11: FEMNIST character pairs. Spearman correlation of the n-weighted merge with the pooled OOD-wrapper vector.

Wrapper output	$K = 5$	$K = 10$	$K = 20$	$K = 30$
RF domain classifier	0.795	0.819	0.843	0.830
MMD-LOCO	0.781	0.887	0.903	0.909



(a)

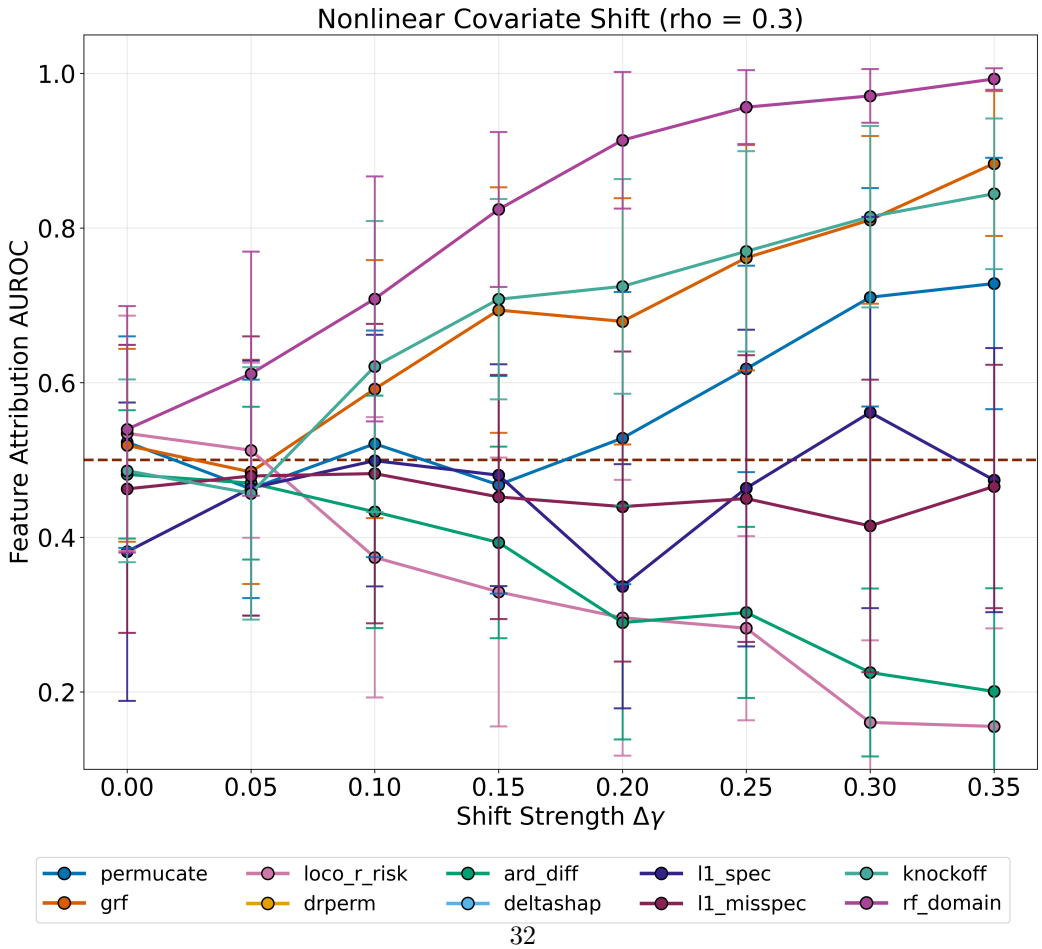
Figure 6: Feature Selection for the nonlinear Concept Drift

Table 12: FEMNIST covariate-shift OOD attribution. Entries are Spearman correlations with the pooled ranking, averaged over the four adjacent pairs among the five most frequent characters (1–7, 7–3, 3–0, and 0–2). Each pair is projected to 30 principal components fit on the pooled images; every site reuses the same coordinates and uses at most 800 images per class. Single-site is the mean correlation of the site-level rankings. Mean and Median are the correlations of the coordinate-wise mean and median of the site-level scores. With a common cap of 800 images, the sample-size-weighted mean coincides with the unweighted mean.

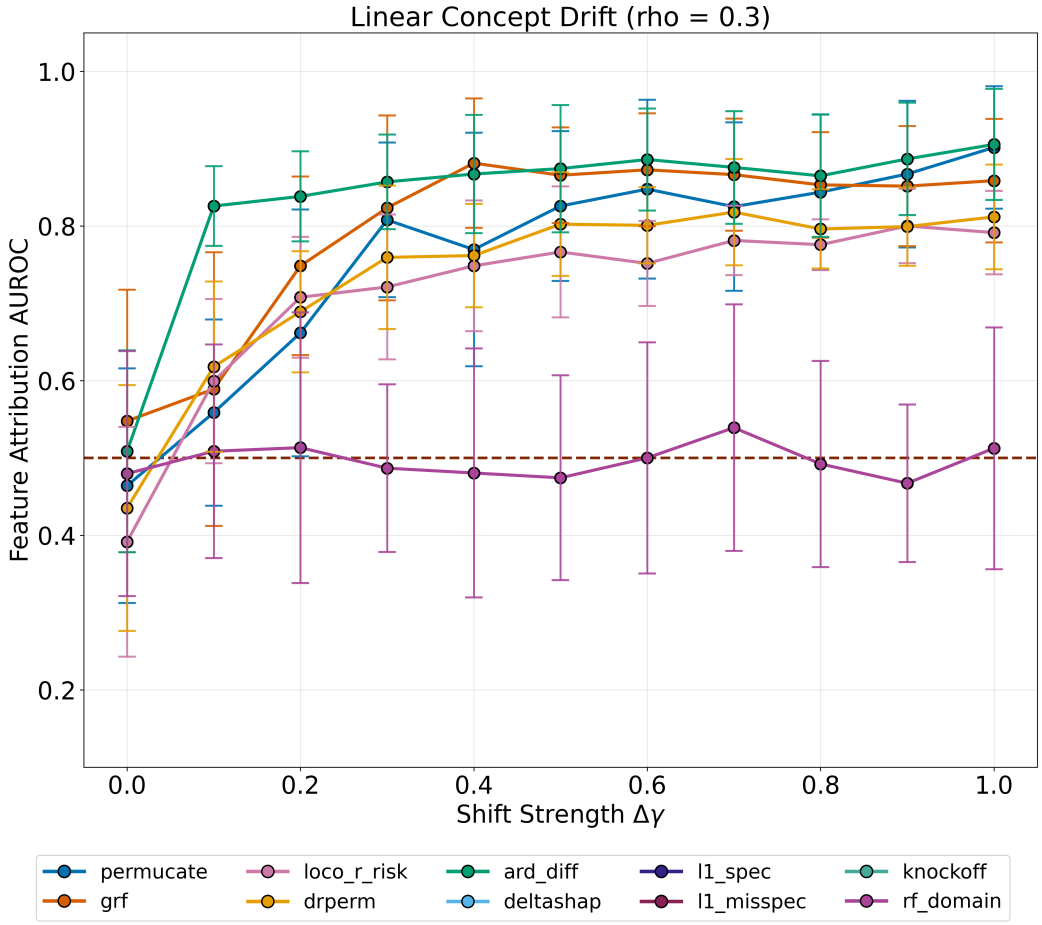
Sites	Method	Single-site	Mean	Median
5	RF domain	0.74	0.80	0.78
	MMD-LOCO	0.79	0.78	0.81
10	RF domain	0.73	0.82	0.81
	MMD-LOCO	0.83	0.89	0.89
20	RF domain	0.74	0.84	0.84
	MMD-LOCO	0.84	0.90	0.90
30	RF domain	0.74	0.83	0.83
	MMD-LOCO	0.83	0.91	0.88

Table 13: Homogeneous distribution shifts across different subsets(clients). Spearman correlation of ranking the aggregated permutation importance with the centralized(aggregated) VIMP

Shift	(δ_X, δ_C)	Uniform	sample size-weighted	IVW	Median
None	$(0, 0)$	0.891	0.900	0.903	0.885
Mean	$(0.5, 0)$	0.815	0.794	0.791	0.815
Mean	$(1.5, 0)$	0.839	0.827	0.833	0.861
Covariance	$(0.6, 0)$	0.842	0.836	0.833	0.845
Mixture	$(0.8, 0)$	0.809	0.845	0.848	0.848
Linear CD	$(0, 1)$	0.903	0.933	0.921	0.897
Nonlinear CD	$(0, 1)$	0.867	0.894	0.888	0.873
Cubic CD	$(0, 0.4)$	0.842	0.845	0.852	0.888
Interaction CD	$(0, 1)$	0.803	0.830	0.855	0.830
Threshold CD	$(0, 1)$	0.867	0.867	0.873	0.873
Mean + nonlinear	$(1, 1)$	0.836	0.830	0.821	0.848
Mean + interaction	$(1, 1)$	0.836	0.848	0.845	0.909

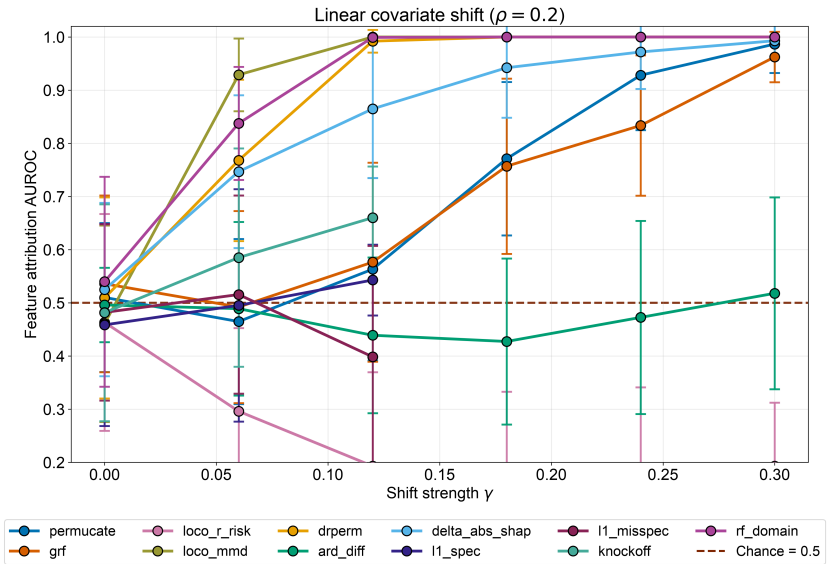


(a)



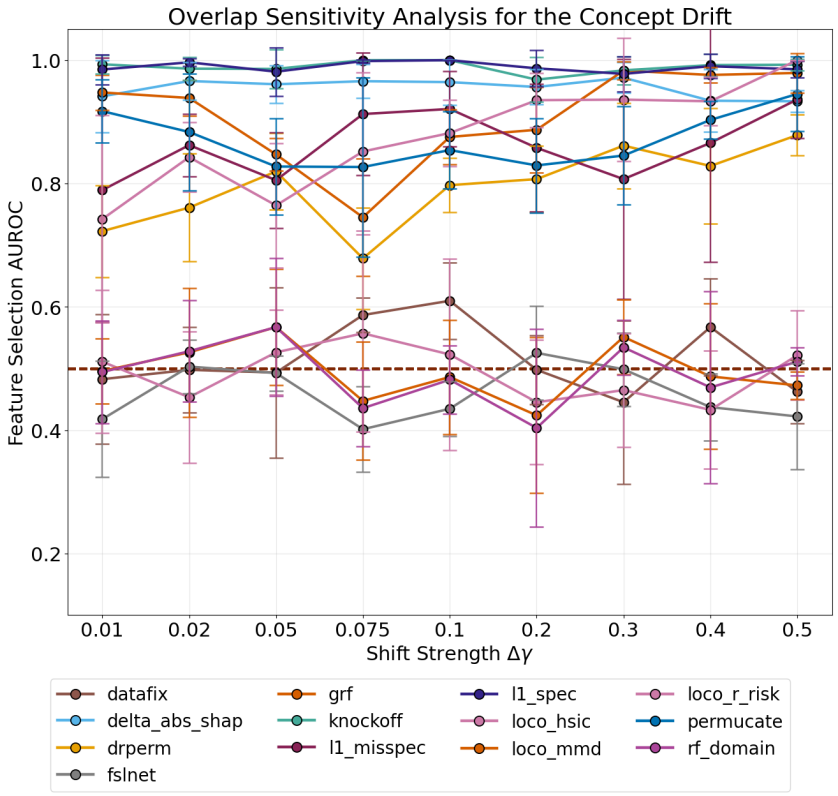
(a)

Figure 8: Feature Selection for the Linear Concept Drift



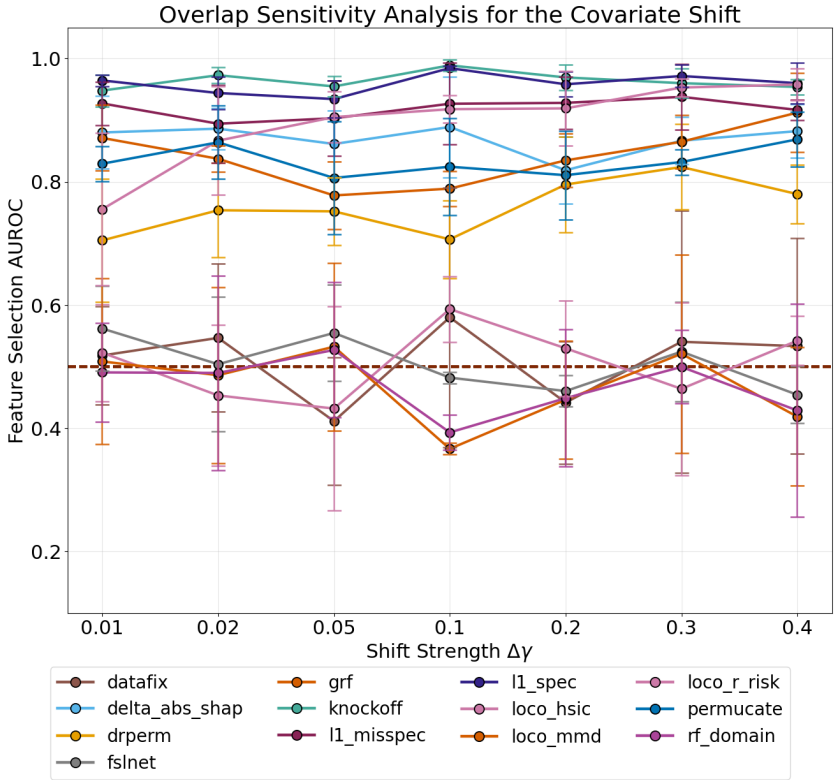
(a)

Figure 9: Feature Selection for the linear covariate shift



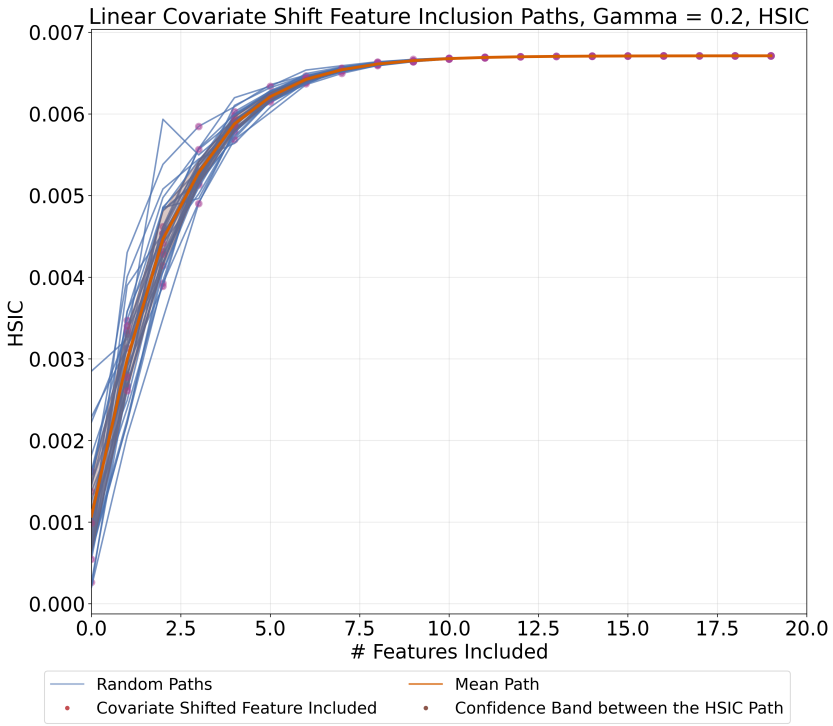
(a)

Figure 10: Overlap Sensitivity Analysis for Concept Drift



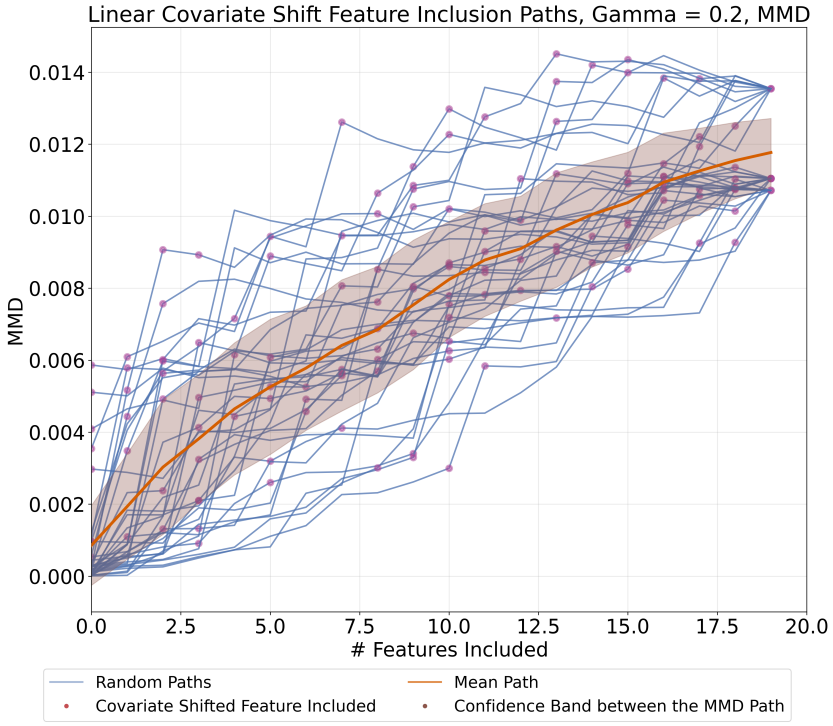
(a)

Figure 11: Overlap Sensitivity Analysis for Covariate Shift



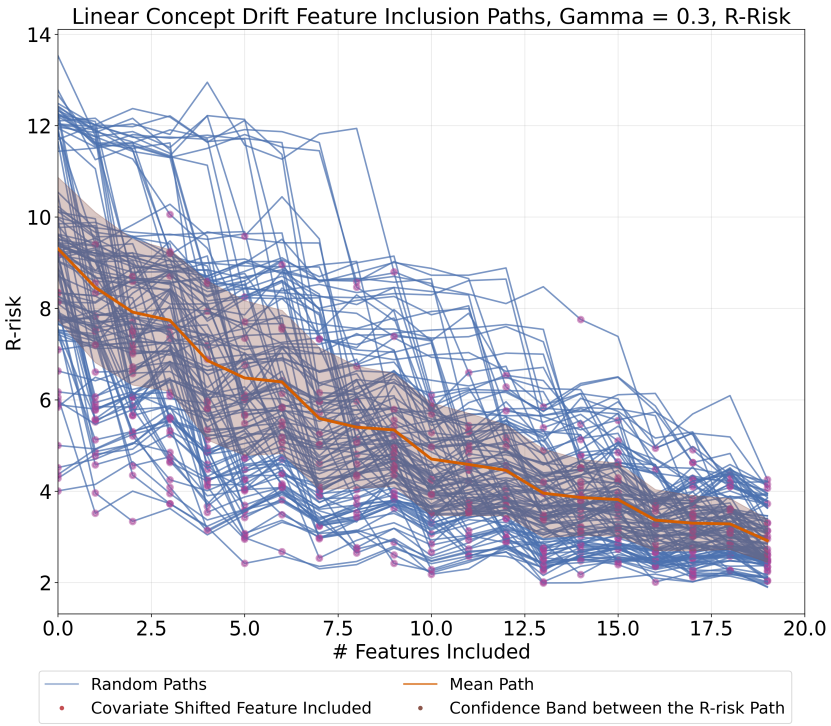
(a)

Figure 12: Feature Path for the HSIC measure for Covariate Shift, the feature inclusion paths are permuted and the the incremental HSIC is recorded(the first 4 features are those with covariate shift)



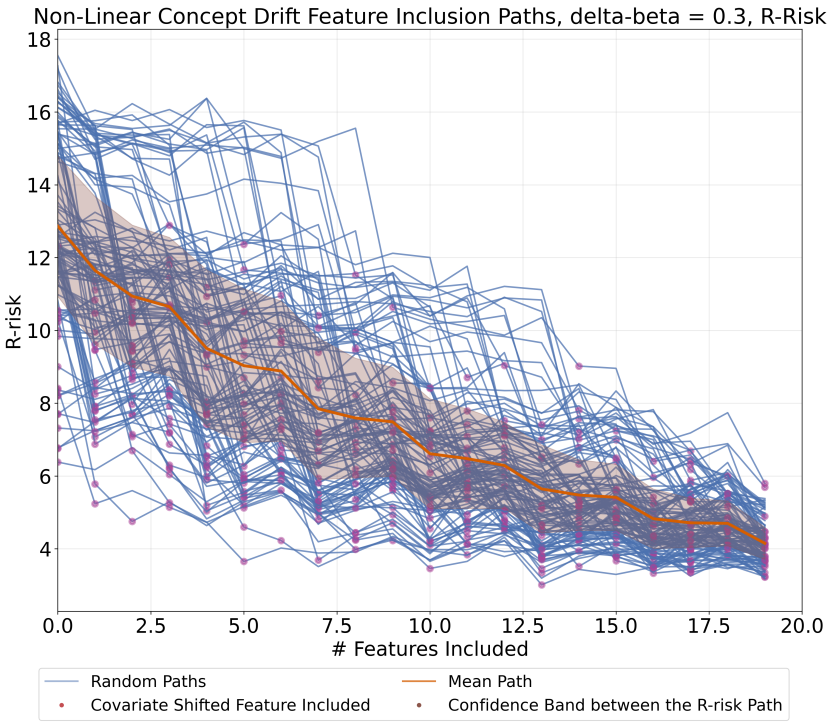
(a)

Figure 13: Feature Path for the MMD measure for Covariate Shift, the feature inclusion paths are permuted and the the incremental MMD is recorded(the first 4 features are those with covariate shift)



(a)

Figure 14: Feature Path for the R-risk measure for Linear Concept Drift, the feature inclusion paths are permuted and the R-risk is recorded(the first 4 features are those with concept drift)



(a)

Figure 15: Feature Path for the R-risk measure for Nonlinear Concept Drift, the feature inclusion paths are permuted and the R-risk is recorded(the first 4 features are those with concept drift)

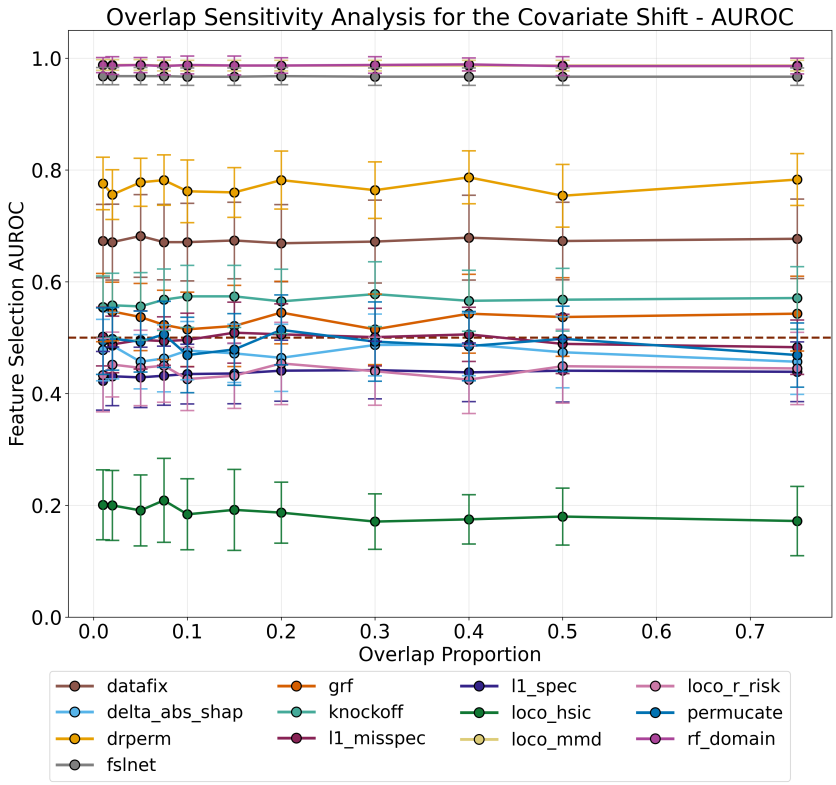


Figure 16: AUC vs. overlap sensitivity for the ChronoBerg Dataset with the Synthetic Covariate Shift

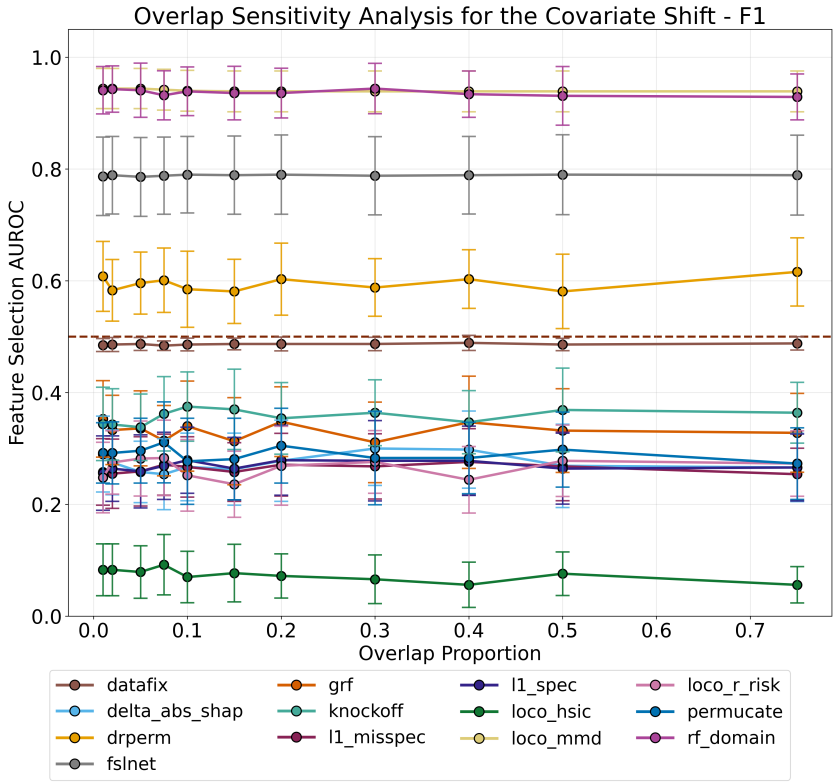


Figure 17: F1 vs. overlap sensitivity for the ChronoBerg Dataset with the Synthetic Covariate Shift

Table 14: Summary of the datasets we used - scales, feature types and the detailed descriptions

	0	1	2	3	4	5	6	7
dataset	dfadult	dfhouse16H	dfelevators	dfpol	dfcpu	dfkin	dfAil	dfbank32nh
nsamples	48842	22784	16599	15000	8192	8192	13750	8192
nfeatures	13	16	18	48	21	8	40	32
ncontinuousfeatures	0	16	18	26	21	8	40	32
ndiscretenumericfeatures	5	0	0	22	0	0	0	0
ncategoricalfeatures	8	0	0	0	0	0	0	0
featuremix	mixed	all continuous	all continuous	mixed	all continuous	all continuous	all continuous	all continuous
responsename	Y	Y	Goal	Y	usr	y	goal	rej
responsetype	continuous	continuous	continuous	discrete	continuous	continuous	continuous	continuous
nresponselevels	28523	2045	61	11	56	8191	35	6284

Table 15: Concept Drift Injection Comparison Procedures for feature selection, aggregation across 8 real-world datasets - F1 & AUC recorded, FGSM50 denotes that 50% of the features are shifted,

method	FGSM30AUC	FGSM30F1	covariateshiftAUC	covariateshiftF1	FGSM50AUC	FGSM50F1
grf	0.883	0.756	0.82	0.682	0.753	0.712
l1misspec	0.835	0.705	0.817	0.675	0.682	0.673
l1spec	0.815	0.668	0.822	0.693	0.682	0.6245
locorrisk	0.827	0.685	0.764	0.592	0.726	0.7
permucate	0.892	0.727	0.785	0.651	0.772	0.734
rfdomain	0.572	0.378	0.896	0.87	0.762	0.734
locommd	0.497	0.336	0.933	0.877	0.713	0.679
deltashap	0.658	0.475	0.733	0.695	0.688	0.621
knockoff	0.796	0.645	0.762	0.715	0.713	0.658
datafix	0.881	0.757	0.866	0.708	0.854	0.768
fslnet	0.866	0.665	0.846	0.732	0.633	0.497
ard	0.703	0.754	0.728	0.726	0.756	0.638

Table 16: Nonlinear Concept Drift for the real-data applications - average across different datasets

	avgauc	avgF1	sdauc	sdF1
deltashap	0.77	0.6151	0.1656	0.1571
grf	0.8085	0.6731	0.1503	0.2155
knockoff	0.7379	0.6781	0.1898	0.258
l1misspec	0.7942	0.7457	0.1803	0.2087
l1spec	0.8019	0.7493	0.1929	0.2146
locommd	0.7612	0.5876	0.134	0.1725
locorrisk	0.6429	0.5683	0.2443	0.2577
permucate	0.7944	0.7096	0.1461	0.2056
rfdomain	0.7568	0.588	0.1027	0.1361
datafix	0.721	0.546	0.194	0.259
fslnet	0.783	0.597	0.185	0.296
ard	0.7675	0.7265	0.176	0.1672

Table 17: Average Performance for the Adversarial Perturbation - alternative methods, mainly focus on covariate shift.

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.58	0.247	0.601	0.079
1	grf	0.819	0.174	0.859	0.174
2	knockoff	0.564	0.174	0.566	0.106
3	l1misspec	0.613	0.183	0.649	0.149
4	l1spec	0.597	0.152	0.626	0.126
5	locommd	0.868	0.187	0.883	0.183
6	locorrisk	0.366	0.162	0.346	0.146
7	permucate	0.817	0.174	0.839	0.068
8	rfdomain	0.888	0.148	0.936	0.129
9	fslnet	0.879	0.173	0.935	0.186
10	datafix	0.862	0.279	0.918	0.262
11	rfoob	0.901	0.178	0.958	0.095
12	xgboob	0.802	0.258	0.899	0.141
13	mlpoob	0.641	0.36	0.739	0.32
14	ard	0.844	0.075	0.789	0.141

Table 18: Adversarial for the covariate shift - 50% features attacked

	method	shifttype	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	cs	0.631	0.212	0.7	0.235
1	drperm	cs	0.656	0.201	0.689	0.251
2	grf	cs	0.74	0.13	0.797	0.171
3	knockoff	cs	0.644	0.228	0.651	0.233
4	l1misspec	cs	0.555	0.115	0.531	0.147
5	l1spec	cs	0.58	0.266	0.606	0.258
6	locommd	cs	0.704	0.243	0.713	0.258
7	locorrisk	cs	0.526	0.146	0.551	0.172
8	permucate	cs	0.745	0.229	0.802	0.186
9	rfdomain	cs	0.85	0.168	0.914	0.152
10	datafix	cs	0.757	0.368	0.854	0.194
11	fslnet	cs	0.711	0.212	0.866	0.259
12	ard	cs	0.656	0.199	0.66	0.253

Table 19: Adversarial for the Case where Concept Drift Dominated - Whyshift Dataset

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.702	0.095	0.763	0.763
1	drperm	0.519	0.056	0.525	0.525
2	grf	0.548	0.122	0.597	0.597
3	knockoff	0.518	0.064	0.52	0.52
4	l1misspec	0.457	0.088	0.47	0.47
5	l1spec	0.373	0.103	0.383	0.383
6	locommd	0.21	0.067	0.188	0.188
7	locorrisk	0.343	0.058	0.293	0.293
8	permucate	0.634	0.091	0.666	0.173
9	rfdomain	0.698	0.101	0.732	0.732
10	fslnet	0.416	0.078	0.541	0.095
11	datafix	0.436	0.059	0.539	0.111
12	ard	0.574	0.061	0.593	0.126

Table 20: Adversarial perturbation - 50% features attacked, mainly covariate shift, minor concept drift

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.639	0.096	0.708	0.107
1	grf	0.703	0.186	0.735	0.188
2	knockoff	0.613	0.115	0.66	0.138
3	l1misspec	0.625	0.174	0.629	0.174
4	l1spec	0.619	0.186	0.655	0.158
5	locommd	0.741	0.228	0.796	0.273
6	locorrisk	0.496	0.124	0.516	0.097
7	permucate	0.72	0.231	0.752	0.271
8	rfdomain	0.732	0.208	0.754	0.189
9	datafix	0.709	0.353	0.824	0.187
10	fslnet	0.658	0.21	0.722	0.251
11	ard	0.56	0.253	0.65	0.199

Table 21: Adversarial Perturbation: FGSM + Concept Drift with Concept Drift Dominated

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.483	0.15	0.696	0.172
1	drperm	0.525	0.248	0.693	0.063
2	grf	0.455	0.198	0.708	0.118
3	knockoff	0.511	0.265	0.703	0.086
4	l1misspec	0.565	0.309	0.725	0.093
5	l1spec	0.582	0.303	0.714	0.084
6	locommd	0.417	0.279	0.647	0.087
7	locorrisk	0.435	0.13	0.556	0.112
8	permucate	0.534	0.164	0.732	0.178
9	rfdomain	0.405	0.387	0.685	0.107
10	fslnet	0.392	0.138	0.578	0.092
11	datafix	0.317	0.104	0.529	0.106
12	ard	0.512	0.186	0.728	0.088

Table 22: FMoW dataset benchmark, covariate shift dominated

	method	avgF1	sdF1	avgAUC	sdAUC
0	deltashap	0.3115	0.0865	0.4992	0.0847
1	drperm	0.3881	0.171	0.5857	0.1543
2	grf	0.2462	0.1156	0.4442	0.1119
3	knockoff	0.4035	0.141	0.6247	0.1033
4	l1misspec	0.4244	0.137	0.6256	0.1071
5	l1spec	0.4027	0.1431	0.5921	0.1225
6	locommd	0.7029	0.2823	0.8222	0.2158
7	locorrisk	0.2762	0.0967	0.4619	0.0925
8	permucate	0.3029	0.0981	0.5039	0.1035
9	rfdomain	0.5426	0.1854	0.8125	0.1625
10	fslnet	0.5625	0.2469	0.8096	0.139
11	datafix	0.4273	0.1736	0.6822	0.1195

Table 23: ChronoBerg Dataset, average over the many comparisons with all types of shifts - covariate shifts dominates

	method	avgF1	sdF1	avgAUC	sdAUC
0	deltashap	0.2666	0.1076	0.488	0.1012
1	drperm	0.4963	0.184	0.6924	0.1527
2	grf	0.3667	0.168	0.5485	0.1445
3	knockoff	0.3407	0.1303	0.5792	0.1294
4	l1misspec	0.2296	0.1127	0.503	0.1163
5	l1spec	0.2444	0.1182	0.4586	0.1011
6	locommd	0.6963	0.3561	0.7577	0.321
7	locorrisk	0.3	0.1373	0.4762	0.1373
8	permucate	0.3222	0.125	0.5443	0.1151
9	rfdomain	0.8056	0.319	0.7073	0.2665
10	fslnet	0.8169	0.1783	0.6772	0.2317
11	datafix	0.7229	0.1791	0.6289	0.2734
12	ard	0.4382	0.1527	0.5367	0.0534

Table 24: Whyshift Dataset, average over the many comparisons with all types of shifts - covariate shift & Concept Drift

	method	avgF1	sdF1	avgAUC	sdAUC
0	deltashap	0.7081	0.1273	0.7779	0.0883
1	drperm	0.5513	0.0678	0.555	0.0424
2	grf	0.4919	0.1169	0.5382	0.1309
3	knockoff	0.5189	0.0483	0.5636	0.0604
4	l1misspec	0.3784	0.0427	0.4221	0.0393
5	l1spec	0.3568	0.082	0.3544	0.1024
6	locommd	0.2486	0.0353	0.2245	0.0389
7	locorrisk	0.3622	0.0491	0.3215	0.0213
8	permucate	0.6703	0.0673	0.7091	0.076
9	rfdomain	0.7297	0.0689	0.7561	0.0483
10	fslnet	0.4159	0.1863	0.5388	0.095
11	datafix	0.5731	0.0301	0.5414	0.111
12	rfoob	0.6843	0.1165	0.7251	0.1174
13	mlpoob	0.6282	0.1578	0.6095	0.0143
14	ard	0.4571	0.1741	0.5361	0.1239

Table 25: Comparison of feature selection methods on SECOM and Diabetes130 datasets under Covariate (Cov) and Concept (Con) shifts, evaluated by AUC and Precision@10 (P@10). Two split strategies are reported: random split and `max_feature` split. Best FSDS results are in bold.

Method	SECOM (random)				Diabetes130 (random)				SECOM (`max_feature`)				Diabetes130 (`max_feature`)			
	Cov AUC	Cov P@10	Con AUC	Con P@10	Cov AUC	Cov P@10	Con AUC	Con P@10	Cov AUC	Cov P@10	Con AUC	Con P@10	Cov AUC	Cov P@10	Con AUC	Con P@10
PO	0.452	0.200	0.619	0.400	0.308	0.633	0.892	0.867	0.487	0.267	0.379	0.200	0.492	0.733	0.758	0.767
FSDS	**0.716**	**0.433**	**0.598**	**0.400**	**0.792**	**0.867**	**0.742**	**0.800**	**0.650**	**0.267**	**0.442**	**0.200**	**0.483**	**0.733**	**0.713**	**0.767**
deg	0.950	0.833	0.552	0.333	0.992	0.967	0.467	0.733	0.916	0.767	0.429	0.167	0.850	0.900	0.583	0.733
RF	1.000	1.000	0.549	0.300	1.000	1.000	0.558	0.800	0.920	0.733	0.459	0.200	0.917	0.967	0.617	0.767
RF-LOCO	0.622	0.367	0.447	0.267	0.758	0.767	0.475	0.733	0.631	0.367	0.418	0.200	0.575	0.767	0.433	0.667
LOCO-MMD	0.987	0.967	0.496	0.233	0.983	0.967	0.475	0.733	0.897	0.800	0.517	0.233	0.700	0.767	0.408	0.700
HSIC-LOCO	0.402	0.167	0.577	0.367	0.575	0.767	0.475	0.767	0.505	0.367	0.612	0.367	0.558	0.767	0.475	0.700
GRF	0.543	0.267	0.774	0.633	0.442	0.700	0.692	0.833	0.526	0.300	0.667	0.433	0.433	0.700	0.592	0.733
ℓ_1 mis	0.580	0.267	0.803	0.667	0.458	0.700	0.725	0.767	0.718	0.500	0.697	0.500	0.533	0.767	0.700	0.767
ℓ_1 spec	0.587	0.367	0.796	0.633	0.583	0.767	0.550	0.700	0.657	0.467	0.749	0.567	0.550	0.733	0.683	0.767
ΔSHAP	0.587	0.267	0.778	0.633	0.475	0.733	0.750	0.800	0.478	0.233	0.744	0.600	0.333	0.633	0.792	0.800
knockoff	0.578	0.367	0.796	0.600	0.500	0.733	0.996	0.967	0.626	0.367	0.749	0.633	0.329	0.667	0.883	0.900
PermuCATE*	0.523	0.350	0.748	0.600	0.492	0.733	0.808	0.800	0.539	0.367	0.709	0.567	0.450	0.667	0.717	0.767

* PermuCATE on SECOM random covariate is the mean of 2 reps; rep 0 had a one-class label batch and the fit was skipped.

Table 26: Planted-support recovery under covariate vs. concept drift on five low-SNR panels (K=10, SNR=1.5). Each cell is AUC / P@10.

Dataset	Method	Covariate		Concept	
		AUC	P@10	AUC	P@10
SECOM	PO-risk	0.452	0.200	0.619	0.400
	FSDS	0.716	0.433	0.598	0.400
	LOCO-MMD	0.987	0.967	0.496	0.233
	DEG $\|\Delta\mu\|$	0.950	0.833	0.552	0.333
	SHIFT ℓ_1-mis	0.580	0.267	0.803	0.667
	GRF	0.543	0.267	0.774	0.633
	RF-domain	1.000	1.000	0.549	0.300
TCGA-PANCAN	PO-risk	0.503	0.267	0.772	0.633
	FSDS	0.944	0.800	0.718	0.467
	LOCO-MMD	1.000	1.000	0.461	0.300
	DEG $\|\Delta\mu\|$	1.000	1.000	0.408	0.200
	SHIFT ℓ_1-mis	0.541	0.300	0.684	0.467
	GRF	0.702	0.433	0.584	0.233
	RF-domain	1.000	1.000	0.543	0.233
PBMC3k	PO-risk	0.496	0.267	0.638	0.567
	FSDS	0.904	0.767	0.714	0.533
	LOCO-MMD	1.000	1.000	0.506	0.300
	DEG $\|\Delta\mu\|$	1.000	1.000	0.556	0.333
	SHIFT ℓ_1-mis	0.531	0.333	0.477	0.267
	GRF	0.547	0.300	0.494	0.233
	RF-domain	1.000	1.000	0.493	0.267
Diabetes130	PO-risk	0.308	0.633	0.892	0.867
	FSDS	0.792	0.867	0.742	0.800
	LOCO-MMD	0.983	0.967	0.475	0.733
	DEG $\|\Delta\mu\|$	0.992	0.967	0.467	0.733
	SHIFT ℓ_1-mis	0.458	0.700	0.725	0.767
	GRF	0.442	0.700	0.692	0.833
	RF-domain	1.000	1.000	0.558	0.800
TEP	PO-risk	0.448	0.267	0.671	0.467
	FSDS	0.777	0.500	0.704	0.533
	LOCO-MMD	1.000	1.000	0.560	0.267
	DEG $\|\Delta\mu\|$	0.938	0.767	0.489	0.300
	SHIFT ℓ_1-mis	0.530	0.267	0.553	0.333
	GRF	0.579	0.300	0.607	0.300
	RF-domain	1.000	1.000	0.539	0.333

Table 27: Predictive permutation importance on FEMNIST digits. A random forest classifies the digit from 30 pooled PCA scores on 200,000 images. Importance is the drop in accuracy when a component is permuted. The same rows are cut into 20 sites of 10,000, and the site vectors are merged. Sites have equal size, so the uniform average and the sample-size weighted average coincide. Correlation is with the importance computed on all 200,000 rows. The leading four components are 2, 3, 6, and 9 in both the pooled vector and the merge.

Merge	Spearman	Pearson
Single site (mean of 20)	0.975	0.958
Uniform / n-weighted	0.996	0.996
Inverse-variance	0.993	0.997

Table 28: Time Profiling Comparison w.r.t. Sample-sizes

n	drpermmean	drpermstd	grfmean	grfstd	knockoffmean	knockoffstd	l1misspecmean	l1misspecstd	locomean	locostd	mmdmean	mmdstd	permucatemean	permucatestd	rfmean	rfstd	shapmean	shapstd	walltotalmean	walltotalstd	locorriskmean	locorriskstd
200	21.108	3.965	23.831	12.755	9.693	1.419	5.71	0.81	3.69	0.39	1.115	0.315	32.077	10.723	1.267	0.829	3.469	1.263	92.155	23.294	3.106	2.341
300	34.798	6.308	4.371	1.214	11.992	1.648	11.59	0.82	14.2	2.2	2.148	0.613	52.135	9.848	0.823	0.334	3.78	0.864	112.727	15.816	12.684	3.835
500	35.92	4.886	9.408	1.325	25.587	11.257	11.59	1.82	12.48	2.84	5.465	1.093	52.092	7.098	0.95	0.203	5.299	0.842	128.329	12.327	16.412	2.934
750	46.4	4.691	10.887	1.391	38.117	5.116	10.86	1.36	22.8	3.74	9.128	1.225	48.578	7.501	1.027	0.208	6.823	1.496	144.457	14.339	18.834	4.072
1000	55.497	8.733	14.071	3.503	35.059	8.77	26.66	4.99	34.21	6.84	16.634	2.287	60.771	10.703	0.948	0.229	8.412	1.117	182.075	24.771	22.037	5.097
1500	100.609	14.274	24.966	5.693	22.16	2.3	35.19	4.02	38.23	6.4	21.006	4.923	94.789	16.942	1.766	0.464	16.223	3.114	302.051	45.736	35.085	7.408
2000	93.15	31.83	21.53	6.604	36.444	37.504	52.01	11.48	56.58	16.31	12.781	4.452	75.932	23.157	1.349	0.421	14.678	5.892	274.069	73.085	28.277	10.795
3000	69.154	41.886	43.589	11.409	256.954	102.464	67.37	14.15	75.01	16.31	14.694	4.727	145.383	42.229	2.87	0.975	29.649	9.653	689.306	171.061	18.002	11.802

Table 29: Time Profiling Comparison w.r.t. Sample-sizes - with neural network based methods, n = 2000

p	20	50	150	200
datafixmean	37.71	54.61	70.62	101.93
datafixsd	11.11	18.3	22.49	27.93
rfoobmean	10.57	25.68	46.87	93.67
rfoobsd	1.65	9.1	22.1	41.38
fslnetmean	9.02	15.83	24.11	59.87
fslnetsd	3.33	4.13	7.64	26.95

Table 30: Overlap Sensitivity for Concept Drift

	overlap	method	nreps	F1mean	F1std	precisionmean	precisionstd	recallmean	recallstd	selectionAUCmean	selectionAUCstd
0	0.01	datafix	2	0.0	0.0	0.0	0.0	0.0	0.0	0.564	0.038
13	0.1	datafix	2	0.0	0.0	0.0	0.0	0.0	0.0	0.58	0.211
26	0.4	datafix	2	0.0	0.0	0.0	0.0	0.0	0.0	0.567	0.292
1	0.01	deltashap	2	0.8	0.094	0.8	0.094	0.8	0.094	0.862	0.082
14	0.1	deltashap	2	0.767	0.141	0.767	0.141	0.767	0.141	0.889	0.082
27	0.4	deltashap	2	0.833	0.047	0.833	0.047	0.833	0.047	0.9	0.009
2	0.01	drperm	2	0.667	0.094	0.667	0.094	0.667	0.094	0.749	0.141
15	0.1	drperm	2	0.633	0.047	0.633	0.047	0.633	0.047	0.707	0.063
28	0.4	drperm	2	0.733	0.0	0.733	0.0	0.733	0.0	0.744	0.041
3	0.01	fslnet	2	0.059	0.083	0.25	0.354	0.033	0.047	0.584	0.129
16	0.1	fslnet	2	0.111	0.157	0.333	0.471	0.067	0.094	0.482	0.009
29	0.4	fslnet	2	0.0	0.0	0.0	0.0	0.0	0.0	0.464	0.053
4	0.01	grf	2	0.767	0.047	0.767	0.047	0.767	0.047	0.869	0.009
17	0.1	grf	2	0.733	0.094	0.733	0.094	0.733	0.094	0.789	0.028
30	0.4	grf	2	0.767	0.047	0.767	0.047	0.767	0.047	0.896	0.06
5	0.01	knockoff	2	0.9	0.047	0.9	0.047	0.9	0.047	0.93	0.011
18	0.1	knockoff	2	0.933	0.0	0.933	0.0	0.933	0.0	0.989	0.009
31	0.4	knockoff	2	0.933	0.0	0.933	0.0	0.933	0.0	0.948	0.017
6	0.01	l1misspec	2	0.9	0.047	0.9	0.047	0.9	0.047	0.916	0.031
19	0.1	l1misspec	2	0.9	0.047	0.9	0.047	0.9	0.047	0.927	0.066
32	0.4	l1misspec	2	0.867	0.0	0.867	0.0	0.867	0.0	0.902	0.0
7	0.01	l1spec	2	0.9	0.047	0.9	0.047	0.9	0.047	0.967	0.009
20	0.1	l1spec	2	0.933	0.0	0.933	0.0	0.933	0.0	0.984	0.003
33	0.4	l1spec	2	0.933	0.0	0.933	0.0	0.933	0.0	0.962	0.022
8	0.01	locohsic	2	0.5	0.047	0.5	0.047	0.5	0.047	0.489	0.05
21	0.1	locohsic	2	0.533	0.0	0.533	0.0	0.533	0.0	0.593	0.053
34	0.4	locohsic	2	0.533	0.094	0.533	0.094	0.533	0.094	0.576	0.016
9	0.01	locommd	2	0.5	0.141	0.5	0.141	0.5	0.141	0.513	0.229
22	0.1	locommd	2	0.433	0.047	0.433	0.047	0.433	0.047	0.367	0.009
35	0.4	locommd	2	0.4	0.0	0.4	0.0	0.4	0.0	0.418	0.101
10	0.01	locorrisk	2	0.7	0.047	0.7	0.047	0.7	0.047	0.802	0.072
23	0.1	locorrisk	2	0.867	0.0	0.867	0.0	0.867	0.0	0.918	0.022
36	0.4	locorrisk	2	0.933	0.0	0.933	0.0	0.933	0.0	0.96	0.031
11	0.01	permucate	2	0.733	0.094	0.733	0.094	0.733	0.094	0.824	0.041
24	0.1	permucate	2	0.767	0.141	0.767	0.141	0.767	0.141	0.824	0.079
37	0.4	permucate	2	0.767	0.047	0.767	0.047	0.767	0.047	0.836	0.038
12	0.01	rfdomain	2	0.533	0.094	0.533	0.094	0.533	0.094	0.482	0.097
25	0.1	rfdomain	2	0.4	0.0	0.4	0.0	0.4	0.0	0.393	0.028
38	0.4	rfdomain	2	0.4	0.283	0.4	0.283	0.4	0.283	0.349	0.211

Table 31: Sensitivity Analysis of the choice of the propensity score model for the Meta-Learner Based methods in terms of the feature selection efficiency - Synthetic Datasets

	shift	method	model	avgAUC	stdAUC	avgF1	stdF1
0	linearconceptdrift	drperm	logistic	1.0	0.0	1.0	0.0
1	linearconceptdrift	drperm	rf	1.0	0.0	1.0	0.0
2	linearconceptdrift	drperm	xgb	1.0	0.0	1.0	0.0
3	linearconceptdrift	permucate	logistic	0.9969	0.007	0.95	0.1118
4	linearconceptdrift	permucate	rf	0.95	0.1118	0.95	0.1118
5	linearconceptdrift	permucate	xgb	0.9	0.1369	0.9	0.1369
6	linearcovariateshift	drperm	logistic	1.0	0.0	1.0	0.0
7	linearcovariateshift	drperm	rf	1.0	0.0	1.0	0.0
8	linearcovariateshift	drperm	xgb	1.0	0.0	1.0	0.0
9	linearcovariateshift	permucate	logistic	0.4156	0.189	0.35	0.2236
10	linearcovariateshift	permucate	rf	1.0	0.0	1.0	0.0
11	linearcovariateshift	permucate	xgb	0.1437	0.2029	0.1	0.1369
12	nonlinearconceptdrift	drperm	logistic	1.0	0.0	1.0	0.0
13	nonlinearconceptdrift	drperm	rf	1.0	0.0	1.0	0.0
14	nonlinearconceptdrift	drperm	xgb	1.0	0.0	1.0	0.0
15	nonlinearconceptdrift	permucate	logistic	0.9875	0.028	0.95	0.1118
16	nonlinearconceptdrift	permucate	rf	1.0	0.0	1.0	0.0
17	nonlinearconceptdrift	permucate	xgb	0.9187	0.1118	0.9	0.1369
18	nonlinearcovariateshift	drperm	logistic	1.0	0.0	1.0	0.0
19	nonlinearcovariateshift	drperm	rf	1.0	0.0	1.0	0.0
20	nonlinearcovariateshift	drperm	xgb	1.0	0.0	1.0	0.0
21	nonlinearcovariateshift	permucate	logistic	1.0	0.0	1.0	0.0
22	nonlinearcovariateshift	permucate	rf	1.0	0.0	1.0	0.0
23	nonlinearcovariateshift	permucate	xgb	0.6219	0.1823	0.6	0.2236

Table 32: Sensitivity Analysis of the choice of the propensity score model for the Meta-Learner Based methods in terms of the feature selection efficiency - Real-Datasets

	scenario	method	estimator	avgAUC	stdAUC	avgF1	stdF1
0	chronoberg	drperm	logistic	0.4633	0.0802	0.3022	0.099
1	chronoberg	drperm	rf	0.8027	0.0988	0.6222	0.1283
2	chronoberg	drperm	xgb	0.6635	0.0644	0.5067	0.0854
3	chronoberg	permucate	logistic	0.5173	0.1281	0.3422	0.1395
4	chronoberg	permucate	rf	0.4686	0.1103	0.2666	0.0962
5	chronoberg	permucate	xgb	0.495	0.0807	0.2755	0.0793
6	tabular	drperm	logistic	0.5894	0.2998	0.5545	0.2864
7	tabular	drperm	rf	0.6877	0.2853	0.6536	0.2754
8	tabular	drperm	xgb	0.6824	0.2684	0.6458	0.2653
9	tabular	permucate	logistic	0.7397	0.276	0.7075	0.2622
10	tabular	permucate	rf	0.7947	0.2262	0.7464	0.236
11	tabular	permucate	xgb	0.7555	0.2654	0.7074	0.2621

Table 33: Sensitivity Analysis of the choice of the outcome model the Meta-Learner Based methods in terms of the feature selection efficiency: Synthetic Data

	$shift_type$	method	estimator	avgAUC	stdAUC	avgF1	stdF1
0	jitter	drperm	rf	0.3756	0.1497	0.1667	0.0786
1	jitter	permucate	rf	0.3915	0.1795	0.1667	0.0786
2	linear	drperm	mlp	0.6001	0.2853	0.5723	0.2678
3	linear	drperm	rf	0.6583	0.2877	0.6233	0.267
4	linear	drperm	ridge	0.6568	0.2914	0.6181	0.2748
5	linear	drperm	xgb	0.6744	0.3152	0.621	0.2959
6	linear	permucate	mlp	0.6818	0.3089	0.6317	0.2829
7	linear	permucate	rf	0.7733	0.2397	0.7198	0.2365
8	linear	permucate	ridge	0.7742	0.2634	0.7119	0.2597
9	linear	permucate	xgb	0.7238	0.3028	0.6766	0.2816
10	linearconcept	drperm	mlp	0.9469	0.0506	0.775	0.2189
11	linearconcept	drperm	rf	1.0	0.0	1.0	0.0
12	linearconcept	drperm	ridge	1.0	0.0	1.0	0.0
13	linearconcept	drperm	xgb	1.0	0.0	1.0	0.0
14	linearconcept	permucate	mlp	1.0	0.0	1.0	0.0
15	linearconcept	permucate	rf	0.9953	0.0148	0.975	0.0791
16	linearconcept	permucate	ridge	1.0	0.0	1.0	0.0
17	linearconcept	permucate	xgb	1.0	0.0	1.0	0.0
18	linearcovariate	drperm	mlp	0.9578	0.0683	0.85	0.2415
19	linearcovariate	drperm	rf	1.0	0.0	1.0	0.0
20	linearcovariate	drperm	ridge	1.0	0.0	1.0	0.0
21	linearcovariate	drperm	xgb	1.0	0.0	1.0	0.0
22	linearcovariate	permucate	mlp	0.5	0.1622	0.275	0.1419
23	linearcovariate	permucate	rf	0.4953	0.1828	0.4	0.2415
24	linearcovariate	permucate	ridge	0.5094	0.1485	0.2	0.1581
25	linearcovariate	permucate	xgb	0.586	0.2585	0.475	0.2486
26	miFGSM	drperm	rf	0.4815	0.1715	0.2592	0.1283
27	miFGSM	permucate	rf	0.4656	0.2378	0.2592	0.1697
28	nonlinearconcept	drperm	mlp	1.0	0.0	1.0	0.0
29	nonlinearconcept	drperm	rf	1.0	0.0	1.0	0.0
30	nonlinearconcept	drperm	ridge	1.0	0.0	1.0	0.0
31	nonlinearconcept	drperm	xgb	1.0	0.0	1.0	0.0
32	nonlinearconcept	permucate	mlp	0.8641	0.1527	0.775	0.1845
33	nonlinearconcept	permucate	rf	0.9812	0.0593	0.975	0.0791
34	nonlinearconcept	permucate	ridge	0.8016	0.2105	0.675	0.2372
35	nonlinearconcept	permucate	xgb	1.0	0.0	1.0	0.0
36	nonlinearcovariate	drperm	mlp	0.5798	0.3141	0.5358	0.2882
37	nonlinearcovariate	drperm	rf	0.6081	0.3087	0.5843	0.3036
38	nonlinearcovariate	drperm	ridge	0.6253	0.3132	0.6053	0.3014
39	nonlinearcovariate	drperm	xgb	0.569	0.3304	0.5677	0.3126
40	nonlinearcovariate	permucate	mlp	0.7091	0.255	0.6501	0.272
41	nonlinearcovariate	permucate	rf	0.8234	0.2271	0.7687	0.2376
42	nonlinearcovariate	permucate	ridge	0.6887	0.2594	0.6132	0.2801
43	nonlinearcovariate	permucate	xgb	0.7112	0.2911	0.6765	0.2894
44	nonlinearinteraction	drperm	mlp	0.6462	0.2747	0.6145	0.2623
45	nonlinearinteraction	drperm	rf	0.6653	0.3093	0.6387	0.2927
46	nonlinearinteraction	drperm	ridge	0.6515	0.2895	0.6274	0.2733
47	nonlinearinteraction	drperm	xgb	0.7452	0.3464	0.7063	0.3347
48	nonlinearinteraction	permucate	mlp	0.8017	0.2557	0.7685	0.2425
49	nonlinearinteraction	permucate	rf	0.8754	0.1602	0.8379	0.1858
50	nonlinearinteraction	permucate	ridge	0.8472	0.1648	0.8121	0.1929
51	nonlinearinteraction	permucate	xgb	0.8463	0.1927	0.81	0.2019
52	rFGSM	drperm	rf	0.4074	0.1123	0.2222	0.1571
53	rFGSM	permucate	rf	0.3783	0.0486	0.1111	0.0

Table 34: Random Forest Binary Classifier in comparison with other procedures - Boruta, ShadowVIMP and VITA-PIMP for synthetic data and tabular data

	$shift_type$	method	scenario	avgAUC	stdAUC	avgF1	stdF1
14	beta=0.2	boruta	synthetic	0.4305	0.137	0.175	0.1687
15	beta=0.2	dccondperm	synthetic	0.4922	0.1494	0.175	0.1208
16	beta=0.2	dcloco	synthetic	0.5734	0.2035	0.325	0.2372
17	beta=0.2	dcoob	synthetic	0.4625	0.172	0.2	0.1972
18	beta=0.2	rfvanilla	synthetic	0.4531	0.2233	0.175	0.2648
19	beta=0.2	shadowVIMP	synthetic	0.4984	0.157	0.175	0.1687
20	beta=0.2	vitaPIMP	synthetic	0.4625	0.217	0.175	0.2058
21	gamma=0.2	boruta	synthetic	0.9828	0.029	0.9	0.1291
22	gamma=0.2	dccondperm	synthetic	0.8953	0.1192	0.775	0.2486
23	gamma=0.2	dcloco	synthetic	0.8	0.1441	0.6	0.1748
24	gamma=0.2	dcoob	synthetic	0.9359	0.0753	0.75	0.1179
25	gamma=0.2	rfvanilla	synthetic	0.9422	0.074	0.8	0.1581
26	gamma=0.2	shadowVIMP	synthetic	0.9172	0.0807	0.775	0.1845
27	gamma=0.2	vitaPIMP	synthetic	0.8547	0.1162	0.675	0.2648
28	interaction	boruta	tabular	0.7233	0.2149	0.6953	0.2211
29	interaction	dccondperm	tabular	0.9966	0.0107	0.9846	0.0436
30	interaction	dcloco	tabular	0.8421	0.1649	0.7961	0.1891
31	interaction	dcoob	tabular	0.9971	0.01	0.9864	0.0377
32	interaction	rfvanilla	tabular	0.9993	0.0032	0.9948	0.0255
33	interaction	shadowVIMP	tabular	1.0	0.0	1.0	0.0
34	interaction	vitaPIMP	tabular	0.9948	0.0134	0.9763	0.0493
49	linear	boruta	tabular	0.4721	0.2216	0.4519	0.2255
50	linear	dccondperm	tabular	0.5009	0.241	0.4801	0.2258
51	linear	dcloco	tabular	0.5246	0.2425	0.4998	0.2358
52	linear	dcoob	tabular	0.5682	0.2336	0.5784	0.2207
53	linear	rfvanilla	tabular	0.4623	0.2251	0.468	0.2379
54	linear	shadowVIMP	tabular	0.5101	0.2138	0.5011	0.2059
55	linear	vitaPIMP	tabular	0.4879	0.2564	0.5024	0.2394
84	shift	boruta	tabular	0.7416	0.2254	0.7153	0.2691
85	shift	dccondperm	tabular	0.989	0.0237	0.9733	0.0542
86	shift	dcloco	tabular	0.8475	0.1562	0.8088	0.1754
87	shift	dcoob	tabular	0.9887	0.0236	0.9691	0.0559
88	shift	rfvanilla	tabular	0.9964	0.0103	0.9802	0.0454
89	shift	shadowVIMP	tabular	0.9957	0.0132	0.9802	0.0454
90	shift	vitaPIMP	tabular	0.9848	0.0325	0.9565	0.0698

Table 35: Random Forest Binary Classifier in comparison with other procedures - Boruta, ShadowVIMP and VITA-PIMP for Chronoberg and FMoW data

	shift$_type$	method	scenario	avgAUC	stdAUC	avgF1	stdF1
0	FGSM	boruta	chronoberg	0.5545	0.0418	0.0667	0.0574
1	FGSM	boruta	fmow	0.569	0.0769	0.1495	0.1125
2	FGSM	dccondperm	chronoberg	0.664	0.0858	0.4222	0.1148
3	FGSM	dccondperm	fmow	0.6642	0.049	0.391	0.0753
4	FGSM	dcloco	chronoberg	0.554	0.1315	0.3333	0.1283
5	FGSM	dcloco	fmow	0.5538	0.0667	0.3468	0.1146
6	FGSM	dcoob	chronoberg	0.6667	0.1044	0.4333	0.1223
7	FGSM	dcoob	fmow	0.6231	0.0416	0.3744	0.0916
8	FGSM	rfvanilla	chronoberg	0.6577	0.1184	0.4222	0.1463
9	FGSM	rfvanilla	fmow	0.6353	0.0596	0.366	0.0742
10	FGSM	shadowVIMP	chronoberg	0.6704	0.0841	0.4667	0.1463
11	FGSM	shadowVIMP	fmow	0.6483	0.0837	0.3458	0.099
12	FGSM	vitaPIMP	chronoberg	0.6635	0.1192	0.4667	0.1262
13	FGSM	vitaPIMP	fmow	0.645	0.0936	0.3897	0.0749
35	jitter	boruta	chronoberg	0.5952	0.0445	0.1111	0.0
36	jitter	boruta	fmow	0.7037	0.1642	0.0991	0.0476
37	jitter	dccondperm	chronoberg	0.982	0.0346	0.9333	0.0994
38	jitter	dccondperm	fmow	0.8768	0.1924	0.7179	0.3109
39	jitter	dcloco	chronoberg	0.7619	0.0571	0.6	0.0609
40	jitter	dcloco	fmow	0.7276	0.1642	0.5778	0.1401
41	jitter	dcoob	chronoberg	0.9767	0.039	0.9111	0.1449
42	jitter	dcoob	fmow	0.8462	0.2189	0.6923	0.3525
43	jitter	rfvanilla	chronoberg	0.9714	0.0481	0.9111	0.1449
44	jitter	rfvanilla	fmow	0.8412	0.2415	0.718	0.3635
45	jitter	shadowVIMP	chronoberg	0.9841	0.0254	0.9111	0.1449
46	jitter	shadowVIMP	fmow	0.9098	0.1519	0.7692	0.3353
47	jitter	vitaPIMP	chronoberg	0.9852	0.0252	0.9333	0.0994
48	jitter	vitaPIMP	fmow	0.9041	0.1556	0.7692	0.3353
56	miFGSM	boruta	chronoberg	0.6	0.0516	0.1111	0.0
57	miFGSM	boruta	fmow	0.6858	0.1335	0.1169	0.074
58	miFGSM	dccondperm	chronoberg	0.9778	0.0468	0.9333	0.0994
59	miFGSM	dccondperm	fmow	0.8948	0.1641	0.7928	0.2544
60	miFGSM	dcloco	chronoberg	0.7619	0.0958	0.5556	0.1361
61	miFGSM	dcloco	fmow	0.7575	0.2067	0.6072	0.2321
62	miFGSM	dcoob	chronoberg	0.9873	0.0229	0.9333	0.0994
63	miFGSM	dcoob	fmow	0.9073	0.1496	0.7795	0.2423
64	miFGSM	rfvanilla	chronoberg	0.9767	0.037	0.9111	0.1449
65	miFGSM	rfvanilla	fmow	0.8939	0.1794	0.7795	0.2423
66	miFGSM	shadowVIMP	chronoberg	0.9884	0.0206	0.9333	0.0994
67	miFGSM	shadowVIMP	fmow	0.9408	0.102	0.8195	0.2536
68	miFGSM	vitaPIMP	chronoberg	0.9862	0.021	0.9556	0.0609
69	miFGSM	vitaPIMP	fmow	0.9127	0.1411	0.8062	0.2522
70	rFGSM	boruta	chronoberg	0.6048	0.0361	0.1111	0.0
71	rFGSM	boruta	fmow	0.7091	0.1687	0.0991	0.0476
72	rFGSM	dccondperm	chronoberg	0.9831	0.0268	0.8889	0.1361
73	rFGSM	dccondperm	fmow	0.871	0.2086	0.7436	0.3795
74	rFGSM	dcloco	chronoberg	0.7566	0.0639	0.5556	0.0786
75	rFGSM	dcloco	fmow	0.7468	0.1537	0.6513	0.1344
76	rFGSM	dcoob	chronoberg	0.9746	0.0377	0.8889	0.1361
77	rFGSM	dcoob	fmow	0.8594	0.235	0.718	0.3635
78	rFGSM	rfvanilla	chronoberg	0.9714	0.0442	0.9111	0.1449
79	rFGSM	rfvanilla	fmow	0.8495	0.2437	0.718	0.3635
80	rFGSM	shadowVIMP	chronoberg	0.9852	0.0232	0.9333	0.0994
81	rFGSM	shadowVIMP	fmow	0.9082	0.159	0.7949	0.3553
82	rFGSM	vitaPIMP	chronoberg	0.9778	0.036	0.8889	0.1361
83	rFGSM	vitaPIMP	fmow	0.8966	0.1769	0.7436	0.3795

Table 36: Mixed blocks: two covariate-shift clients, two concept-drift clients, one mixed client, and one clean client. No attack. Spearman and Pearson correlation with the centralized importance.

Aggregator	Spearman	Pearson
Uniform	0.838	0.920
n-weighted	0.867	0.952
Median	0.842	0.921
Shapley (client agreement)	0.879	0.922
Top-3, then n-weighted	0.925	0.957
Top-5, then n-weighted	0.929	0.951

Table 37: Mixed synthetic blocks with corrupted client vectors. Spearman correlation with the centralized importance.

Attack	# bad	Uniform	n-weighted	Median	Shapley
Random	1	0.208	0.539	0.798	0.802
Random	2	0.055	0.156	0.741	0.737
Extreme	1	0.503	0.576	0.830	0.814
Extreme	2	0.624	0.669	0.794	0.762
Collude	2	0.572	0.612	0.899	0.842

Table 38: FSDS `whyshift dataset`, ten folds. Spearman correlation with the pooled OOD-wrapper vector.

Wrapper output	Single fold	n-weighted	IVW
RF domain classifier	0.848	0.887	0.918
MMD-LOCO	0.891	0.985	—
PO-risk	0.410	0.185	0.431

ct $n_{\text{bad}} \times \text{SNR}_{\text{low}} \times \alpha \times$ aggregator on the high-dimensional stress test. Entries are means over five replicates.

n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
0	1	0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.5	0.959	0.977	0.962	0.844	1.000	1.000	1.000	1.00	1.00	1.00	0.17	0.44
2	1	0.5	0.925	0.955	0.928	0.772	1.000	1.000	1.000	1.00	1.00	1.00	0.31	0.43
3	1	0.5	0.894	0.935	0.891	0.730	1.000	1.000	1.000	1.00	1.00	1.00	0.43	0.43
4	1	0.5	0.864	0.912	0.857	0.709	1.000	1.000	1.000	1.00	1.00	1.00	0.53	0.44
5	1	0.5	0.836	0.889	0.823	0.692	1.000	1.000	1.000	1.00	1.00	1.00	0.61	0.45
6	1	0.5	0.812	0.866	0.795	0.689	1.000	1.000	1.000	1.00	1.00	1.00	0.69	0.45
7	1	0.5	0.792	0.842	0.759	0.689	1.000	1.000	1.000	1.00	1.00	1.00	0.76	0.44
8	1	0.5	0.776	0.822	0.732	0.691	1.000	1.000	1.000	1.00	1.00	1.00	0.82	0.44
9	1	0.5	0.761	0.800	0.702	0.699	1.000	1.000	1.000	1.00	1.00	1.00	0.87	0.43
10	1	0.5	0.745	0.779	0.674	0.698	1.000	1.000	1.000	1.00	1.00	1.00	0.92	0.42
11	1	0.5	0.734	0.761	0.655	0.709	1.000	1.000	1.000	1.00	1.00	1.00	0.97	0.38
12	1	0.5	0.719	0.737	0.631	0.710	1.000	1.000	1.000	1.00	1.00	1.00	1.00	—
0	1	0.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.15	0.938	0.973	0.957	0.768	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0.15	0.890	0.946	0.911	0.691	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0.15	0.852	0.920	0.867	0.657	1.000	1.000	1.000	1.00	1.00	1.00	0.45	0.39
4	1	0.15	0.818	0.894	0.828	0.636	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.40
5	1	0.15	0.791	0.870	0.788	0.630	1.000	1.000	1.000	1.00	1.00	1.00	0.63	0.40
6	1	0.15	0.767	0.845	0.751	0.635	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0.15	0.746	0.818	0.708	0.637	1.000	1.000	1.000	1.00	1.00	1.00	0.77	0.40
8	1	0.15	0.730	0.794	0.672	0.646	1.000	1.000	1.000	1.00	0.97	1.00	0.83	0.40
9	1	0.15	0.713	0.767	0.641	0.652	1.000	0.999	1.000	1.00	0.91	1.00	0.88	0.39
10	1	0.15	0.699	0.742	0.609	0.654	1.000	0.995	1.000	1.00	0.86	0.99	0.93	0.39
11	1	0.15	0.686	0.716	0.582	0.664	1.000	0.987	1.000	1.00	0.77	0.92	0.97	0.35
12	1	0.15	0.675	0.692	0.562	0.668	0.988	0.971	0.988	0.75	0.64	0.77	1.00	—
0	1	0.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.05	0.936	0.973	0.955	0.759	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0.05	0.887	0.947	0.911	0.686	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0.05	0.844	0.919	0.862	0.645	1.000	1.000	1.000	1.00	1.00	1.00	0.46	0.39
4	1	0.05	0.808	0.891	0.818	0.623	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.39
5	1	0.05	0.780	0.866	0.775	0.614	1.000	1.000	1.000	1.00	1.00	1.00	0.63	0.40
6	1	0.05	0.755	0.838	0.730	0.620	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0.05	0.732	0.809	0.691	0.620	1.000	0.989	1.000	1.00	0.87	1.00	0.77	0.40
8	1	0.05	0.715	0.782	0.649	0.627	1.000	0.968	1.000	1.00	0.68	1.00	0.83	0.40
9	1	0.05	0.701	0.754	0.615	0.638	1.000	0.920	1.000	1.00	0.45	1.00	0.88	0.39
10	1	0.05	0.689	0.722	0.577	0.643	1.000	0.828	0.999	1.00	0.25	0.92	0.93	0.39
11	1	0.05	0.677	0.685	0.538	0.651	0.998	0.721	0.953	0.92	0.14	0.53	0.97	0.35
12	1	0.05	0.622	0.642	0.506	0.615	0.602	0.593	0.603	0.04	0.03	0.04	1.00	—
0	1	0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0	0.936	0.973	0.955	0.761	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0	0.886	0.946	0.910	0.681	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0	0.842	0.918	0.860	0.640	1.000	1.000	1.000	1.00	1.00	1.00	0.46	0.38
4	1	0	0.806	0.890	0.819	0.619	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.39
5	1	0	0.779	0.865	0.773	0.613	1.000	1.000	1.000	1.00	1.00	1.00	0.64	0.40
6	1	0	0.754	0.837	0.730	0.619	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0	0.731	0.808	0.691	0.620	1.000	0.990	1.000	1.00	0.80	1.00	0.77	0.40
8	1	0	0.713	0.778	0.647	0.627	1.000	0.950	1.000	1.00	0.58	1.00	0.83	0.40

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
9	1	0	0.698	0.747	0.608	0.637	1.000	0.886	1.000	1.00	0.34	0.99	0.88	0.39
10	1	0	0.689	0.711	0.567	0.645	1.000	0.763	0.999	1.00	0.17	0.93	0.93	0.39
11	1	0	0.677	0.668	0.524	0.650	0.996	0.621	0.928	0.89	0.06	0.44	0.97	0.35
12	1	0	0.606	0.624	0.491	0.601	0.479	0.478	0.478	0.00	0.02	0.00	1.00	—
0	0.75	0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.5	0.958	0.976	0.964	0.855	1.000	1.000	1.000	1.00	1.00	1.00	0.16	0.45
2	0.75	0.5	0.926	0.954	0.931	0.785	1.000	1.000	1.000	1.00	1.00	1.00	0.31	0.44
3	0.75	0.5	0.896	0.930	0.896	0.746	1.000	1.000	1.000	1.00	1.00	1.00	0.43	0.44
4	0.75	0.5	0.866	0.907	0.859	0.722	1.000	1.000	1.000	1.00	1.00	1.00	0.52	0.44
5	0.75	0.5	0.840	0.885	0.825	0.705	1.000	1.000	1.000	1.00	1.00	1.00	0.61	0.45
6	0.75	0.5	0.818	0.863	0.794	0.697	1.000	1.000	1.000	1.00	1.00	1.00	0.69	0.45
7	0.75	0.5	0.796	0.842	0.763	0.690	1.000	1.000	1.000	1.00	1.00	1.00	0.76	0.44
8	0.75	0.5	0.775	0.816	0.735	0.688	1.000	1.000	1.000	1.00	1.00	1.00	0.82	0.44
9	0.75	0.5	0.759	0.794	0.702	0.694	1.000	1.000	1.000	1.00	1.00	1.00	0.87	0.43
10	0.75	0.5	0.742	0.771	0.672	0.696	1.000	1.000	1.000	1.00	1.00	1.00	0.92	0.42
11	0.75	0.5	0.723	0.748	0.654	0.696	1.000	1.000	1.000	1.00	1.00	1.00	0.96	0.44
12	0.75	0.5	0.712	0.729	0.633	0.703	1.000	1.000	1.000	1.00	1.00	1.00	1.00	—
0	0.75	0.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.15	0.942	0.972	0.957	0.799	1.000	0.998	1.000	1.00	0.99	1.00	0.17	0.42
2	0.75	0.15	0.898	0.946	0.918	0.718	1.000	0.998	1.000	1.00	0.99	1.00	0.32	0.41
3	0.75	0.15	0.858	0.919	0.874	0.674	1.000	0.997	1.000	1.00	0.98	1.00	0.44	0.41
4	0.75	0.15	0.827	0.892	0.834	0.657	1.000	0.996	1.000	1.00	0.98	1.00	0.54	0.41
5	0.75	0.15	0.795	0.865	0.790	0.641	1.000	0.993	1.000	1.00	0.94	1.00	0.62	0.42
6	0.75	0.15	0.770	0.839	0.756	0.636	1.000	0.986	1.000	1.00	0.89	1.00	0.70	0.42
7	0.75	0.15	0.748	0.815	0.722	0.635	1.000	0.975	1.000	1.00	0.85	1.00	0.77	0.41
8	0.75	0.15	0.729	0.787	0.690	0.639	1.000	0.969	1.000	1.00	0.80	0.99	0.83	0.42
9	0.75	0.15	0.716	0.760	0.652	0.649	0.999	0.952	0.999	0.98	0.72	0.98	0.88	0.41
10	0.75	0.15	0.701	0.737	0.618	0.655	0.997	0.938	0.995	0.94	0.63	0.93	0.93	0.40
11	0.75	0.15	0.688	0.712	0.602	0.659	0.985	0.915	0.984	0.86	0.55	0.80	0.96	0.41
12	0.75	0.15	0.672	0.685	0.571	0.663	0.945	0.893	0.945	0.60	0.44	0.59	1.00	—
0	0.75	0.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.05	0.941	0.972	0.956	0.798	1.000	0.993	1.000	1.00	0.98	1.00	0.17	0.42
2	0.75	0.05	0.895	0.945	0.913	0.712	1.000	0.992	1.000	1.00	0.99	1.00	0.32	0.41
3	0.75	0.05	0.855	0.916	0.869	0.672	1.000	0.989	1.000	1.00	0.94	1.00	0.44	0.41
4	0.75	0.05	0.821	0.890	0.830	0.650	1.000	0.981	1.000	1.00	0.89	1.00	0.54	0.41
5	0.75	0.05	0.789	0.861	0.786	0.631	1.000	0.973	1.000	1.00	0.78	1.00	0.63	0.41
6	0.75	0.05	0.767	0.836	0.749	0.633	1.000	0.943	1.000	1.00	0.72	1.00	0.70	0.41
7	0.75	0.05	0.744	0.807	0.707	0.633	1.000	0.919	1.000	1.00	0.58	0.99	0.77	0.41
8	0.75	0.05	0.723	0.776	0.674	0.634	1.000	0.885	0.999	1.00	0.43	0.96	0.83	0.41
9	0.75	0.05	0.709	0.744	0.631	0.642	0.993	0.802	0.991	0.98	0.28	0.89	0.88	0.41
10	0.75	0.05	0.693	0.713	0.593	0.646	0.968	0.723	0.958	0.92	0.13	0.77	0.93	0.40
11	0.75	0.05	0.674	0.680	0.565	0.644	0.890	0.638	0.877	0.76	0.06	0.61	0.96	0.41
12	0.75	0.05	0.630	0.639	0.524	0.622	0.589	0.527	0.589	0.04	0.01	0.05	1.00	—
0	0.75	0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0	0.941	0.972	0.956	0.796	1.000	0.991	1.000	1.00	0.98	1.00	0.17	0.42
2	0.75	0	0.895	0.946	0.917	0.712	1.000	0.994	1.000	1.00	0.98	1.00	0.32	0.41
3	0.75	0	0.855	0.916	0.871	0.669	1.000	0.990	1.000	1.00	0.95	1.00	0.44	0.41
4	0.75	0	0.822	0.890	0.830	0.646	1.000	0.983	1.000	1.00	0.86	1.00	0.54	0.41
5	0.75	0	0.790	0.861	0.788	0.629	1.000	0.966	1.000	1.00	0.79	1.00	0.63	0.41

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
6	0.75	0	0.767	0.835	0.748	0.632	1.000	0.933	1.000	1.00	0.67	1.00	0.70	0.41
7	0.75	0	0.745	0.805	0.704	0.631	1.000	0.890	1.000	1.00	0.58	0.99	0.77	0.41
8	0.75	0	0.726	0.774	0.672	0.635	1.000	0.855	0.998	0.99	0.35	0.96	0.83	0.41
9	0.75	0	0.709	0.738	0.629	0.642	0.990	0.775	0.987	0.98	0.22	0.90	0.88	0.41
10	0.75	0	0.693	0.705	0.586	0.646	0.962	0.673	0.948	0.91	0.10	0.75	0.93	0.40
11	0.75	0	0.671	0.669	0.557	0.641	0.866	0.569	0.850	0.76	0.02	0.52	0.96	0.41
12	0.75	0	0.616	0.626	0.515	0.608	0.480	0.457	0.480	0.01	0.00	0.01	1.00	—
0	0.5	0.5	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.5	0.963	0.975	0.961	0.894	1.000	0.972	1.000	1.00	0.80	1.00	0.15	0.51
2	0.5	0.5	0.930	0.951	0.924	0.826	1.000	0.972	1.000	1.00	0.80	1.00	0.28	0.50
3	0.5	0.5	0.901	0.928	0.890	0.785	1.000	0.968	1.000	1.00	0.81	1.00	0.39	0.51
4	0.5	0.5	0.876	0.906	0.853	0.758	1.000	0.972	1.000	1.00	0.81	1.00	0.49	0.51
5	0.5	0.5	0.855	0.883	0.823	0.748	1.000	0.968	1.000	1.00	0.81	1.00	0.58	0.51
6	0.5	0.5	0.833	0.861	0.788	0.735	1.000	0.967	1.000	1.00	0.81	1.00	0.66	0.50
7	0.5	0.5	0.816	0.842	0.761	0.735	1.000	0.969	1.000	1.00	0.82	1.00	0.72	0.52
8	0.5	0.5	0.796	0.821	0.733	0.727	1.000	0.971	1.000	1.00	0.81	1.00	0.78	0.53
9	0.5	0.5	0.781	0.802	0.711	0.728	1.000	0.974	1.000	1.00	0.82	1.00	0.85	0.51
10	0.5	0.5	0.763	0.777	0.678	0.725	1.000	0.975	1.000	1.00	0.82	1.00	0.90	0.51
11	0.5	0.5	0.746	0.756	0.652	0.722	1.000	0.973	1.000	1.00	0.83	1.00	0.96	0.48
12	0.5	0.5	0.732	0.734	0.630	0.723	1.000	0.973	1.000	1.00	0.81	1.00	1.00	—
0	0.5	0.15	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.15	0.954	0.974	0.958	0.862	1.000	0.977	1.000	1.00	0.78	1.00	0.15	0.48
2	0.5	0.15	0.915	0.947	0.917	0.787	1.000	0.972	1.000	1.00	0.78	1.00	0.29	0.48
3	0.5	0.15	0.885	0.925	0.877	0.747	1.000	0.964	1.000	1.00	0.74	1.00	0.40	0.49
4	0.5	0.15	0.857	0.900	0.837	0.720	1.000	0.959	1.000	1.00	0.66	1.00	0.50	0.49
5	0.5	0.15	0.833	0.874	0.796	0.708	1.000	0.944	1.000	0.99	0.62	0.99	0.59	0.48
6	0.5	0.15	0.811	0.850	0.758	0.699	1.000	0.934	1.000	0.99	0.58	0.98	0.67	0.48
7	0.5	0.15	0.795	0.828	0.728	0.701	0.999	0.917	0.999	0.97	0.54	0.96	0.73	0.49
8	0.5	0.15	0.777	0.804	0.690	0.697	0.985	0.902	0.985	0.94	0.49	0.94	0.79	0.51
9	0.5	0.15	0.760	0.781	0.662	0.698	0.980	0.893	0.980	0.90	0.42	0.89	0.86	0.48
10	0.5	0.15	0.741	0.756	0.635	0.695	0.962	0.868	0.963	0.80	0.36	0.80	0.91	0.49
11	0.5	0.15	0.724	0.730	0.608	0.694	0.936	0.850	0.937	0.64	0.31	0.66	0.96	0.46
12	0.5	0.15	0.706	0.708	0.581	0.693	0.890	0.817	0.890	0.37	0.27	0.37	1.00	—
0	0.5	0.05	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.05	0.954	0.973	0.956	0.862	1.000	0.970	1.000	1.00	0.76	1.00	0.15	0.48
2	0.5	0.05	0.916	0.946	0.913	0.788	1.000	0.951	1.000	1.00	0.70	1.00	0.29	0.48
3	0.5	0.05	0.885	0.923	0.872	0.749	1.000	0.928	1.000	1.00	0.62	0.99	0.40	0.48
4	0.5	0.05	0.857	0.896	0.829	0.722	1.000	0.911	1.000	1.00	0.55	0.99	0.50	0.48
5	0.5	0.05	0.833	0.869	0.788	0.710	0.997	0.885	0.997	0.99	0.48	0.98	0.59	0.48
6	0.5	0.05	0.808	0.841	0.750	0.698	0.997	0.856	0.996	0.98	0.38	0.96	0.67	0.48
7	0.5	0.05	0.791	0.818	0.720	0.697	0.987	0.824	0.986	0.96	0.31	0.94	0.73	0.49
8	0.5	0.05	0.772	0.790	0.678	0.691	0.966	0.785	0.965	0.94	0.25	0.89	0.79	0.51
9	0.5	0.05	0.754	0.764	0.644	0.691	0.946	0.730	0.944	0.89	0.18	0.86	0.86	0.48
10	0.5	0.05	0.731	0.732	0.610	0.684	0.892	0.670	0.891	0.77	0.10	0.74	0.91	0.49
11	0.5	0.05	0.703	0.698	0.577	0.672	0.787	0.621	0.787	0.53	0.06	0.50	0.96	0.46
12	0.5	0.05	0.664	0.668	0.547	0.650	0.561	0.568	0.561	0.02	0.02	0.01	1.00	—
0	0.5	0	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0	0.954	0.973	0.956	0.860	1.000	0.970	1.000	1.00	0.78	1.00	0.16	0.48
2	0.5	0	0.916	0.946	0.913	0.787	1.000	0.948	1.000	1.00	0.68	1.00	0.29	0.47

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
3	0.5	0	0.884	0.921	0.870	0.747	1.000	0.908	1.000	1.00	0.64	1.00	0.40	0.48
4	0.5	0	0.855	0.894	0.826	0.717	1.000	0.885	1.000	1.00	0.57	1.00	0.50	0.48
5	0.5	0	0.831	0.866	0.785	0.706	0.996	0.853	0.996	0.99	0.46	0.99	0.59	0.48
6	0.5	0	0.806	0.838	0.745	0.694	0.996	0.809	0.994	0.98	0.37	0.97	0.67	0.48
7	0.5	0	0.789	0.815	0.715	0.694	0.985	0.773	0.983	0.96	0.30	0.94	0.73	0.49
8	0.5	0	0.771	0.785	0.673	0.689	0.966	0.737	0.964	0.94	0.22	0.90	0.79	0.51
9	0.5	0	0.753	0.757	0.639	0.689	0.946	0.670	0.943	0.89	0.12	0.84	0.86	0.48
10	0.5	0	0.729	0.723	0.604	0.683	0.890	0.611	0.888	0.77	0.08	0.70	0.91	0.49
11	0.5	0	0.699	0.688	0.569	0.667	0.760	0.537	0.757	0.52	0.04	0.47	0.96	0.46
12	0.5	0	0.653	0.655	0.536	0.639	0.488	0.483	0.488	0.02	0.03	0.02	1.00	—
0	0.25	0.5	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.5	0.978	0.981	0.967	0.949	1.000	0.769	1.000	0.98	0.30	0.99	0.12	0.62
2	0.25	0.5	0.956	0.962	0.938	0.910	1.000	0.764	1.000	0.98	0.29	0.99	0.24	0.61
3	0.25	0.5	0.939	0.946	0.907	0.882	1.000	0.765	1.000	0.98	0.30	0.98	0.34	0.62
4	0.25	0.5	0.922	0.928	0.870	0.864	1.000	0.761	1.000	0.97	0.32	0.97	0.44	0.61
5	0.25	0.5	0.905	0.910	0.844	0.846	1.000	0.762	1.000	0.97	0.32	0.97	0.53	0.60
6	0.25	0.5	0.889	0.892	0.818	0.834	1.000	0.772	1.000	0.96	0.32	0.96	0.61	0.61
7	0.25	0.5	0.876	0.877	0.792	0.825	0.998	0.780	0.998	0.94	0.30	0.95	0.69	0.60
8	0.25	0.5	0.861	0.859	0.766	0.817	0.998	0.779	0.998	0.93	0.29	0.93	0.76	0.62
9	0.25	0.5	0.842	0.837	0.740	0.806	0.998	0.771	0.998	0.92	0.28	0.92	0.82	0.62
10	0.25	0.5	0.827	0.819	0.715	0.799	0.998	0.772	0.999	0.92	0.29	0.92	0.89	0.60
11	0.25	0.5	0.815	0.802	0.691	0.800	0.998	0.772	0.998	0.91	0.30	0.91	0.94	0.67
12	0.25	0.5	0.802	0.783	0.671	0.797	0.998	0.769	0.998	0.91	0.30	0.90	1.00	—
0	0.25	0.15	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.15	0.973	0.979	0.964	0.938	0.991	0.765	0.991	0.98	0.29	0.98	0.13	0.61
2	0.25	0.15	0.949	0.960	0.933	0.893	0.990	0.758	0.989	0.98	0.23	0.98	0.25	0.60
3	0.25	0.15	0.929	0.942	0.900	0.862	0.984	0.745	0.983	0.97	0.21	0.97	0.35	0.61
4	0.25	0.15	0.909	0.922	0.864	0.844	0.979	0.741	0.979	0.95	0.21	0.95	0.45	0.60
5	0.25	0.15	0.888	0.902	0.835	0.823	0.968	0.734	0.969	0.92	0.19	0.92	0.54	0.60
6	0.25	0.15	0.874	0.883	0.804	0.812	0.953	0.738	0.953	0.86	0.18	0.86	0.61	0.61
7	0.25	0.15	0.856	0.866	0.779	0.801	0.930	0.733	0.929	0.79	0.13	0.79	0.69	0.60
8	0.25	0.15	0.837	0.847	0.750	0.789	0.905	0.710	0.907	0.72	0.14	0.72	0.76	0.61
9	0.25	0.15	0.819	0.824	0.717	0.780	0.880	0.697	0.882	0.65	0.10	0.67	0.83	0.61
10	0.25	0.15	0.801	0.802	0.690	0.770	0.857	0.674	0.859	0.52	0.10	0.54	0.89	0.59
11	0.25	0.15	0.785	0.784	0.662	0.767	0.799	0.657	0.800	0.37	0.08	0.37	0.94	0.66
12	0.25	0.15	0.768	0.765	0.639	0.760	0.744	0.646	0.744	0.19	0.05	0.18	1.00	—
0	0.25	0.05	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.05	0.973	0.978	0.963	0.936	0.988	0.761	0.986	0.98	0.26	0.98	0.13	0.61
2	0.25	0.05	0.949	0.958	0.929	0.891	0.987	0.727	0.984	0.98	0.17	0.98	0.25	0.59
3	0.25	0.05	0.928	0.941	0.898	0.859	0.980	0.710	0.978	0.97	0.15	0.97	0.35	0.60
4	0.25	0.05	0.908	0.920	0.861	0.840	0.974	0.690	0.972	0.95	0.12	0.95	0.45	0.60
5	0.25	0.05	0.886	0.897	0.830	0.819	0.956	0.679	0.953	0.92	0.12	0.92	0.54	0.59
6	0.25	0.05	0.869	0.876	0.798	0.806	0.927	0.660	0.923	0.84	0.09	0.84	0.62	0.60
7	0.25	0.05	0.852	0.859	0.771	0.797	0.895	0.639	0.893	0.76	0.07	0.77	0.69	0.60
8	0.25	0.05	0.832	0.838	0.740	0.783	0.858	0.610	0.856	0.67	0.06	0.68	0.76	0.61
9	0.25	0.05	0.812	0.813	0.705	0.773	0.811	0.575	0.812	0.58	0.04	0.58	0.83	0.61
10	0.25	0.05	0.791	0.791	0.676	0.760	0.761	0.534	0.763	0.47	0.02	0.47	0.89	0.59
11	0.25	0.05	0.766	0.767	0.645	0.748	0.637	0.512	0.639	0.27	0.02	0.27	0.94	0.66
12	0.25	0.05	0.741	0.744	0.620	0.734	0.519	0.487	0.522	0.01	0.02	0.01	1.00	—

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
0	0.25	0	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0	0.973	0.978	0.964	0.935	0.987	0.757	0.985	0.98	0.27	0.98	0.13	0.61
2	0.25	0	0.948	0.958	0.929	0.890	0.986	0.726	0.983	0.98	0.18	0.98	0.25	0.59
3	0.25	0	0.928	0.940	0.897	0.858	0.979	0.697	0.977	0.97	0.13	0.97	0.35	0.60
4	0.25	0	0.907	0.918	0.857	0.839	0.973	0.670	0.970	0.95	0.11	0.95	0.45	0.59
5	0.25	0	0.886	0.897	0.828	0.819	0.954	0.648	0.950	0.92	0.08	0.92	0.54	0.59
6	0.25	0	0.869	0.874	0.795	0.807	0.922	0.627	0.918	0.83	0.07	0.83	0.62	0.60
7	0.25	0	0.852	0.858	0.770	0.797	0.888	0.602	0.885	0.76	0.06	0.76	0.69	0.60
8	0.25	0	0.830	0.835	0.738	0.781	0.844	0.564	0.843	0.67	0.02	0.67	0.76	0.61
9	0.25	0	0.810	0.810	0.701	0.770	0.791	0.527	0.792	0.58	0.03	0.58	0.83	0.61
10	0.25	0	0.789	0.788	0.673	0.758	0.741	0.503	0.744	0.46	0.02	0.45	0.89	0.59
11	0.25	0	0.763	0.764	0.642	0.744	0.610	0.476	0.611	0.25	0.02	0.25	0.94	0.66
12	0.25	0	0.736	0.741	0.617	0.728	0.480	0.455	0.482	0.00	0.00	0.00	1.00	—

Table 40: Simulation grid (245 cells; size $\in \{0.1, \ldots, 0.9\}$, corr $\in \{0, 0.15, \ldots, 0.6\}$, hetero $\in \{0, \ldots, 1\}$): mean ρ by size-mismatch bin. Pairwise concordance is landscape only.

size bin	n	pair	unif	n-wt	$n\|S\|$	rank	$\Delta(nS - \text{unif})$
low (≤ 0.3)	70	0.241	0.651	0.660	0.655	0.658	+0.004
mid	105	0.191	0.658	0.786	0.784	0.647	+0.126
high (≥ 0.7)	70	0.105	0.620	0.905	0.905	0.580	+0.285

Table 41: Fine grid slice: mean $\rho(n|S|)$, pairwise, and $\Delta(n|S| - \text{unif})$ by size bin $\times$ hetero fraction (averaged over feature correlation). Under high size mismatch, Δ *rises* with hetero while pairwise falls—alignment to the mixed pool strengthens even as sites disagree.

	low size			mid			high size	
hetero	$\rho_{n\|S}$	pair	Δ	$\rho_{n\|S}$	pair	Δ	$\rho_{n\|S}$	Δ
0	0.653	0.339	+0.01	0.768	0.297	+0.11	0.893	+0.24
1/6	0.639	0.283	+0.01	0.791	0.245	+0.12	0.886	+0.25
1/3	0.656	0.262	−0.00	0.769	0.215	+0.11	0.888	+0.27
1/2	0.640	0.250	+0.00	0.773	0.197	+0.13	0.892	+0.27
2/3	0.657	0.213	−0.00	0.793	0.145	+0.13	0.921	+0.29
5/6	0.664	0.183	+0.01	0.790	0.134	+0.12	0.925	+0.31
1	0.675	0.157	+0.01	0.804	0.103	+0.15	0.927	+0.37

Table 42: High size-mismatch cells only: $\Delta(n|S| - \text{unif})$ on the hetero $\times$ feature-correlation grid. The size-weighting gain is large and stable across correlation; it grows further as hetero $\to 1$.

hetero \ corr	0	0.15	0.30	0.45	0.60
0	+0.19	+0.23	+0.25	+0.27	+0.26
1/6	+0.25	+0.26	+0.27	+0.26	+0.23
1/3	+0.24	+0.27	+0.26	+0.33	+0.23
1/2	+0.22	+0.28	+0.26	+0.28	+0.31
2/3	+0.29	+0.28	+0.29	+0.31	+0.30
5/6	+0.29	+0.36	+0.30	+0.28	+0.28
1	+0.40	+0.36	+0.37	+0.35	+0.36

Table 43: TabReD federated FSDS: ρ on nine task$\times$split cells. Site sizes are nearly balanced (max-frac $\equiv 0.24$), so $\Delta(n|S| - \text{unif})$ is small; $n|S|$ remains a safe default. The cooking / `num_105` cell is consensus-misaligned (high pair, negative site$\leftrightarrow$whole); see Table 44.

task	split	pair	site$\leftrightarrow$whole	unif	n	$n\|S\|$	rank	best
cooking	num_105	0.884	−0.198	−0.200	−0.191	−0.204	−0.209	n
cooking	num_80	0.383	0.161	0.250	0.305	0.329	0.251	$n\|S\|$
cooking	num_69	0.835	0.468	0.520	0.525	0.530	0.520	$n\|S\|$
ecom	num_109	0.787	0.545	0.642	0.645	0.647	0.635	$n\|S\|$
ecom	num_100	0.825	0.532	0.598	0.596	0.610	0.605	$n\|S\|$
ecom	num_103	0.851	0.470	0.498	0.505	0.516	0.501	$n\|S\|$
weather	num_38	0.598	0.428	0.554	0.514	0.531	0.546	unif
weather	num_44	0.716	0.725	0.836	0.840	0.846	0.827	$n\|S\|$
weather	num_37	0.519	0.283	0.378	0.399	0.389	0.401	med
mean	—	0.711	0.379	0.453	0.460	0.466	0.453	—

Table 44: cooking-time / `num_105`: feature-rank versus random array-split (mean±SD of ρ). The defect is the split axis, not $n|S|$ (aggregator headroom ≈ 0).

protocol	n	pair	unif	n	$n\|S\|$	Δ
feature-rank	20	0.88 ± 0.01	0.07 ± 0.17	0.07 ± 0.17	0.07 ± 0.17	+0.01
random	200	0.26 ± 0.06	0.80 ± 0.06	0.81 ± 0.06	0.80 ± 0.06	+0.00

Table 45: TabReD method comparison: F1 vs linreg top-$|\beta|$ (PermuCATE $n_{\text{perm}} = 50$, RF outcome). Single-batch only—not the federated metric.

task	split	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	$L1_m$	$L1_s$	KO
cooking-time	num_184	0.100	0.200	0.500	0.200	0.400	0.200	0.200	0.400	0.300	0.200
cooking-time	num_38	0.100	0.200	0.600	0.200	0.400	0.200	0.400	0.400	0.000	0.000
cooking-time	num_123	0.100	0.200	0.300	0.200	0.300	0.200	0.200	0.600	0.000	0.000
ecom-offers	num_67	0.500	0.500	0.400	0.500	0.500	0.500	0.500	0.200	0.200	0.000
ecom-offers	num_104	0.500	0.500	0.400	0.400	0.400	0.500	0.500	0.300	0.300	0.200
ecom-offers	num_60	0.400	0.500	0.200	0.400	0.400	0.500	0.400	0.300	0.400	0.100
weather	num_5	0.400	0.300	0.400	0.200	0.400	0.500	0.200	0.300	0.200	0.400
weather	num_84	0.300	0.300	0.500	0.200	0.300	0.500	0.400	0.300	0.500	0.300
weather	num_98	0.400	0.300	0.600	0.300	0.500	0.500	0.500	0.300	0.300	0.400
mean	—	0.311	0.333	0.433	0.289	0.400	0.400	0.367	0.344	0.244	0.178

Table 46: TabReD: mean selection AUC by task (same run as the F1 table).

task	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	$L1_m$	$L1_s$	KO
cooking-time	0.537	0.588	0.673	0.500	0.510	0.583	0.596	0.758	0.416	0.319
ecom-offers	0.797	0.792	0.700	0.626	0.642	0.613	0.806	0.511	0.572	0.361
weather	0.430	0.561	0.729	0.588	0.682	0.687	0.629	0.512	0.510	0.536
mean	0.588	0.647	0.701	0.571	0.611	0.628	0.677	0.594	0.499	0.405

Table 47: TabReD efficiency: mean wall seconds per cell (9 cells). PermuCATE / LOCO / GRF share one causal block (time split ≈ equally).

	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	$L1_m$	$L1_s$	KO	total
sec	0.6	0.6	7.9	20.9	56.0	56.0	56.0	7.5	7.5	7.5	220.5
× vs first	1.0	1.0	14.2	37.6	100.9	100.9	100.9	13.5	13.5	13.5	—

Table 48: TabReD federated FSDS: primary metric is Spearman(agg, whole-batch), not AUC. Whole-batch = FSDS on the pooled site sample.

task	split	pair	ρ_{unif}	ρ_n	$\rho_{n\|S\|}$	ρ_{rank}	best
cooking-time	num_184	0.875	0.114	0.111	0.127	0.119	ns_weighted
cooking-time	num_38	0.885	0.399	0.403	0.400	0.393	n_weighted
cooking-time	num_123	0.812	0.403	0.433	0.433	0.397	ns_weighted
ecom-offers	num_67	0.769	0.321	0.362	0.356	0.300	n_weighted
ecom-offers	num_104	0.768	0.222	0.222	0.233	0.215	ns_weighted
ecom-offers	num_60	0.649	0.099	0.106	0.107	0.071	ns_weighted
weather	num_5	0.791	−0.054	−0.061	−0.057	−0.064	median
weather	num_84	0.767	−0.111	−0.118	−0.108	−0.131	median
weather	num_98	0.440	0.615	0.623	0.631	0.559	ns_weighted

Table 49: TabReD FSDS leave-one-site-out (LOSO): $\Delta_k = \rho(\text{agg}, \text{whole}) - \rho(\text{agg}_{-k}, \text{whole})$ with whole fixed. Report ρ and $\max_k |\Delta_k|$. Random = deployment partition; feature-rank = sensitivity.

protocol	task	n	$\rho_{n\|S\|}$	$\|\Delta\|_{n\|S\|}$	ρ_{unif}	$\|\Delta\|_{\text{unif}}$	$\|\Delta\|_{\text{med}}$
feature_rank	ALL	9	0.243 ± 0.241	0.057 ± 0.017	0.231 ± 0.237	0.047 ± 0.016	0.052 ± 0.029
feature_rank	cooking-time	3	0.300 ± 0.183	0.053 ± 0.008	0.299 ± 0.199	0.045 ± 0.015	0.036 ± 0.013
feature_rank	ecom-offers	3	0.264 ± 0.093	0.057 ± 0.006	0.245 ± 0.088	0.057 ± 0.005	0.061 ± 0.019
feature_rank	weather	3	0.165 ± 0.419	0.060 ± 0.031	0.148 ± 0.400	0.040 ± 0.024	0.060 ± 0.047
random	ALL	45	0.889 ± 0.106	0.054 ± 0.052	0.884 ± 0.115	0.049 ± 0.042	0.064 ± 0.048
random	cooking-time	15	0.807 ± 0.144	0.101 ± 0.068	0.794 ± 0.159	0.090 ± 0.049	0.114 ± 0.047
random	ecom-offers	15	0.921 ± 0.049	0.023 ± 0.013	0.923 ± 0.044	0.022 ± 0.012	0.030 ± 0.017
random	weather	15	0.939 ± 0.033	0.040 ± 0.014	0.934 ± 0.034	0.035 ± 0.011	0.048 ± 0.024

Table 50: Feature-rank cells: LOSO $|\Delta|_{\max}$ by aggregator (one draw per linreg top split).

task	split	$\rho_{n\|S\|}$	$\|\Delta\|_{n\|S\|}$	$\|\Delta\|_n$	$\|\Delta\|_{\text{unif}}$	$\|\Delta\|_{\text{med}}$	$\|\Delta\|_{\text{rank}}$
cooking-time	num_184	0.092	0.054	0.047	0.061	0.027	0.031
cooking-time	num_38	0.437	0.045	0.054	0.042	0.031	0.048
cooking-time	num_123	0.373	0.060	0.062	0.032	0.051	0.053
ecom-offers	num_67	0.347	0.062	0.057	0.062	0.081	0.073
ecom-offers	num_104	0.283	0.059	0.067	0.053	0.042	0.056
ecom-offers	num_60	0.163	0.050	0.050	0.055	0.059	0.077
weather	num_5	−0.025	0.031	0.029	0.022	0.029	0.059
weather	num_84	−0.126	0.055	0.038	0.031	0.037	0.039
weather	num_98	0.645	0.093	0.068	0.067	0.114	0.054